College Physics Experiment

Introduction

The study of physics experiment course in college is different from the experiment course in middle school. Because the physical experiment in the middle school is mainly to expand the vision, enrich the perceptual knowledge and increase the hands-on opportunity; Then help students understand and consolidate the theoretical knowledge taught in class. It is only an adjunct teaching link of physics course teaching. However, the physics experiment course set up in the university stage is an independent and compulsory basic experiment course, besides "university physics", which carries on the scientific experiment basic training for the students, and is the beginning of the systematic experiment skill training for the students in colleges and universities. It is in training students to use experimental means to analysis, observation, discovery and even research, problem-solving ability; It plays an important role in improving students' scientific experiment quality. At the same time, it will also lay a good experimental foundation for students' future study and work.

1. Basic requirements for university physics laboratory courses

How can we make the students master the basic skills of physics experiment through the teaching of physics experiment and achieve the purpose of cultivating high-quality innovative talents? In summary, the following three basic requirements should be achieved through the physics experiment course:

(1) Getting strict and systematic training in the basic knowledge, basic methods and basic skills of physical experiments is the basis for doing physical experiments well.

Basic knowledge includes experimental principles, structure and working mechanism of various instruments, error analysis and uncertainty assessment, expression methods of experimental results, and how to analyze and judge

experimental results, etc.

The basic methods include how to determine the experimental idea and scheme according to the experimental purpose, how to choose and use the instrument correctly, how to reduce all kinds of errors, how to use some special methods to get the results that are usually difficult to get, etc.

Basic skills include various adjustment and testing techniques (rough tuning, fine tuning, collimation, zero setting, reading, scaling...) , electrical technology (identification of components, welding, troubleshooting, safe use of electricity,...) , electronic technology (microcurrent detection, weak signal amplification. .....) , sensor technology (force sensor, displacement sensor, temperature sensor, magnetic sensor, light sensor...) As well as the ability to consult the literature, self-study ability, the ability to work together, the ability to summarize and conclude.

This kind of three basic training may sometimes be boring, but it is absolutely necessary, it reflects the most basic practical ability, so we must first ensure the realization of this requirement. Without this kind of strict basic training, it is difficult to become a high-quality talent.

(2) Learn to use experimental methods to study physical phenomena, verify physical laws, deepen the understanding and grasp of physical theories, and improve the ability to find, analyze and solve problems in practice.

The fundamental purpose of physical experiments is to study physical phenomena and verify physical laws. Consciously learn this ability in the process of learning the "three bases". Although the general "verification experiment" is arranged by the teacher, but the students should carefully understand the mystery of it, should not only according to the prescribed steps to operate, remember the data, get the results even if completed. Ask more questions about why, what would be the problem if you didn't follow the prescribed steps, or could you think of other ways to achieve the same goal?

Under certain conditions, with the consent of teachers, they can also do experiments designed by themselves.

Some unexpected problems are often encountered in the experiment. Although these questions may not be the main subjects of experimental research, they should not be lightly dismissed. This is often a good opportunity to improve your analytical and problem-solving skills. Attention should be paid to observation and timely record. Analyze carefully and conduct further research if necessary. In fact, many of the most important discoveries in the history of science have come about by "accident" under unexpected circumstances.

(3) Develop a scientific attitude of seeking truth from facts and a scientific spirit of active innovation

This is the requirement that should be carried out in the whole teaching process, and it is particularly important in the physics experiment teaching. The most practical and realistic, rigorous and practical scientific attitude can be cultivated in the physics experiment class. Any falsification, manipulation or even falsification of data is not allowed and is relatively easy to detect. There are no "good" or "bad" results. Experimental results that do not conform to the original expectations should not be dismissed, but should also be given special attention, it may be the beginning of a new discovery. As a matter of fact, if you do your experiments carefully, you are bound to find many problems, some of which even teachers may not be able to solve. Therefore, the laboratory should and can be the best place to cultivate students' realistic attitude and innovative spirit.

2. The basic teaching link of college physics experiment course

Physical experiment was conducted under the guidance of the students in the teachers' independent experiment a kind of practice, experiment teaching arrangement may not like books teaching makes all of the students in accordance with the same

content in the same progress, teaching methods are mainly students themselves, completes the experiment content, the task to learn, teachers is the key place to give tips and guidance, therefore to learn physics experiment requires that the students spend more effort, has the strong ability to work independently. The key to learn physics experiment lesson well is to grasp the following three basic teaching links:

(1) Preview before the experiment

The experimental textbook is the instruction book for conducting experiments. It clearly expounds the purpose and requirement of each experiment and the experimental principle. Therefore, we should read carefully before the experiment class and read relevant reference materials if necessary. Basic understanding of the principles and methods used in the experiment, and learn to sort out the main experimental conditions, the key problems in the experiment and the experimental matters needing attention, according to the experimental tasks in the experimental data record book to draw a table for recording data. Some experiments also require students to design their own experimental scheme before class, design their own circuit diagram or optical diagram, self-drawn data tables. For the measuring instrument involved, the introduction of the instrument can be read in the textbook during preview to understand its construction principle, working conditions and operating procedures. Some experiments have software for preview, students can from the computer more clearly see the experiment overview and principle, equipment, etc., some experiments have no preview software, students had better go to the laboratory to observe the object in the specified time, and on this basis to write a preview report, answer preview questions. The preview report mainly includes the following aspects: ① the name of the experiment; ② Purpose of the experiment; ③ Principle abstract :(including the main principle formula, lists the conditions related to the measurement and the rules to be verified. Which physical quantity should be defined as the direct measurement quantity; Which physical quantity is indirect measurement quantity; The methods and measuring instruments used; Electrical experiment should

draw electrical schematic diagram, optical experiment should draw optical path diagram); ④ main instruments and equipment; (5) list the data record form in the experimental record book; ⑥ Answer preview questions. In short, the quality of preview before class is the key to the initiative in this experiment.

(2) Operation in the experiment

The biggest difference between the laboratory and the classroom is that the laboratory has a large number of instruments and equipment and experimental materials. In different laboratories, there are also high-power power supply, water, gas, compressed air, radioactive substances, lasers, inflammable and explosive items or other toxic and harmful items. Therefore, before entering the laboratory, we must understand in detail and strictly abide by the rules and regulations of the laboratory. These rules and regulations are for the protection of personal safety and equipment safety and regulations, violation may lead to accidents, this must be borne in mind first!

Experimental operation is the main content of the experiment. During the experiment, carefully read the notes and instructions for the use of the instrument; Use the instrument correctly under the guidance of the teacher, take care of the instrument, hold it properly and put it safely to prevent damage. For electromagnetics experiment, the instructor must check the correct connection of the circuit before connecting the power supply for the experiment. For those who seriously violate the laboratory rules, the instructor should stop the experiment and deal with them according to relevant regulations.

After entering the laboratory, students should abide by the rules of the laboratory and behave as a scientist would. In order to arrange the instrument, safe operation, pay attention to carefully observe the experimental phenomenon, seriously study and explore the problems in the experiment. Don't expect your experiment to be smooth sailing. When you encounter a problem in your experiment, you should treat it as an

opportunity to learn, analyze and deal with it calmly. When the instrument fails, it is also necessary to learn the troubleshooting method under the guidance of the teacher. In a word, we should focus on the cultivation of experimental ability, rather than simply measure a few data and think that the task has been completed.

Making records of experiments is a basic skill of scientific experiments. During observation and measurement, correct readings should be achieved and objective phenomena and data should be recorded realistically. The name of the experiment, the date of the experiment, the members of the group, and, if necessary, the weather, room temperature, atmospheric pressure, temperature and other environmental conditions shall be indicated in the numbered experimental record book. Then to write down the name, model, specification, number and performance of the instrument used in the experiment, so that it can be used to repeat the measurement and use the accuracy of the instrument to check the experimental results. Take experimental data seriously. Write down raw data with pen and ballpoint pen. If you do remember something wrong, don't correct it. Should be gently on a line, next to write down the correct value (more errors, need to record), so that the correct and false data can be clearly discernible, for the analysis of measurement results and errors when reference.

To learn to analyze the experiment step by step, to eliminate a variety of simple failures in the experiment. At the end of an experiment, there's always data, and how do we know if that data is correct or not? What about the data? Are the results correct? These questions mainly depend on the analysis of the experiment itself to judge, that is, we must analyze the experimental method is correct, how much error does it bring? How much error does the instrument make? How much the experimental environment is affected and so on. The discussion after the experiment is an important link to give play to students' intelligence and improve students' ability to analyze and solve problems. But we should pay attention to, do not make empty comments, should make every effort to quantitatively analyze the problem, do what is justified. Often some students, when the experimental data and theoretical calculation are consistent, will be

satisfied, simply think that they have learned the experiment; Once there is a big difference between the data and calculation, people will feel disappointed, complain about the equipment and even piece together the data. These two attitudes are not desirable in experimental teaching and all experimental research activities. In fact, any theoretical formula is a certain theoretical abstraction and simplification, while the objective reality and the environmental conditions of the experiment are much more complex. The experimental results will inevitably bring the difference from the theoretical formula, and the problem lies in whether the difference is reasonable. So whether the data is good or bad, we should gradually learn to analyze the experiment, find out the reason for success or failure.

The knowledge of error and data processing is the special language of physics experiment. Whether the experiment is done well or poorly; Whether the results of the same physical quantity measured by two methods are consistent; Whether the experiment verifies or does not verify the theory, etc., this cannot be based on feeling, but must use the experimental data and experimental error to make the assertion. Only by understanding and applying this language can we truly immerse ourselves in the experiment and personally feel the joy of success and the confusion of failure.

I hope students can correct their bad habits and cultivate good scientific style from the very beginning. At the end of the experiment, the measured data should be handed over to the instructor for review and signature, and the unreasonable or wrong experimental results should be made up or redone after analysis. Before leaving the laboratory, we should arrange the used instruments and do a good job of cleaning.

(3) Report after the experiment

The experiment report can be written on the basis of the preview report, or it can be rewritten.

As for the experimental report, some students tend to only pay attention to data

processing and experimental results, but pay little attention to the recording of the experiment and the writing of principles and steps. This is very wrong. The writing of experimental report is an important link in training experimental research talents.

Engaging in experimental research generally requires an experimental research record book to record various phenomena and data occurring in the experiment, which is a valuable data for scientific research and is generally kept in the laboratory for a long time. In order to develop a good habit of complete recording, so as to learn the basic skills of engaging in experimental research work, in the experimental report, it is required to record the experimental conditions, experimental instruments, experimental environment, experimental phenomena and measured data in detail.

The results of research work are usually published in the form of papers. In order to train the ability to express the experimental results in a literal way, in the experimental report, it is required to briefly state the purpose, principle and steps of the experiment and discuss them appropriately in their own language.

After the experiment, the experimental data should be processed in time. If there are many corrections to the original records, they should be sorted out and important data should be re-listed. Data processing includes calculation, mapping, error analysis and so on. The calculation should have formula (or calculation examples), and the data substituted should be based, so that others can understand and check easily. Drawing should be in accordance with the rules of drawing, drawing line to rules, beautiful. Experimental results should be given after data processing. The final requirement to write a concise, clear, neat, insightful experimental report.

The purpose of writing laboratory reports is to cultivate and train students' ability to summarize their work or report scientific results in written form. The report is a text report of experimental results, so at least it should be clear handwriting, smooth writing, correct charts, complete data and clear conclusions. The report should give peers with clear ideas, insights and new enlightenment can be considered a successful

report. This is the ability that every college student must have to report work results. Should write on special experiment report paper commonly, give guidance teacher to review when next experiment. The content of the experimental report should generally include:

① Name of the experiment.

② Purpose of the experiment.

③ Experimental principle. Should be based on the understanding of principles in their own language to brief, concise, it figure (light path diagram, schematic circuit diagram or experiment equipment) and texts, lists based on measurement and calculation of the main formula, indicate the formula in the physical meaning and the amount of units, which should satisfy formula established experimental conditions, etc.

④ Content of the experiment. Write out the main process of the experiment and the key points of safety attention. Design experiments should include key steps and precautions.

⑤ Data forms and data processing. The record should contain the serial number, specification and complete experimental data of the main experimental instrument. It is generally required to reflect the complete and clear original measurement data in the form of a list. Requires to write the main process of data processing, curve drawing and error analysis. After the computational processing is complete, the experimental results must be presented in an eye-catching and complete way.

⑥ Summary or discussion. Generally, the content of discussion is not limited. It can be the analysis of observed experimental phenomena, the analysis of results and error reasons, the discussion and comment on the experimental scheme and its improvement opinions, and the experience of doing the experiment and the criticism and suggestions to teachers or textbooks. This is the most open and flexible part of the experiment report, which focuses on reasoning, so it can reflect the level of the

experimenter's observation and analysis ability.

The report should undoubtedly be written in accordance with one's own thinking, and I am particularly appreciated for my own experience.

3. How to learn college physics experiment lesson well

College physics experiment is a practical course, students are in their own independent work process to increase knowledge, improve ability. Therefore, whether the above teaching purpose can be achieved depends to a great extent on the efforts of the students themselves. In view of the current situation that the training of students' experiments is generally weak in middle schools in China, it is not only necessary to spend more effort and hard work to learn physics experiment course well in college. Particular attention should also be paid to improving one's own learning methods.

(1) It is necessary to master the basic experimental methods and measurement techniques

Basic experimental methods and measurement techniques are often used in practical work and are the basis of complex methods and techniques. When learning not only to make clear their basic truth, but also should be gradually familiar with and remember them, and can use these methods and techniques to design some simple experiments. Any kind of experimental method and measurement technology has its application conditions, advantages and disadvantages and limitations, only after doing a certain number of experiments, we can have personal experience of these conditions, advantages and disadvantages and limitations.

(2) to consciously cultivate good experimental habits

Before starting the experiment, you should carefully read the experimental

materials and relevant instrument data, so that you are likely to have a specific and clear understanding of the experimental work to be done; And when you complete an experiment at the same time, must have a complete and real experiment record, so that you can be at any time when you need to refer to these records, so as to process data, analysis results, have enough first-hand information; To help you understand exactly what you're doing. In the process of the experiment, where necessary, should be repeated a number of measurements, read several times, generally always better than read only once (at least to ensure that no wrong reading!) . We should pay attention to recording the environmental conditions of the experiment (such as room temperature, air pressure, temperature, instrument name, specification, range and accuracy, etc.), pay attention to the requirements and characteristics of the placement and use of the experimental instruments, and sometimes even pay attention to correct their incorrect operating habits and posture. Good habits can only be formed after a lot of experimentation, reflection and review. And good experimental habits, to ensure the normal conduct of the experiment, to ensure the safety of the experiment, to prevent the occurrence of errors, have a very good role. If a single experimental habit, because it is easy to understand, and not difficult to master, but easy to be ignored by beginners. Countless practices have proved that the formation of good habits, only in the process of the experiment to consciously exercise their own.

In specific experimental subjects, some experiments need to be completed in cooperation with other students. Discuss and analyze the experimental results with them together, which will make you gain more benefits than you do alone. Sometimes, when you are doing an experiment, if you are limited by time or conditions, you can only finish half or less of the task in time, but as long as you persist in doing it seriously, you will benefit more than rushing and sloppily to finish all the task.

(3) to pay attention to the formation of good analytical habits

In the experiment, we should be good at capturing and analyzing experimental phenomena, strive to independently eliminate all kinds of possible failures in the

experiment, and exercise our ability to independently discover, analyze and solve problems. For example, are the experimental data reasonable and correct? How to judge? What does "good" or "bad" data mean? How reliable and correct are the experimental results? The solution of these problems mainly depends on the analysis of the experiment itself and the process of the experiment to judge; In other words, is the experimental method correct and reasonable? How much error is it likely to introduce? How much error will the experimental equipment bring? The experimental environment, the influence of the conditions will be how.

In order to help beginners overcome the practical situation such as lack of experimental experience and lack of a complete set of analytical methods, physics laboratories, as basic teaching experiments in universities, often arrange a few experimental subjects with very exact theoretical conclusions in the experimental textbooks, so that beginners can easily contact and judge the correctness of the experimental results. But don't be fooled into thinking that the purpose of an experiment is to get a standard result. No matter the experimental results or data are good or bad, we should develop the habit of analysis, and do not jump to conclusions. First, check your own operations and readings; Pay attention to experimental equipment and environmental conditions. If the operation and readings are checked to be correct, then faults may occur in the instrument and the device itself. Small faults, small problems experimenter should strive to eliminate, at least also should pay attention to the teacher or laboratory staff is how to start to solve; The instrument failure, also want to learn the teacher how to determine the instrument failure or fault? How to fix it? It should also be emphasized that whether the fault of the instrument can be found and repaired in time and quickly is also an important performance of a person's experimental ability. Beginners should ask themselves to gradually improve their ability in this respect.

(4) to master the key points of each experiment

Each experiment has more content, first should complete the basic content, which

is not only the basis, but also the focus. So we must pay attention to the purpose of the experiment, which can improve the learning efficiency. After completing the basic content, if time permits, you can try to analyze some possible problems in the following experiment according to your actual situation, such as the accuracy and reliability of the instrument used, and whether the experimental conditions have been met. How to give confirmation? Or further put forward suggestions to improve the experiment, try to do some new experimental content. Of course, it should not be simply repeated.

In a word, physics experiment course has its own characteristics and rules, to learn this course is not an easy thing. It is hoped that students will continue to improve their interest in it in the process of learning.

1. Measurement of physical quantity, measurement error and data processing

Physics experiments are based on measurement. To study physical phenomena, to understand the properties of matter, and to verify the principles of physics all require measurement. Practice has proved that: any measurement result has error, error from beginning to end in all scientific experiments and measurement process. This is because any measuring instruments, measuring environment, measuring personnel and measuring methods can not be absolutely rigorous, which makes the measurement inevitably accompanied by errors. Therefore, it is an essential work in physical experiments and many scientific experiments to analyze the possible errors in the measurement, eliminate their effects as far as possible, and estimate the errors that cannot be eliminated in the measurement results. To this end, we must understand the concept of error, characteristics, causes and estimation methods of relevant knowledge.

This chapter mainly introduces the preliminary knowledge of the measurement of physical quantity, the theory of measurement error, experimental data processing, the expression of experimental results and experimental design, as the basic preparation before entering the experiment. These knowledge should not only be used in every physical experiment, but also be understood and mastered in future scientific experiment work. It is beyond the scope of the course to discuss them in depth because of the wide range of these topics and the number of new concepts. Therefore, we can only focus on introducing some basic concepts and quoting some conclusions and formulas to meet the teaching needs of this course. Because students do not have enough basic knowledge, learning this part of the content will feel some difficulties, coupled with more content, so it is impossible to learn through one or two can master. This part of the content is very important, requiring students to carefully read the textbook on the basis of a preliminary understanding of the problems mentioned, and then combined with each specific experiment to carefully read the relevant paragraphs,

through the use of to master. It should be noted that further discussion of these topics is the task of general metrology and mathematical statistics. For the time being, we can only quote some of their conclusions and formulas. Detailed discussions and proofs are reserved for mathematical statistics.

1.1 Measurement and error

1.1.1 Measurement and units

1.1.1.1 quantities

All the quantities that describe the state and motion of matter are physical quantities. These quantities can only be determined by measurement. The so-called measurement, is to determine the physical quantity to be measured directly or indirectly and take as a standard unit of the same kind of physical quantity for comparison. The process of getting the ratio is called "measurement". This ratio is the value of the physical quantity to be measured, and the corresponding units constitute a complete "physical quantity".

In different periods of human history, different countries and even different regions, there are many different units of measurement for the same physical quantity. For example, the units of length are yards, feet, feet and meters. In order to facilitate the exchange of international trade, science and technology and culture, the unification of units has become widely expected. So international metrology conference in 1960 established the international system of units (SI), which established A seven basic units: length - meters (m), time (s) - seconds, quality - kilogram (kg), current - ampere (A), the thermodynamic temperature - kelvin (K), amount of substance, Moore (mol) and luminous intensity, candela (CD), also provides two auxiliary units: plane Angle - arc (rad) and solid Angle (sr). All other physical quantities (such as force, energy, voltage, magnetic induction, etc.) are derived from these basic units and auxiliary units.

1.1.1.2 Direct measurement and indirect measurement

According to the definition of metrology: measurement is the whole operation

process for the purpose of determining the value of the measured object. Measurement is divided into direct measurement and indirect measurement.

"Direct measurement" refers to a measurement that directly compares the physical quantity to be measured with the selected standard units of similar physical quantities to obtain the size of the measured value. It doesn't have to do any functional calculations. For example, the length is measured with a steel ruler, the mass of the object is measured with a balance, and the current in the line is measured with an ammeter.

"Indirect measurement" refers to the measurement of other quantities that have a functional relationship with the measurement and then obtain the size of the measured value through calculation. For example, by measuring the length to determine the rectangular area; Indirect measurements are made by measuring the voltage at both ends of a conductor with a voltmeter, by measuring the current passing through the conductor with an ammeter, and by calculating the resistance of a conductor with a known formula $\boldsymbol{R} = \boldsymbol{U}/\boldsymbol{I}$.

As can be seen from the above example of measuring the resistance of a conductor, some physical quantities can be measured either directly or indirectly, depending on the instrument and measurement method used. With the development of measurement technology, more and more instruments are used for direct measurement. But in the physical experiment, there are many physical quantities still need to be measured indirectly.

The measurement result should give the measured value, which includes two parts: the value and the unit (the value without the unit can not be the quantity value). In fact, the instrument is the physical embodiment of the unit in the measurement.

1.1.1.3 Equal accuracy measurement and unequal accuracy measurement

If a physical quantity is repeatedly measured, and the conditions of each measurement are the same (such as a group of instruments, the same measurement method, the same observer and the same environmental conditions, etc.), a set of

measured data are respectively $x_1, x_2, \mathrm{L}, x_n$. There is no good reason to think that one measurement is more accurate than the other, even though the results are not exactly the same. It can only be assumed that they measure exactly the same degree of accuracy. Therefore, the measurement with the same degree of accuracy is called equal precision measurement. Such a set of numbers is called the equal-precision measurement series (abbreviated as the measurement series). In all the measurement conditions, as long as one of the changes, the measurement is an unequal precision measurement.

In the physical experiment, all the requirements of multiple measurements refer to the measurement of equal precision, and the measurement conditions of equal precision should be kept unchanged as far as possible. Strictly speaking, it is very difficult to keep the measurement conditions unchanged during the experiment, but when the change of a certain condition has little effect on the measurement results, it can still be regarded as equal precision measurement. In this chapter we discuss all measures of equal accuracy unless otherwise specified.

1.1.2 Definition and classification of errors

1.1.2.1. Definition of error

The size of the physical quantity to be measured (i.e. the truth value) is an objective existence. However, when it comes to specific measurements. After certain program design, the use of certain experimental methods, under certain conditions, with the help of instruments by the experimenter to carry out and complete. Although we did everything possible to improve the design of the experimental scheme, improve the accuracy of the instrument and the level of the surveyor. However, there is always a limit to the improvement of the accuracy of the instrument. The experimental method cannot be perfect, and the technical level of the measuring personnel cannot be improved infinitely, which makes the measurement results differ from the objective truth value to a certain extent. The measurement error is the difference between the measurement result and the measured true value (or the agreed

true value). The size of the measurement error reflects the accuracy of the measurement result. Measurement error can be expressed in terms of absolute error or relative error. The absolute error Δx is equal to the difference between the measured result (x) and the truth value to be measured (or the convention truth value, denoted by x_0). That is:

$$\Delta x = x - x_0 \qquad (1)$$

Measurement error exists in all measurement, throughout the measurement process. With the continuous development of science and technology, measurement error can be controlled less and less.

The truth value x_0 to be measured refers to the value objectively possessed by the physical quantity to be measured at a certain time and in a certain state, called the truth value. It is an ideal concept and is generally unknowable. To do this. There are three types of truth values we usually refer to:

(1) The theoretical truth value or the defined truth value, such as the sum of the three interior angles of a triangle is equal to 180^0 and so on;

(2) The agreed truth value of metrology: the value agreed by the resolution of the International Metrology Conference. As the unit standard of the basic physical quantity introduced in front, as well as the basic physical constant agreed by the General Assembly. It should be pointed out that, since these fundamental constants can only reflect the measurement level of the general assembly at that time, obviously they also contain some error; Because the error is much smaller than the error of ordinary laboratory measurements, they are accepted as convention truth values. With the passage of time and the continuous improvement of measurement technology, these basic constants will be more perfect and closer to their true values.

(3) the relative true value (or actual value) of the standard: usually when measuring, it is impossible to use the measuring instruments one by one to directly calibrate with the national or international standards. It is a series of step-by-step

calibration through a multi-stage metrological verification network. Therefore, the value of the standard one level higher than the calibrated instrument is taken as the relative truth value of the standard (also known as the actual value). For example, if the current of A circuit measured by A 0.5 ammeter is 1.200A and A 0.2 ammeter is 1.202A, the latter can be considered the actual value of the former.

In the actual measurement work, the actual value of the measurement or the modified arithmetic mean value is often used to replace the truth value, also known as the conventional truth value.

1.1.2.2. Error expression method

The measurement error not only reflects the deviation of the measurement result from the true value (that is, reflects the accuracy of the measurement result), but also reflects whether the measurement result is larger or smaller than the true value, and has the same units as the measured one. Because it is compared with the true value, it is also called absolute error, error for short.

In order to comprehensively evaluate the merits and demerits of the measurement results, it is necessary to consider the size of the measurement itself. For example, to compare two different physical quantities, such as 20mm and 2mm thick plate, the absolute error measured by the micrometer is 0.004mm. If the absolute error is used to evaluate, the measurement error is the same. Obviously, the absolute error does not reflect their essential characteristics. In addition, to compare the advantages and disadvantages of the measurement of two different physical quantities, such as a length of 20.000mm, the absolute error is 0.005mm; The mass of an object is 17.03g and the absolute error is 0.02g, which cannot be compared because the absolute error and units are not the same. Based on the above situation, we need to introduce the concept of relative error. Relative error E is defined as the ratio of absolute error Δx to the best value measured $\bar{x}$, that is:

$$E = \frac{\Delta x}{\bar{x}} \times 100\% \qquad (2)$$

Relative error E_0 is often expressed as a percentage. As can be seen from the above equation, relative error is a pure number without units, so it can not only evaluate the measurement of the same kind of physical quantity with different quantity values, but also evaluate the measurement of different kinds of physical quantity, so as to judge the advantages and disadvantages between them.

Sometimes there are accepted values or agreed values measured, then percentage error is used to express, which is defined as:

$$E_0 = \frac{|\text{测量最佳值} - \text{公认值}|}{\text{公认值}} \times 100\% \qquad (3)$$

1.1.2.3 Classification of errors

There are many reasons for the generation of measurement error. According to the nature and causes of measurement error, it can be divided into systematic error and random error.

(1) Systematic error

The component of measurement error that remains constant or changes in a predictable manner over the course of multiple measurements of the same measurement. The main reasons for the system error are as follows:

1) Inherent deficiencies of the instrument. For example, the calibration is wrong; Zero is not adjusted; The level or alignment of the instrument is not adjusted; Weight not calibrated, etc.

For example, according to the national standard of measurement, the error $\pm 2mg$ of the third class weight of 50g is allowed. When the actual value of a weight is 49.998g, it is a qualified product. But when the experimenter made the measurement using this weight, which has a nominal value of 50mg, it introduced an error of 0.002g.

2) The experimental method is not perfect or the theory based on which the method is based is approximate. For example, the buoyancy of the air is not

considered when weighing, and the influence of the internal resistance of the meter is not considered when measuring the resistance by voltammetric method.

3) Environmental influence or failure to use the instrument in accordance with the specified conditions. For example, the standard battery is based on electromotive force value at 20^0C as the nominal value, using at 30^0C,without being corrected, will introduce the system error.

4) the physical or psychological characteristics of the experimenter, or the error introduced by lack of experience. For example, some people used to squint reading, it will make the value of the estimation of large or small.

The changes of many systematic errors are very complicated. Whether the identification and elimination of systematic errors is closely related to the experience of the experimentors. For beginners, it is important to gain experience in this area from the beginning.

(2) Random error

The component of measurement error in which the absolute value and sign change in an unpredictable way during multiple measurements of the same measurement. Random error is impossible to correct. The generation of random error is caused by small changes in various factors in the experiment. For example, the variability of the experimental device and measuring mechanism in each adjustment operation, the variability of the indicated value of the measuring instrument, and the variability of the observer in judging and estimating the reading, etc. The combined effect of these factors makes the measurement value fluctuate around the mean value of the measurement, and this variation is the random error of each measurement. The occurrence of random error is not regular for a certain measurement value, and its size and direction are unpredictable. However, when a certain physical quantity is measured for many times enough, it will be found that their random error obeys a certain statistical law. A common situation is that the probability of positive direction error and negative direction error travel is roughly equal, the probability of error of

small value is larger, the error of large value usually does not appear in the absence of error. This rule is more obvious when the number of measurements is more, which is the most typical distribution law of random error - the normal distribution law. So we can use statistical method to estimate its influence on the measured value.

In the whole process of measurement, in addition to the errors of the above two properties, there may also be errors in reading, recording, instrument damage, improper operation and other measurement errors. Errors are not errors, and these erroneous data should be eliminated during processing.

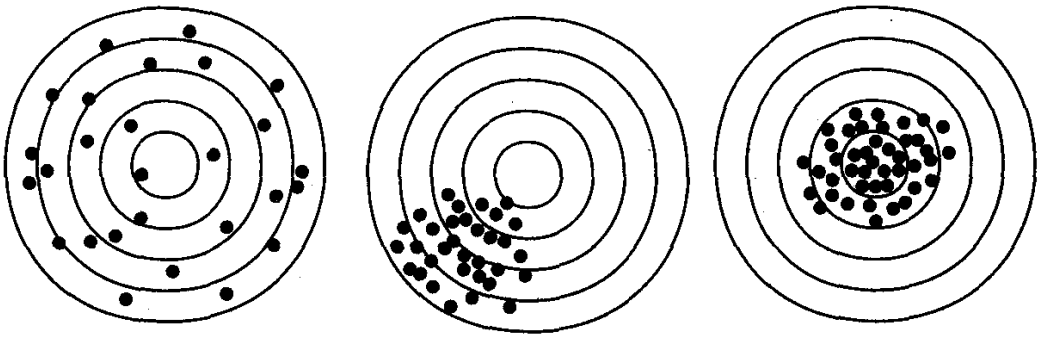
We usually use precision to reflect the degree of error in the measurement results. The accuracy of small error is high, the accuracy of large error is low. But here precision is a general concept, it is not clear what kind of error is described. In order to make precision concrete, precision can be divided into:

Precision: Indicates the degree of random error in a measurement result. It refers to the degree of conformity between the results (the degree of dispersion of the measurement value) when the measurement is made for many times under the specified measurement conditions. Accuracy for short.

Correctness: Indicates the degree of systematic error in measurement results. It reflects the synthesis of all systematic errors in the measurement results under the specified conditions.

Accuracy: Indicates the degree of agreement between the measured result and the "true" value being measured. It reflects the synthesis of systematic error and random error in the measurement results. Measurement accuracy is also called measurement accuracy.

Figure 01-1 illustrates the comparison by taking target shooting as an example. The bull 's-eye in the picture is the target.



(a) High correctness & low precision (b) Low correctness & high precision (c) High Correctness & precision & accuracy

Figure 01—1

1.2 Error treatment

This paper introduces the basic concept of measurement error and emphasizes the inevitability and universality of error. According to the nature of the influence of error on the measurement results, it can be divided into two categories: systematic error and random error. This section mainly introduces the treatment of these two types of errors.

1.2.1 System error

The causes of systematic errors may or may not be known. The system error includes the determinate system error component and the undeterminate system error component.

1.2.1.1. Determined system error

Refers to the absolute value and sign are determined, can be estimated component of the system error; For example, for a third weight with a nominal value of 50mg, there is no way to know what the error value of that weight is. We only know that the limit of undetermined system error caused by it to the measurement results is 2mg. However, if the higher weight is calibrated before use, the determined system error value can be obtained.

1.2.1.2 Undetermined system error

A systematic component of error whose sign or absolute value is not determined. You can

generally only estimate the limit. For example, the accuracy index of the instrument at the factory is indicated by the symbol meter. It only gives the limit range of the error of this type of instrument. However, when using the instrument, the experimenter does not know the exact size, positive and negative error of the instrument, but only knows that the accuracy of the instrument will not exceed the limit of the instrument (for example, the limit of error of the three class weight with a nominal value of 50mg is 2mg, as mentioned above). Therefore, this kind of systematic error is usually limited to its limit range, and cannot be corrected because it is impossible to know its exact size and positive and negative. For the undetermined system error, we generally only consider the (maximum) allowable error meter (instrument error for short) of the measuring instrument in the physical experiment.

Generally speaking, for systematic errors, the instrument can be calibrated before the experiment, and the experimental method can be improved. In the experiment, some methods are adopted to compensate and eliminate the system error. Modify the experimental results after the experiment. We should anticipate and analyze all possible factors of system error and try to reduce them. The quality of an experimental result often depends on whether the systematic error has been found or eliminated as far as possible. In the future experiment, the measurement results should be corrected for the determined system error. For the undetermined system error, the error limit is estimated as far as possible to master its influence on the measurement results. We will analyze and discuss the systematic error according to the specific situation of each experiment in the future experiment class.

1.2.2 Random error and its estimation

The properties of random error and system error are different, and the methods to deal with them are also different. Suppose that we have cut the system error in the experiment to negligible levels, such as through measuring accuracy, caused by various factors change small unpredictable differences between measured value and get a series of measurements for ($x_1, x_2, \mathrm{L}, x_n$) we care about is the most close to the true value of the

value of the (called the best estimate of true value known as the columns of the measurement results in metrology expected estimate)? And how to make a proper evaluation of the quality of the measurement data in the measurement column?

1.2.2.1 Statistical law of random error

Both practice and theory prove that most of the random errors of measurement obey statistical law. The error distribution is shown in Figure 01-2. The abscissa $\Delta x = x_i - x_0$ represents the error, where x_0 is the measured truth value. The ordinate $f(\Delta x)$ is a probability density function related to the probability of errors. Applying the mathematical method of probability theory, we can derive:

$$f(\Delta x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{\Delta x^2}{2\sigma^2}}$$

This distribution is called the normal distribution. The characteristic quantity σ in the above equation is:

$$\sigma = \lim_{n\to\infty}\sqrt{\frac{\sum(x - x_0)^2}{n}} \qquad (4)$$

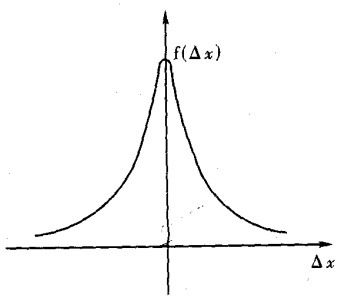


Figure 01—2

σ is called the standard error of a single measurement.

Typical random errors that follow a normal distribution have the following characteristics:

(1) Unimodality. The probability of an error with a small absolute value is greater than the probability of an error with a large absolute value;

(2) Symmetry. Positive and negative errors with equal absolute values have the same probability of occurrence;

(3) Boundedness. Under certain measurement conditions, the absolute value of the error will not exceed a certain limit.

(4) Compensation. The arithmetic mean of random error tends to zero as the number of measurements increases. That is:

$$\lim_{n\to\infty}\frac{\sum \Delta x_i}{n}=0$$

190/5000

What is the best value that is close to the truth, given that the measurement is inevitably subject to random error, so that the value of each measurement is different?

1.2.2.2 The best estimate of truth value -- arithmetic mean

Random error has an extremely important characteristic: compensatory. That is, in a series of measurements of equal precision, because the error of each measurement value is sometimes large and sometimes small, and is positive and negative, the arithmetic mean value of the error tends to 0 more and more with the infinite increase of the measurement number n. According to this property, we can obtain the best estimate of the truth value.

Set a series of equal precision measurement values as: $(x_1, x_2, \mathrm{L}, x_n)$

Then the arithmetic mean of the measured values in this column is:

$$\bar{x}=\frac{1}{n}\sum_{i=1}^{n}x_i \qquad (i=1,2,\mathrm{L},n) \qquad (5)$$

The (absolute) error of each measurement is: $\Delta x_i = x_i - x_0$

Where x_i is the ith measurement value, and the sum of the (absolute) errors of n measurements is obtained as follows:

$$\sum_{i=1}^{n}\Delta x_i=\sum_{i=1}^{n}x_i-nx_0$$

Divide both sides of the equation by n to get:

$$\frac{1}{n}\sum_{i=1}^{n}\Delta x_i = \frac{1}{n}\sum_{i=1}^{n}x_i - x_0 = \bar{x} - x_0$$

When the measuring times $n \to \infty$, because the random error is compensatory, there is $\lim_{n\to\infty}\sum_{i=1}^{n}\Delta x_i \to 0$, thus $\bar{x} \to x_0$

Therefore, under the premise that the systematic error has been eliminated, the more times of measurement, the closer the arithmetic mean value is to the true value. Therefore, we can consider the arithmetic mean $\bar{x}$ of multiple measurements to be the best estimate of the truth value. That is, the expected estimate of the measurement result of the measurement column. But how much is the random error in the measurement? And how do we express that?

1.2.2.3 The standard error from the probability distribution

（1）Measuring times $n \to \infty$

As shown in Fig. 01-3, the value of σ equal to the magnitude of Δx value at the abscissa of the inflection point of the curve. It is easy to see from the figure that it is an important parameter representing the dispersion of measured values. A small σ value means a small dispersion degree of measured values, while a large σ value means a large dispersion degree of measured values. σ is called the standard error of a single measurement, or the standard error of a normal distribution. The random errors of most measurements follow this distribution. It means that the probability of the random error of the measurement value between [Δx , $\Delta x + d(\Delta x)$] cells is $f(\Delta x)\, d(\Delta x)$, so the probability of the random error of the measurement value is between (a, b) region is:

$$p(a < \Delta x < b) = \int_a^b f(\Delta x)\, d(\Delta x)$$

If a curve is used to illustrate the probability density distribution function, we can get:

In the Δx $f(\Delta x)$ curve, the shape of the curve is related to the σ value, and it is easy to see in the figure that the curve with small σ value is more sharp. A curve with a high σ value is relatively flat. The sharper the curve peak is, the smaller the error set of the measured values is, the smaller the dispersion of the measured values is and the better the repeatability is. The precision of measurement is high. On the contrary, if the σ value is large, the curve is flat, and the dispersion of the measured values is large. The precision of measurement is low. Therefore, as the standard error of the eigenvalue of the probability density distribution function in the random measurement, Its statistical significance is relatively clear. At present, standard error σ has been widely used to evaluate the quality of measurement.

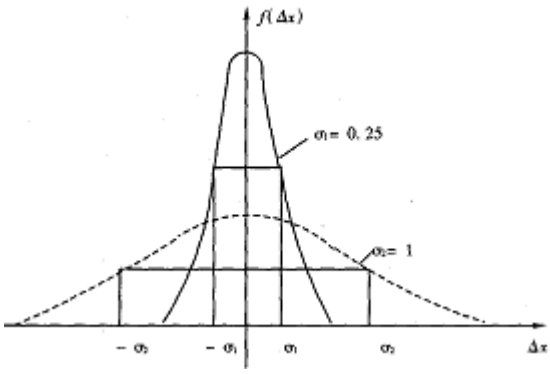


Figure 01—3

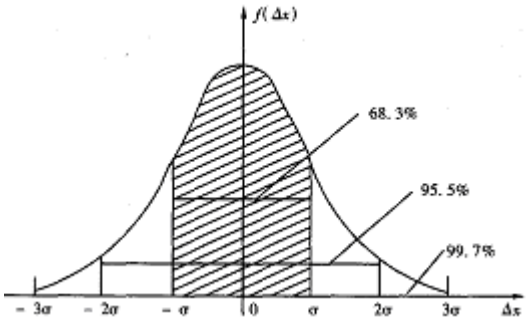


Figure 01—4

The probability meaning of standard error σ is that the number of measurements whose error is within the range of $(-\sigma,+\sigma)$ accounts for the total number:

$$p(-\sigma < \Delta x < +\sigma) = \int_{-\sigma}^{+\sigma} f(\Delta x)d(\Delta x) = 68.3\%$$

From the graph of the probability density distribution function, let the area under the curve be 1, that is, 100%. The area under the curve between $(-\sigma,+\sigma)$ is 68.3%.

Thus, the standard error $\boldsymbol{\sigma}$ of a single measurement means that the probability of the error of any measurement value in the measurement column falling within the interval $(-\sigma,+\sigma)$ is, or the probability (or possibility) of any measurement value in the measurement column falling within the interval $(x_0 - \boldsymbol{\sigma}, x_0 + \boldsymbol{\sigma})$ is 68.3%. The

interval [-σ, +σ] is called the confidence interval, and the corresponding probability P (σ) = 68.3% is called the confidence probability. Obviously, if the confidence interval expands, the confidence probability increases. The confidence interval is [-2σ, +2σ] and [-3σ, +3σ], and the corresponding confidence probability P (2σ) = 95.4% and P (3σ) = 99.7%. Obviously the probability that the absolute value of the measurement error is greater than is only. Therefore, it is also called limit error, as shown in Fig. 01-4.

It should be pointed out that the probability density distribution in most experimental measurements is normal distribution, but the normal distribution is not the only distribution in experimental measurements. Here we are only dealing with the case where the random error is normally distributed. Since x_0 can't be measured, $\Delta x_i = x_i - x_0$ and σ can't be calculated, how do we estimate the magnitude of random error?

(2) When the number of measurements is finite

Since the number of measurements in an experiment is always finite, in the case of a finite number of measurements (in university physics experiments, usually $5 < n \leq 10$), the truth value is unknown and the error $\boldsymbol{\Delta x_i = x_i - x_0}$ cannot be calculated. According to mathematical statistical theory. We replace the true value $\boldsymbol{x}_0$ with the arithmetic mean of the corresponding measurement column $\bar{x}$, and introduce the difference between each measurement value in the measurement column and the arithmetic mean -- deviation $\boldsymbol{\nu}_i = x_i - \bar{x}$ to estimate the standard deviation of the finite measurement column. Then, the parameters S_x representing the dispersity of measurement are calculated with the following Bessel formula, that is:

$$S_x = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n-1}} = \sqrt{\frac{\sum \boldsymbol{\nu}_i^{\,2}}{n-1}} \qquad (6)$$

S_x is called standard deviation of a single measurement, or standard

deviation for short. Used to characterize the dispersion degree of the results when the same measurement is made n times. Apparently as $x \to \infty, S_x \to \boldsymbol{\sigma}$. So, the S_x value is an assessment of the $\boldsymbol{\sigma}$. But it should be noted that S_x is only an estimate of σ as $n \to \infty$. At present, all kinds of function calculators have the function of calculating statistical error, which can directly obtain the required values, such as the arithmetic mean and standard deviation of the measurement column. Students should read the instructions of the calculator carefully and skillfully use the functional calculator, which will bring great convenience to the data processing work of the physical experiment.

It can be proved that:

$$S_{\bar{x}} = \frac{S_x}{\sqrt{n}} = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n(n-1)}} \qquad (7)$$

1.3 Representation of direct measurement results and assessment of total uncertainty

1.3.1 General representation of measurement results

The final expression of the experimental measurement result must have ① the measured value; ② unit of measurement; ③ Total uncertainty of measurement results. The three elements mentioned above are called the three elements for the correct representation of measurement results.

Assume that equal precision measurement is carried out for n repeats of X to be measured: $x_1, x_2, x_3, \cdots\cdots x_n$

Then the complete expression form of measurement results is as follows:

$$X = x \pm \Delta \qquad (8)$$

The above formula indicates that the probability that the measured truth value falls outside the range of ($x - \Delta, x + \Delta$) is small. Δ means the degree of uncertainty of

the measured value due to measurement error, which is characterized by measuring the true value of the value of the scope of the assessment. It can be simply understood as the error range corresponding to a certain confidence probability.

For example, when $n \to \infty$, there is no systematic error in the measured data, but only random error. When the measured value is normally distributed, if the confidence probability $p = 0.683$ (i.e. 68.3%) is used, then $\Delta = \boldsymbol{\sigma}$, the probability that the measured true value or the measured value exists within the interval ($x \pm \boldsymbol{\sigma}$) is 68.3%; If the confidence probability $p = 0.955$ (i.e. 95.5%) is used, then $\Delta = 2\boldsymbol{\sigma}$; If the confidence probability $p = 0.997$ (i.e. 97.7%) is used, then $\Delta = 3\sigma$, and vice versa.

Uncertainty can only make an appropriate evaluation of the reliability of the measurement results on an order of magnitude, so its value does not need to be calculated too accurately. It is usually agreed that the uncertainty and relative uncertainty are usually one or two digits, and when the first digit of the uncertainty is small, two digits are often taken.

In direct measurement, the measured quantity x is generally the arithmetic mean of the measurement column $\bar{x}$. In some experiments, only a single measurement is made, or only a single measurement can be made due to the limitations of experimental conditions (such as the temperature measured at a certain moment in the cooling process of the object), then the measurement result shall be the measurement value of a single measurement x_t. Finally, when the measured quantity x in the direct measurement result is expressed, it is usually necessary to subtract the determined systematic error component (that is, the estimable error component with both absolute value and sign determined) from the arithmetic mean of the measurement column or the single measurement value x_t to obtain x. Such as screw micrometer zero correction, voltammetry measurement resistance in the meter resistance correction.

1.3.2 The concept of uncertainty

As expressed in Equation (8), it is difficult to determine the true value to be measured due to the existence of measurement error. An estimate representing the degree of uncertainty of the measured true value within a range of values, known as the uncertainty of the measured value. That is to say, uncertainty is the part that cannot be corrected in the measurement result, which reflects the error range that the truth value of the measurement cannot be confirmed. According to the document "Guidelines for Measurement Uncertainty Representation ISO1993 (E)" jointly published by 7 international organizations including the International Organization for Quantification, the uncertainty of measurement results generally contains several components, and there are two kinds of evaluation methods for each component of uncertainty:

①Repeat measurement of class A components Δ_A calculated by statistical methods,

② Class B components Δ_B estimated by other methods.

Statistical method refers to the data obtained through certain experiments, using statistical principles to obtain a certain uncertainty component. For example, a series of measurement data are obtained through repeated measurements, and the standard deviation of a single measurement S_X is estimated according to Bessel's formula, which reflects the uncertainty of the value of measurement results, that is, the degree of dispersion of possible values. If you multiply the standard deviation S_X by some coefficient t, the limit of error tS_x will also reflect the uncertainty of the value of the measurement. Both of these parameters can describe the degree of dispersion of measurement results. So. The value of the multiplied factor should also be given when the uncertainty component is given.

The non-statistical method is to estimate the uncertainty based on experience or other information through careful analysis of the measurement process. For example,

the error limit of measurement is given by using high precision standard physical comparison. Then the corresponding uncertainty components are estimated. The standard deviation of non-statistical uncertainty can be obtained according to the error limit and the selected confidence probability. There is no fixed way to follow the non-statistical method, and it is inevitably affected by people's subjective factors, so it is difficult to estimate its reliability.

The uncertainty components of the measurement results were combined by square and root methods. It's called the resultant uncertainty. The uncertainty expressed by the experimental results should be the combined uncertainty, or the total uncertainty obtained by multiplying the combined uncertainty by a certain factor.

In 1986, the Chinese Academy of Metrology issued a notice to adopt uncertainty as the index name of error numbers. On October 1, 1992, China began to implement the national metrology technical specification JJG1027-91 "Measurement Error and Data Processing (Trial)", which stipulated that the final expression form of measurement results should be expressed by total uncertainty or relative uncertainty.

1.3.3 Simplified treatment of total uncertainty evaluation in university physics experiments

This is seen in the above statement of the concept of uncertainty. In fact, it is not very simple to give the uncertainty scientifically and completely. Its requirements go far beyond the scope of this course. In order to have the basic concept of uncertainty, so as to facilitate the connection with scientific experiments, but not too complicated and difficult to operate. Therefore, it is very necessary to simplify the teaching experiment. Quantitative reference international plan organization seven jointly published by the international organization of the measurement uncertainty said guide ISO1993 (E) and standard JJG1027 measurement technology in China - 91, combining the reality of university physics experiment teaching, in university physics experiment teaching, we use a simplified has certain uncertainty estimation method of approximation.

In the representation of the measurement results of physical experiment, the total uncertainty Δ is divided into two components from the estimation method. That is, the class A components Δ_A calculated by statistical methods and the class B components Δ_B estimated by other methods are repeatedly measured. They are synthesized into the total uncertainty Δ (the uncertainty and its components below generally refer to the total uncertainty and its components) by the method of square and root:

$$\Delta = \sqrt{\Delta_A^2 + \Delta_B^2} \qquad (9)$$

1.3.3.1 Class A components of the total uncertainty Δ_A

In the actual measurement of physical experiment, only a limited number of measurements can be made, and the measurement error does not completely obey the law of normal distribution, but the law known as t distribution. In this case, the estimate of the measurement error is multiplied by a factor based on Equation (6). Is measured in the same conditions of the same measurement as A n time, if only to calculate the total uncertainty of Δ class A component Δ_A, then it is equal to the standard deviation of the measured value S_x multiplied by a factor $t_p(n-1)/\sqrt{n}$, namely:

$$\Delta_A = \frac{t_p(n-1)}{\sqrt{n}} S_x \qquad (10)$$

Where, $t_p(n-1)$ is a quantity related to the number of measurements n and the confidence probability P. After the probability P and the number of measurements n are determined, $t_p(n-1)$ is determined. The value of the factor $t_p(n-1)$ can be retrieved from a special table. When p=0.95, part of the data of $t_p(n-1)/\sqrt{n}$ can be found in the following table:

Measuring times n	2	3	4	5	6	7	8	9	10
$t_p(n-1)/\sqrt{n}$	8.98	2.48	1.59	1.24	1.05	0.93	0.84	0.77	0.72

In the physical experiment, when the same measurement is repeated for several times, the general measurement number n is not more than 10. As long as the measurement number n>5, it can be seen from the above table that, when the factor $t_p(n-1)/\sqrt{n}$ is approximated to 1, the error is not very large. In this case, Equation (10) can be simplified as:

$$\Delta_A = S_x \qquad (11)$$

That is, the standard deviation of a single measurement S_x is taken as the total uncertainty component of multiple measurements calculated by statistical methods. Class A components in standard deviation S_x and total uncertainty Δ_A are two different concepts. In the physical experiment, when 5<n≤10, S_x is taken as Δ_A. The values that appear below, unless otherwise noted, represent values. Of course, when the number of measurements n is not in the interval or the error estimation is required to be more accurate, the corresponding factor value should be found from the relevant data table. This is one of the most convenient simplified processing methods, because when you can ignore Δ_B, there is $\Delta = \Delta_A = S_x$, then the probability (probability) of the measured truth value falling within the range of $x_0 \pm S_x$ is greater than or close to 95%. S_x that appears below, unless otherwise noted, represent the values of Δ_A. Of course, when the number of measurements n is not between $5 < n \le 10$ or the error estimation is required to be more accurate, the corresponding factor value should be found from the relevant data table.

1.3.3.2 Class B components of the total uncertainty Δ_B

In college physics experiments, the error or error limit of instruments is often encountered. Δ_E refers to the accuracy level or allowable error range of measuring instruments and instruments according to the national standards, which is given by the manufacturer or simplified by the laboratory in combination with the specific measurement methods and conditions. The error of instrument Δ_E is a simplified expression in college physics experiment teaching, which is usually equal to the error limit or basic error limit of instrument and instrument. The reasons for the errors of many measuring instruments and instruments and the calculation and analysis of the specific error components are mostly beyond the scope of the requirements of this course. When most instruments and instruments in the physical laboratory are used to measure the same measurement under the same conditions for many times, the random error component of the measurement is generally much smaller than the basic error limit or the error limit of the indicated value. Some other instruments and instruments are difficult to guarantee the measurement under the same conditions or the specified normal conditions in the actual use, the measurement error in addition to the basic error or value error also includes variation and other components. Therefore, we agree that in most cases of university physics experiments, the total uncertainty is directly simplified as the class B component ΔB estimated by non-statistical methods.

1.3.3.3 The composition of total uncertainty

From (9)、（10）和（11）we have：

$$\Delta = \sqrt{\Delta_A^2 + \Delta_B^2} = \sqrt{S_x^2 + \Delta_E^2} \qquad （5 < n \le 10） \qquad (12)$$

If $S_x < \frac{1}{3}\Delta_E$, or the estimated Δ_A has little effect on the final experimental result, or if only one measurement is made due to the condition, Δ can be simply expressed by an instrumental error meter. In a single measurement, the standard deviation cannot be calculated using Bessel's formula (6). This does not mean that random error does not exist, but only indicates that the random error is relatively small in this case. The accuracy of the measuring instrument itself is difficult to reflect the existence of random error, that is, the error limit of the instrument has

included the random error. When only one measurement is required in the experiment, taking the value of the instrument does not mean that the value of the instrument is smaller than that of the multiple measurements. It only means that the difference between the results estimated by the instrument Δ_E and the formula $\sqrt{\Delta_A^2+\Delta_E^2}$ is not large, or that the estimation requirements of the measured can or must be relaxed in the whole experiment. N increase in number of times, the measured in (12) estimate Δ although generally little change, but the probability of true value falls within the scope $x_0 \pm \Delta$ is closer to 100%. This means that when n increases, the magnitude range of the truth value is actually smaller, so the measurement result is more accurate.

1.3.4 Calculation examples of uncertainty in direct measurement

Example 1: The initial reading of a micrometer ($\Delta_E = 0.004mm$) is given as -0.010mm, and the diameter of the wire is measured with it for a total of 7 times. The data are shown in the table below. Try to find the measurement result, and write the complete measurement result expression.

i	1	2	3	4	5	6	7
d（mm）	0.507	0.510	0.502	0.511	0.509	0.500	0.512

$$\bar{d} = 0.5073\text{mm}$$

$$S_d = 0.00460848 = 0.0046\text{mm}$$

Correcting fixed system errors $\quad d = \bar{d} - (-0.010) = 0.5173\text{mm}$

$$\Delta = \sqrt{0.0046^2 + 0.004^2} = 0.0061 = 0.006\text{mm}\ ;$$

Note that the significant digits of uncertainty Δ and relative uncertainty E_r are generally one to two digits (when first digit ≤ 3, it is generally one digit; Otherwise take two); The valid number representing the measurement result, whose mantissa should be aligned with the mantissa of the uncertainty:

$$d = 0.517 \pm 0.006\text{mm}$$

$$E_d = \frac{\Delta_d}{\bar{d}} = 0.012 = 1.2\%$$

1.4 The Result of Indirect Measurement and Evaluation of Uncertainty

1.4.1 Two Expressions of Uncertainty

The result of physical experiment is usually obtained by indirect measurement, which is calculated by certain functions based on direct measurement.

Supposing the quantity N of indirect measurement is a function of quantities of direct measurement,

$$N = f(x, y, z, \cdots\cdots)$$

where N is the result of indirect measurement, $x, y, z, \cdots\cdots$ are results of direct measurement. These are independent quantities. Supposing $\Delta_x, \Delta_y, \Delta_z \cdots$ are uncertanties of $x, y, z, \cdots\cdots$, they will effect the result of indirect measurement. Then there is a corresponding uncertainty Δ_N about N. We can use the following formula basing on the root-sum-square method to calculate Δ_N.

$$\Delta_N = \sqrt{(\frac{\partial f}{\partial x})^2(\Delta_x)^2 + (\frac{\partial f}{\partial y})^2(\Delta_y)^2 + (\frac{\partial f}{\partial z})^2(\Delta_z)^2 + \cdots} \qquad (13)$$

If we take the natural logarithm firstly, then use total differential, Δ_N can be expressed as,

$$\frac{\Delta_N}{\bar{N}} = \sqrt{(\frac{\partial \ln f}{\partial x})^2(\Delta_x)^2 + (\frac{\partial \ln f}{\partial y})^2(\Delta_y)^2 + (\frac{\partial \ln f}{\partial z})^2(\Delta_z)^2 + \cdots} \quad . \qquad (14)$$

If the quantity N is an addition or a subtraction function of the quantities $x, y, z, \cdots\cdots$, it is convenient to use formula(13) to calculate Δ_N. If the quantity N is a division function or a multiplication function of the quantities $x, y, z, \cdots\cdots$, we can use formula (14) to calculate.

In some simple measurements we can use the expressions of the sum of the absolute value,

$$\Delta_N = \left|\frac{\partial f}{\partial x}\Delta_x\right| + \left|\frac{\partial f}{\partial y}\Delta_y\right| + \left|\frac{\partial f}{\partial z}\Delta_z\right| + \cdots \qquad (15)$$

$$\frac{\Delta_N}{\overline{N}} = \left|\frac{\partial \ln f}{\partial x}\Delta_x\right| + \left|\frac{\partial \ln f}{\partial y}\Delta_y\right| + \left|\frac{\partial \ln f}{\partial z}\Delta_z\right| + \cdots \quad . \qquad (16)$$

But this method is usually used in occasions of low requirements. In scientific experiments, we generally calculate Δ_N by taking formula (13) or formula (14).

The frequently used formulae are shown in Table01-1.:

Table 01—1

function	the root-sum-square formula	formula of the sum of the absolute value
$N = x \pm y$	$\Delta_N = \sqrt{\Delta_x^2 + \Delta_y^2}$	$\Delta_N = \Delta_x + \Delta_y$
$N = x * y \quad N = x/y$	$\frac{\Delta_N}{\overline{N}} = \sqrt{(\frac{\Delta_x}{x})^2 + (\frac{\Delta_y}{y})^2}$	$\frac{\Delta_N}{\overline{N}} = \frac{\Delta_x}{x} + \frac{\Delta_y}{y}$
$N = kx$ （k is a constant）	$\Delta_N = k\Delta_x$, $\frac{\Delta_N}{\overline{N}} = \frac{\Delta_x}{x}$	$\Delta_N = k\Delta_x$, $\frac{\Delta_N}{\overline{N}} = \frac{\Delta_x}{x}$
$N = x^n \quad n = 1,2,3\cdots$	$\frac{\Delta_N}{\overline{N}} = n\frac{\Delta_x}{x}$	$\frac{\Delta_N}{\overline{N}} = n\frac{\Delta_x}{x}$
$N = \sqrt[n]{x}$	$\frac{\Delta_N}{\overline{N}} = \frac{1}{n}\frac{\Delta_x}{x}$	$\frac{\Delta_N}{\overline{N}} = \frac{1}{n}\frac{\Delta_x}{x}$
$N = \frac{x^p y^n}{z^m}$	$\frac{\Delta_N}{\overline{N}} = \sqrt{p^2(\frac{\Delta_x}{x})^2 + n^2(\frac{\Delta_y}{y})^2 + m^2(\frac{\Delta_z}{z})^2}$	$\frac{\Delta_N}{\overline{N}} = p\frac{\Delta_x}{x} + n\frac{\Delta_y}{y} + m\frac{\Delta_z}{z}$
$N = \sin x$	$\Delta_N = \lvert\cos\rvert\Delta_x$	$\Delta_N = \lvert\cos\rvert\Delta_x$
$N = \ln x$	$\Delta_N = \frac{\Delta_x}{x}$	$\Delta_N = \frac{\Delta_x}{x}$

Example 2 A metal ring is measured to have an outside diameter

$D_2 = (3.600 \pm 0.004)cm$, inner diameter of $D_1 = (2.880 \pm 0.004)cm$, axial depths of $h = (2.575 \pm 0.004)cm$. What is the volume of the ring? What is the uncertainty Δ_V?

Solution: the volume of the ring V is,

$$V = \frac{\pi}{4}(D_2^2 - D_1^2)h = \frac{\pi}{4}(3.600^2 - 2.880^2) \times 2.575 = 9.436cm^3$$

Taking the natural logarithm of both sides：

$$\ln V = \ln\frac{\pi}{4} + \ln(D_2^2 - D_1^2) + \ln h$$

It's complete differential formula is：

$$\frac{\partial \ln V}{\partial D_2} = \frac{2D_2}{D_2^2 - D_1^2}, \frac{\partial \ln V}{\partial D_1} = -\frac{2D_1}{D_2^2 - D_1^2}, \frac{\partial \ln V}{\partial h} = \frac{1}{h}$$

Substituting into formula(12), then：

$$\frac{\Delta_V}{\overline{V}} = \sqrt{(\frac{2D_2\Delta_{D_2}}{D_2^2 - D_1^2})^2 + (\frac{2D_1\Delta_{D_1}}{D_2^2 - D_1^2})^2 + (\frac{\Delta_h}{h})^2}$$

$$= \left[(\frac{2 \times 3.600 \times 0.004}{3.600^2 - 2.880^2})^2 + (\frac{2 \times 2.880 \times 0.004}{3.600^2 - 2.880^2})^2 + (\frac{0.004}{2.575})^2\right]^{1/2}$$

$$= \left[(38.1 + 24.4 + 2.4) \times 10^{-6}\right]^{1/2}$$

$$= 0.0081$$

$$\therefore \quad \Delta_V = \overline{V} * \frac{\Delta_V}{\overline{V}} = 9.436 \times 0.0081 \approx 0.08(cm)^3$$

the volume of the ring can be epressed： $V = (9.44 \pm 0.08)cm^3$

1．4．2 Equipment Used and Preocedure Designed

We can design experimental procedure correctly and select equipments properly by evaluation of errors.

1．5 The Errors of Frequently-used Instruments

1．5．1 Concepts Relating to Instrument

(1) Instrument Accuracy In single measurement or in the measurement in which component A is much less than component B of uncertainty, we usually evaluate the measurement error by instrument error which is calculated by instrument accuracy.

(2) Instrument Error

(3) The Least Count of Instrument The least count of a common instrument is 2~0.5 times the instrument accuracy. The least count of a high precision instrument has a higher request.

(4) Sensitivity Threshold

1．5．2 Instrument Error of Common Instruments

Table01-2 Instrument Error of Common Instruments

Instrument	Scale	The least count of the scale	Instrument error
Micrometer caliper	25mm	0.01mm	±0.004mm
Vernier caliper	125mm	0.02mm	±0.02mm
Meterstick	To 300mm	0．1mm	±0.1mm
	300mm above	0.1mm	±0.2mm
Steel tape	1m	1mm	±0.8mm or given by laboratory
	2m	1mm	±1.2mm or given by laboratory
Physical balance	200g	0.02g	±0.02g
Mechanical		0.1s	0.1s

stopwatch			
ammeter			k%×scale （k is the accuracy of the ammeter）

1.6 Basic Methods of Data Analysis

Data analysis is the process by which experimental data is used to obtain a valid and quantifiable result. Part of this analysis must include an estimation of the accuracy of the result, i.e. how certain are you that you have the correct answer. We refer to this as the uncertainty in the result. Any experimental value that does not have an associated uncertainty is worthless since it indicates 100% uncertainty in the value. There are several aspects to data analysis which will be discussed in this document. Let's start with some general information and move on to more complex issues.

1．6．1 Significant Figures

1.6.1.1 Concept

The significant figures(sometimes called significant digits) of a measured value include all the numbers that can be read directly from the instrument scale, plus one doubtful or estimate number—— the fractional part of the least count.

For example, the length of a rod is 16.2mm. It means 1, 6 are accurate figures and 2 is a doubtful figure. Attention should be given that sometimes the integer is read while the doubtful number "0" is omitted. For example, if the length of a rod measured by a meterstick is 16mm, it should not be read 16mm but 16.0mm.

It should be noted that a number of a physical quantity is different from a number in mathematics. For example, in mathematics, $0.2500\text{m} = 25.000\text{cm}$. But in physical measurement, $0.2500\text{m} \neq 25.000\text{cm}$. Because $0.2500m$ has four significant figures, while $25.000cm$ has five significant figures. In fact, the two different notations represent different precision. It is important to report the results with the proper number of significant figures .

1.6.1.2 The Number of Significant Figures and Notes

⑴The number of significant figures is independent of units. When we take unit conversion, the number of significant figures should not vary. For example, if somewhere the acceleration due to gravity is $980\text{cm}/\text{s}^2$, that is same as $9.80\,m/s^2$. Zeros and the decimal point must be properly dealt with in determining the number of significant figures in a result. Zeros at the beginning of a number are not significant. They merely locate the decimal point. For example, 0.0543m has three significant figures (5,4, and 3).

⑵Scientific Notation It is usually use "powers of 10" notation in writing large or small numbers. The method is called scientific notation. For example, the wavelength of He-Ne laser is $632.8nm$ or $0.6328\mu m$. This can be expressed in scientific notation: 6.328×10^{-7} m or 6.328E-7m. They are four significant figures.

⑶The number of significant figures in a measured value gives an indication of the uncertainty or reliability of a measurement. The mantissa of significant figures should be aligned with the mantissa of the uncertainty. The degree of uncertainty is usually taken to be one or two (If the first number is larger than 3, we take one. If the first number is three or less, we take two).

⑷Reading Rules of Direct Measurement When reading the value from a calibrated scale, only a certain number of figures or digits can properly be obtained. This depends on the least count of the instrument scale, which is the smallest subdivision on the measurement scale. This is the unit of the smallest reading that canbe made without estimating. For example, the least count of a meterstick is usually the millimeter(mm). The length of an object is measured by a meterstick, $L = 1.67cm$, as shown in Fig.01-5(a). The length of another one is read, $\mathbf{L = 2.00cm}$, as shown in Fig.01-5(b). 2.00cm has three significant figures like 1.67cm. Then the length of the third object is measured by the same meterstick, $\text{L} = 90.74\text{cm}$, as shown in Fig.01-5(c) . While 90.74cm has four significant figures. It can be seen that the number of significant figures is depend on not only the fineness of the instrument but

also the object itself.

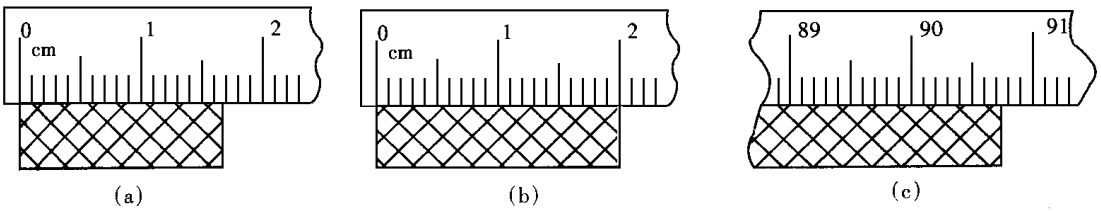


Fig. 01—5

⑸The Relation Between Significant Figures and Accuracy Measured values contain inherent uncertainty because of the estimated figure. The greater the number of significant figures, the greater the reliability of the measurement the number represents.

For example, Measuring the thichness of an object by different instruments with different precision:

Using a meterstick, $d = 6.4mm$, $\Delta_i = 0.1mm$, $E = \frac{0.1}{6.4} = 1.6\%$

Using a vernier caliper, $d = 6.36mm$, $\Delta_i = 0.02\text{mm}$, $E = \frac{0.02}{6.36} = 0.31\%$

Using a micrometer, $d = 6.347mm$, $\Delta_i = 0.004\text{mm}$, $E = \frac{0.004}{6.347} = 0.06\%$,

where Δ_i is the uncertainty of the instrument.

1.6.1.3 Approximate Calculation Rules of Significant Figures

The general principles used for mathematical operations and rounding are as follows. ① There is usually one doubtful figure in the final outcome.②After the last place of the significant figures is determined, the mantissa follows the rounding rule of even. That is, if the first digit to be dropped is less than 5, leave the preceding digit as is.If the first digit to be dropped is greater than 5, increase the preceding digit by one. If the first digit to be dropped is 5, let the preceding digit is an even. For example：Taking three significant figures of “2.455” are “2.46”； Taking two significant figures of “2.455” are “2.4”。

(1)When adding or subtracting quantities, leave the same number of decimal places in the answer as there are in the quantity with the least number of decimal places.

(2)When multiplying and dividing quantities, leave as many significant figures in the answer as there are in the quantity with the least number of significant figures.

(3)Other circumstances: ①When calculating the power or the root of a figure, the number of significant figures is same as that of its base. ②As for some constants, such as π、e、$\sqrt{2}$, it should be contain one more significant figures.

It should be noted that the number of significant figures is not denpend on a mathematical operation but on measurement.

Example 4

$$\begin{aligned} N &= a+b+c+d \\ &= 70.3+6.0-0.8+271 \\ &= 346.5 \end{aligned}$$

∵ d=271 has the least number of decimal places, then rounding off the result, $N = 346$.

Example 5

$$\begin{aligned} N &= \frac{a \times b \times c^2}{d} \\ &= \frac{100.0 \times 50.0 \times 0.20^2}{0.2000} \\ &= 1000 \end{aligned}$$

∵ c=0.20 has the least number of significant figures, after rounding off the result, it can be expressed in scientific notation: $N = 1.0 \times 10^3$.

⑷The Significant Figures of Functional Operation

When we take functional operation, we should calculate the estimated value of the functional obsolute error firstly based on the theroy of error synthesis, leave the

same number of decimal places in the result as there are in the value of obsolute error with the least number of decimal places.

1.6.2 Tabulation Method

Recording the experimental data in a table is called tabulation method. For example, the data of resistances under different temperatures of a resistor are given in Table 01-3.

Table 01—3

times / item	1	2	3	4	5	6	7	8
$\theta(^0C)$	15.5	21.2	27.0	31.1	35.0	40.3	45.0	49.7
R(Ω)	28.09	28.68	29.25	29.68	30.05	30.60	31.08	31.55

There are some notes in using the tabulation method.

1. Each column should be labled with name and unit of the corresponding physical quantity.

2. The data recorded in the label are from measuring and have proper signicant figures. The important immediate results can also be list in the table.

3. If the table is a reflection of certain functional relation, it shluld be arrayed in the order of the values of independent variable.

4. Attatch notes if necessary.

1.6.3 Graphing Method

It is often convenient to represent experimental data in graphical form, not only for reporting but also to obtain information. Graphs are also easy to understand and eye catching.

In order to obtain a complete and proper graph, the basic procedures are as follows: selecting a graph paper; setting axis scales; drawing a best fit curve

corresponding to experimental data points; marking and so on.

1.6.3.1 Selection of Graph Paper and Substitution of Variables

In most cases, the 1-mm grid paper is used in college physical experiment. It is simplest to draw a straight line. Therefore it would be best to change a nonlinear function to a linear function through substitution of variables.

There are four common function expressions in college physical experiment. The linear equations and substitution of variables are as follows:

(2) $y = a + bx$.

(2) $y = a + b/x$, let $n = 1/x$, then $y = a + bn$.

(3) $y = ax^b$, taking the logarithm: $logy = lpga + blogx$, let independent variable is $logx$, then dependent variable is $logy$.

(4) $y = ae^{bx}$, taking the logarithm: $lny = lna + bx$.

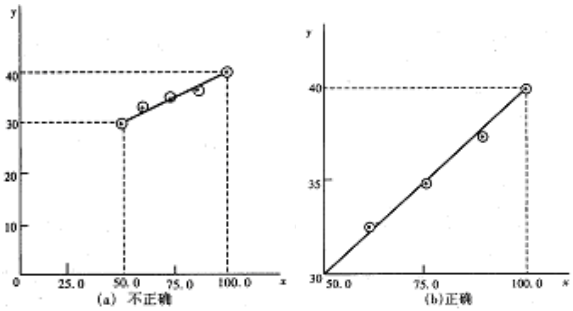
1.6.3.2 Coordinate Scale Division and Label

Axis labels are an essential feature of an informative graph. They must list the name of the variable plotted along each axis and the units of the variable.

The coordinate scale division should accord with the accuracy of the measurement value. As a general rule, it is convenient to choose the unit of the first majorscale division to the right or above the origin or zero point as 1, 2 or 5 so that the minor scale divisions can be easily interpolated and read.

The graph should fill the space allotted for the graph. The graph must be properly scaled. The scale for each axis of the graph is not necessarily beginning at zero. The scale chosen on the axis must be uniform and linear. This means that each square on a given axis must represent the same amount. Obviously each axis for a graph will be scaled independently from the other since they are representing different variables. A given axis must, however, be scaled consistently. An example is shown in

Fig.01-6.



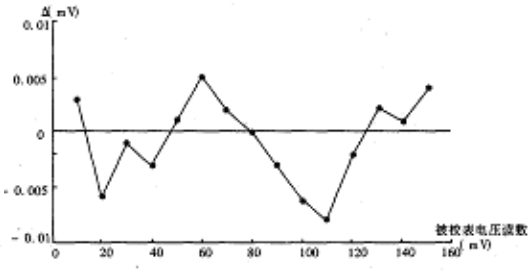


Fig.01-6 Fig.01-7 A calibration line of a voltmeter.

1.6.3.3 Plotting Points and Drawing a line

Each data point should be plotted in the proper position. You should plot a point as a small dot at the position of the data point and you should mark the data point with a “+” so that it will not be obscured by your line of best fit. With the data points plotted, draw a smooth line described by the data points. Smooth means that the line does not have to pass exactly through each point but connects the general areas of significance of the data points.

A line of best fit should show the overall tendency of your data. If the trend is linear, you should draw a straight line which shows that trend using a straight edge. If the trend is a curve, you should sketch a curve which is your best guess as to the tendency of the data. This line (whether straight or curved) does not have to and it may, in some cases, not go through any of them.

When we draw a calibration line of a meter, connect successive data points with a series of straight lines, dot to dot, as shown in Fig.01-7.

1.6.3.4 Notes and Illustrations

There are some notes and illustrations in the graphical representation, such as suitable title, experimental conditions, data sources and so on. If you need to read the values of some points,mark the points you have chosen with a sign(such as △).

1.6.3.5 Graphic Method

The purpose of doing an experiment in physics is to try to find the relationship

between two variables in a controlled experiment. Sometimes the trend of the data can be loosely determined by looking only at the raw data. The trend becomes more clear when one looks at the graph. To more specifically describe the relationship between the variables in an experiment, you will be expected to develop an equation. Graphic method is that one can obtain the equation which describes the behavior of a physical system by using a graph.

It is convenient especially when the graph is a straight line. When the data you collect yields a direct linear graph with a non-zero y-intercept, you will proceed to determine the mathematical equation that describes the relationship between the variables using the slope intercept form of the equation of a line.

1.6.4 Successive Difference Method

The successive difference method is perfoming term-by-term subtraction, or classifying the data into two groups then performing corresponding term-by-term subtraction.

For example, measure the stiffness of a spring. Within the elastic limit, the following formula is satisfied, $F = kx$, where x is the elongation of the spring, F is the tension. The experimental data is shown in Table01-4.

Table 01-4

Weight mass（g）	elongation（cm）	term-by-term subtraction（cm）	Equal-interval subtraction（cm）
0	$L_0 = 10.00$	$L_1 - L_0 = 0.81$	$L_4 - L_0 = 3.22$
10	$L_1 = 10.81$	$L_2 - L_1 = 0.79$	
20	$L_2 = 11.60$	$L_3 - L_2 = 0.83$	$L_5 - L_1 = 3.20$
30	$L_3 = 12.43$	$L_4 - L_3 = 0.79$	
40	$L_4 = 13.22$	$L_5 - L_4 = 0.79$	$L_6 - L_2 = 3.23$
50	$L_5 = 14.01$	$L_6 - L_5 = 0.82$	
60	$L_6 = 14.83$	$L_7 - L_6 = 0.79$	

70	$L_7 = 15.62$		$L_7 - L_3 = 3.19$

It can be seen that the values of "$\Delta L_i = L_{i+1} - L_i$" are almost equal. This can be proved that there is a linear relation between L_i and F.

If we calculate the average by using the following method,

$$\Delta \overline{L} = \frac{\sum \Delta L_i}{n} = \frac{(L_1 - L_0) + (L_2 - L_1) + \cdots + (L_7 - L_6)}{7}$$
$$= \frac{L_7 - L_0}{7} = 0.803\,(mm)$$

We find that there are only two values(beginning and end) work. It is the same as a single measurement of increasing a weight of 70g.

Therefore we usually classify the data into two groups（L_7, L_6, L_5, L_4）and（L_3, L_2, L_1, L_0），performing corresponding term-by-term subtraction and averaging.

$$\overline{\Delta}\,\overline{L} = \frac{1}{4}\left[(L_7 - L_3) + (L_6 - L_2) + (L_5 - L_1) + (L_4 - L_0)\right] = 3.21\,cm$$

Thus all the data can be used and the measurement error can be reduced.

1.6.5 The Least Squares Method and Linear Fitting

The "least squares" method is a form of mathematical regression analysis used to determine the line of best fit for a set of data, providing a visual demonstration of the relationship between the data points. Each point of data represents the relationship between a known independent variable and an unknown dependent variable.

1.6.5.1 An Estimate of the optimum of a fitting line

In this section, only the problem of linear fitting with one variable is discussed by the least square method. A brief introduction of the theory of the least square method is presented as bellow.

Suppose in a certain experiment, the controllable physical quantities are $x_1, x_2, \cdots x_n$, corresponding quantities are $y_1, y_2, \cdots y_n$. Suppose the measurement errors of x_i are very small and the main errors are caused by the measuremet of y_i. The purpose of the linear fitting is to get the best formula $y = a + bx$ from the experimental data by using the method of mathematical analysis. Obviously, there is a deviation δy_i between the measurement y_i and the optimum y. That is,

$$\delta y_i = y_i - y = y_i - (a + bx_i) \qquad (\text{i= 1，2，…，n})$$

The theory of the least square method is that if the error of measurements y_i are mutually independent and comply with a same normal distribution, the best empirical formula can be obtained when the square sum of δy_i is a minimum. Then the constants “a” and “b” can be calculated basing on the theory.

Let the square sum of δy_i is S,

$$S = \sum (\delta y_i)^2 = \sum (y_i - a - bx_i)^2 = \min$$

It can be seen that S is a function of “a” and “b”. Take the partial derivatives respect to a and b respectively and let them zero. Then we can get the values of “a” and “b”.

$$\frac{\partial s}{\partial a} = -2\sum (y_i - a - bx_i) = 0\,, \qquad \frac{\partial s}{\partial b} = -2\sum (y_i - a - bx_i)x_i = 0$$

That is,

$$\sum y_i - na - b\sum x_i = 0\,, \qquad \sum x_i y_i - a\sum x_i - b\sum x^2{}_i = 0$$

The solutions are：

$$a=\frac{\sum x_i\sum y_i-\sum x_i\sum(x_iy_i)}{n\sum x_i^{\ 2}-(\sum x_i)^2}=\frac{\overline{xy}-\bar{x}*\bar{y}}{\overline{x^2}-\bar{x}^2} \qquad (18)$$

$$b=\frac{n\sum(x_iy_i)-\sum x_i\sum y_i}{n\sum x_i^{\ 2}-(\sum x_i)^2}=\bar{y}-b\bar{x} \qquad (19)$$

Substituting a and b into linear equation, the best empirical formula y＝a＋bx can be obtained.

1.6.5.2 Testing the effect of a fitting- line equation

Exercises

1. The side length of a square is measured by a meterstick（$\Delta_i=0.01\,\text{cm}$）. The data is shown in the following table:

a_i（cm）	2.01	2.00	2.04	1.98	1.97	2.01	1.99

What are the circumference “L”and the area “S” of the square? What are their uncertainties?

7. Calculate the following formulae by using the operation rules of significant figures：

(1) $\frac{4000\times1500}{12.60-11.6}$ (2) $\frac{1}{(0.1000)^2}-\frac{1}{(0.5000)^2}$ （“1” is an integer）

8. Draw a line based on the rules of charting and calculate its slope and the acceleration due to gravity “g”（prompt：the length of pendulum “L”is an independent variable, the coordinate origon can be set as: $L=45\text{cm}$, $T^2=1.8\text{s}^2$）。

Length L（cm）	46.1	56.5	67.3	79.0	89.4	99.9
Period $T^2(s^2)$	1.858	2.271	2.706	3.183	3.603	4.032

2. The basic knowledge of physics experiment

2. 1 The basic instrument of physics experiment

Physical experiments need to quantitatively study the relationship between physical quantities. And the appropriate instrument must be selected after the experimental method is determined. Understanding the performance of commonly used instruments and learning how to use them is one of the basic requirements of curriculum teaching. There are many kinds of physical experimental instruments, including force, heat, sound, light, electricity and other types. This section describes only some of the most basic common instruments. Other instruments will be introduced in conjunction with each experimental project.

2.1.1 The basic instrument of mechanics-thermal experiment

The most basic mechanical quantities are length, quality and time. Common instruments include steel tape, steel ruler, slide caliper ruler, spiral micrometer (micrometer), balance, stopwatch and digital millisecond meter, etc.

2.1.1.1 The length measuring instrument

Length is one of the seven basic physical quantities and one of the three basic forces. The measurement method of length is to take a standard length as the measurement standard of length, which is called unit quantity. The length of an object is the magnification between it and this unit quantity, and the unit attached to it. The "Measurement Law of the People's Republic of China" stipulates that the International System of Units (SI system) shall be used as the country's legal unit of measurement. The basic unit of length is the meter (m).

Due to the basic and universal nature of length measurement, a large number of length measurements are required in production and scientific experiments. In measuring instruments, almost all instruments, except digital display meters, eventually convert to length readings, so length measurements are particularly important in basic measurements.

(1) Meters ruler

Steel rulers and steel tapes are commonly used in experiments. The graduation value of the meter stick is 1 mm, and the reading is generally estimated to be 0.1 mm.When measuring with a meter ruler, you should pay attention to the following two points:

1) When the scale of the meter is started by the end edge, do not use the end edge of the meter ruler as the starting point for measurement, so as not to introduce system errors into the measurement results due to wear on the end edge of the meter ruler.

2) In order to reduce the parallax caused by the reading, the object to be measured should be close to the scale line of the meter stick as much as possible, and the eyes should be directly stared to avoid observing from different angles during the measurement, which may lead to differences in the readings.

(2) Vernier and vernier caliper

The vernier caliper is mainly composed of the main ruler and the vernier, as shown in Figure 02-1. The vernier slides close to the main ruler, the outer measuring jaw is used to measure the thickness and outer diameter, the inner measuring jaw is used to measure the inner diameter, the depth gauge is used to measure the depth of the groove, and the fastening screw is used to fix the value reading. When using a vernier caliper, hold the object in one hand and hold the ruler in the other, and gently clamp the object. Special care should be taken to protect the measuring claws from being worn, and it is not allowed to measure rough objects with a vernier, and it is not allowed to move the clamped objects in the bayonet. Before measuring, pay attention to whether the zero line of the vernier is aligned with the zero line of the main ruler. If it is not aligned, it means there is an initial reading. When the zero line of the vernier is on the left of the zero line of the main ruler, the initial reading takes a negative number, otherwise it takes a positive value. In actual measurement, the initial reading should be subtracted from the vernier reading to get the true length of the object.

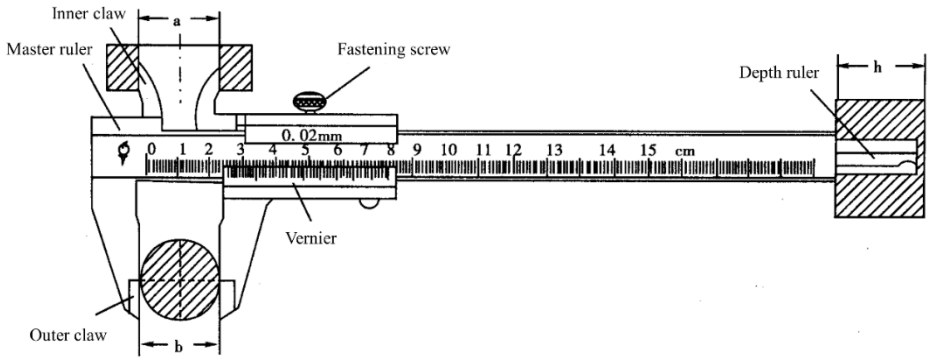


Figure 02-1 Vernier caliper

The method of using vernier to improve measurement accuracy is not only used on vernier caliper, but also widely used in other instruments. Verniers are divided into straight verniers and corner verniers, although verniers have different shapes and lengths, and have different numbers of scales above them, but the basic principles and reading methods are the same.

1) As shown in Figure 02-2, the length of the n-scales on the vernier is equal to the length of $n-1$the scales on the main ruler. Set the minimum score value on the main ruler to a, the minimum score value on the vernier is b, there are:

$$nb=(n-1)a$$

The difference between each grid on the main ruler and the one on the vernier is the accuracy of the vernier:

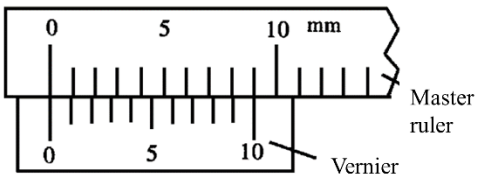
$$vernier\ accuracy = a - b = \frac{a}{n} = \frac{minimum\ graduation\ value}{number\ of\ divisions}$$

It is exactly the minimum value that the vernier can read accurately, also known as the minimum index value of the vernier. For example, in Figure 02-2, $n=10$, $a=1mm$ the accuracy is:

$$\frac{a}{n}=\frac{1}{10}=0.1mm$$

The total length of 10 divisions on the vernier is equal to 9mm , so that the length of

each division on the vernier is 0.9mm. $vernier\ accuracy = a - b = 0.1mm$. When the zero line of the vernier is aligned with the zero line of the main ruler, the distance between the 1st tick mark on the vernier and the 1mm tick line on the main ruler is 0.1mm 0.2mm, and so on, … as shown in Figure 02-3. When the vernier connected to the claw moves to the right, the second tick of the vernier will match the second 0.2mm tick of the main ruler. This means that the distance (the length to be measured) between the vernier "0" line and the main ruler "0" line is 0.2mm.



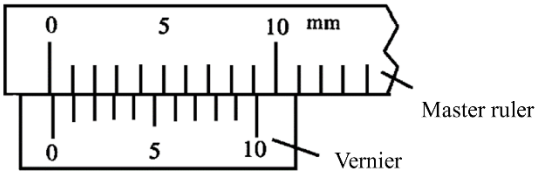


Figure0 2- 2 Figure 02- 3

The vernier only gives a reading below millimeters. When measuring a length greater than 1mm, you should first read the integer scale value on the main ruler corresponding to the vernier “0” scale line, and then read the decimal places in millimeters from the vernier . If the k-th scale line on the vernier coincides with a scale line on the main ruler, the reading of the millimeter decimal place is:

$$k(a-b) = k\frac{a}{n}$$

As shown in Figure 02-4, the accuracy of the vernier is $\frac{a}{n} = \frac{1}{50} = 0.02\text{mm}$, and its reading is:

$$21 + 24 \times \frac{1}{50} = 21.48\text{mm}$$

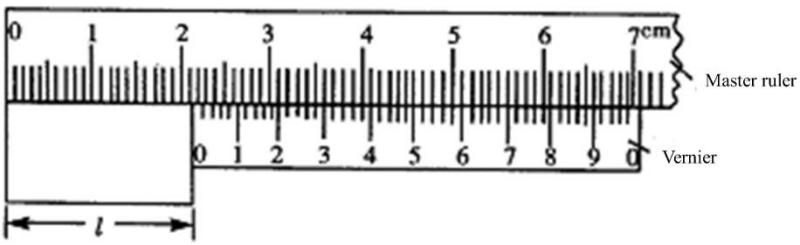


Figure 02-4

2) Angle verniers Are commonly used in many optical instruments, and spectrometers are a typical example.

An angle vernier is a small arc ruler that rotates along a round dial (arc master ruler) and coaxially with it, as shownin Figure0 2-5. The minimum score on the main ruler is $\theta = 0.5^0$,that is 30 points. The vernier is engraved with n scales (typically 30 scales) and its total arc length is equal to that of the scale on the main $n-1$ ruler, i.e.:

$$nRa = (n-1)R\theta$$

So

$$vernier\ accuracy = \theta - a = \frac{\theta}{n} = \frac{minimum\ graduation\ value}{tick\ mark}$$

The reading method is the same as a straight vernier, and in Figure0 2-5, the reading is $165^0 44'$.

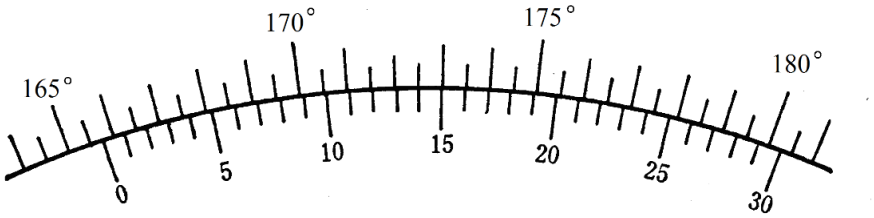


Figure 02-5

(3) Spiral micrometer (micrometer)

The spiral micrometer is designed according to the spiral amplification method and is a length measuring instrument that is more precise than the vernier calliper, and its minimum score value is 0.01mm, which is $1/1000$cm, so it is also called a micrometer. The design of precision length measuring instruments such as optical micrometer eyepiece and reading microscopes is based on the same principle.

The spiral micrometer consists of a precision screw and a nut that is matched to it. The screw back end is connected to a rotatable micro sleeve, as shown in Figure 02-6. Each week each rotation of the micro sleeve, the screw advances (or backs) a

pitch. If there are n scales engraved on the circumference of the micro sleeve and the pitch is amm, the screw will move $\frac{a}{n}mm$ for each index rotation. In Figure 02-6, the pitch is $a = 0.5mm$, the micro sleeve circumferonal is engraved with 50 degrees, each turn of a score, the screw movement distance is $\frac{0.5}{50} = 0.01mm$. When the position of the microcontrote on the fixed sleeve is shown in Figure 02-7, the reading is $10.493mm$. Attention should be paid when using:

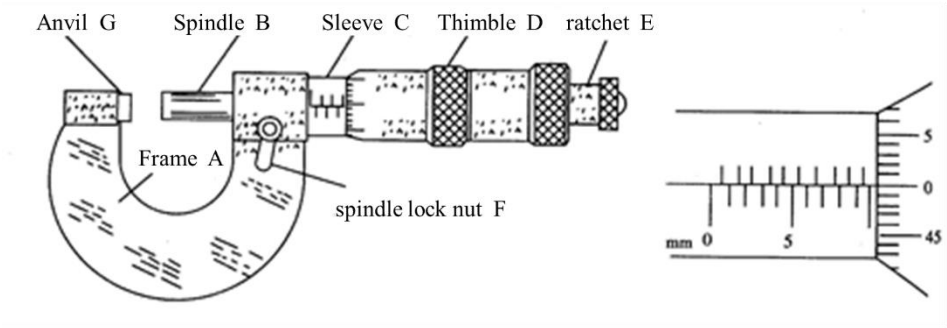


Figure 02- 6 Figure 02- 7

1) The first reading of the micrometer should be recorded. Measure the reading when the anvil G is in contact with the spindle B, and pay attention to whether it is 0 or positive or negative. The length of the object is equal to the measurement reading minus the initial reading.

2) The ratchet E must be used for measuring. The pressure applied to the object being measured when the screw is rotated directly affects the accuracy of the measurement. For this reason, the spiral micro-device adds a ratchet E to the structure as a protective device. When the end face of the micro-screw will come into contact with the measured object, the ratchet E should be rotated until the contact on the measured object, it will slip by itself and make a "click" sound. At this time, the ratchet wheel E should be stopped and the reading should be taken.

3) The thimble D advances 0.5mm in one rotation and 1mm in two rotations. The shaft of the sleeve C is engraved with a millimeter scale (on the lower side) and a half-millimeter scale (on the upper side). Pay special attention when using it.

4) After the measurement is completed, a gap should be left between the anvil G and the spindle B to avoid overpressure of the G and B pressure under thermal expansion, and even damage to the thread.

(4) Digital Calipers

With the development of sensing technology and large-scale integrated circuit technology, digital length measuring devices have been widely used. The digital calipers is one of them, and its structure is shown in Figure 02-8.

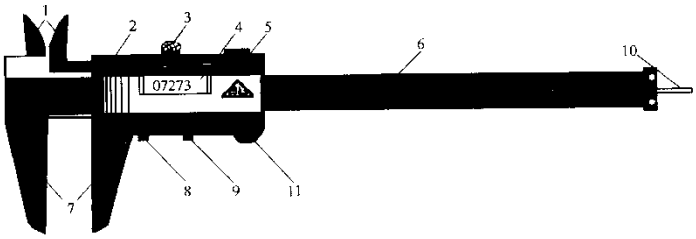


Figure 02-8. 1 Measuring claw inside the knife edge 2 Ruler frame 3 Brake screw 4 Display 5 Data output port 6 Ruler body 7 External measuring claw 8 metric and inch conversion end 9 Zero setting button 10 Depth gauge 11 Replace battery cover

Compared with mechanical caliper, the digital calipers have the advantages of high precision, multiple functions, high measurement efficiency and relative measurement, and can be used as a standard metering device. However, it is relatively high requirements for the environment and conditions of use, if it needs to be moisture-proof, fear of falling, etc., pay attention to it when using it.

2.1.1.2 Measurement of mass

Mass is one of the three basic physical quantities in mechanics. The unit of mass in the International System of Units is kilogram (kg), which is equal to the mass of the International Kilogram Original, which is made of 90% platinum and 10% alloy in a special geometric style (cylinder) and is stored at the International Standards Bureau of Metrology in Paris.

2.1.1.3 Measurements of time

Time is the basic quantity and one of the three basic quantities of mechanics. In

the International System of Units, time is measured in seconds.

2.1.1.4 Measurement of temperature and atmospheric pressure

(1) Temperature measurement

Temperature is one of the seven basic quantities, and it is inseparable from the study of thermal phenomena. Temperature can be simply interpreted as a measure of the heat and cold of an object.

(2) Measurement of atmospheric pressure

In the International System of Units, the unit of pressure is Pa [ska], and thesymbol is Pa, which is $N*m^{-2}$. The atmospheric pressure on the Earth's surface (sea level) is approximately 760mmH(mmHg is listed as a common unit of measurement that should be abolished by the National Bureau of Metrology, $1mmHg = 133.322Pa$). Theoretical calculations show that in the gravity field, the number of gas molecules per unit volume decreases exponentially with the increase of altitude, and the atmospheric pressure decreases exponentially with the increase of altitude, and is related to temperature. The higher the sea level, the lower the air pressure. Between 1000 and 5000m above sea level, the air pressure will drop by about 100m for every increase. Below sea level, the air pressure increases as the depth increases. For example, in a mine, the air pressure increases by about $(0.8\sim1.1)\times10^3 Pa$ for every 100m of altitude drop.

2．1．2 Basic instruments and preparatory knowledge of electromagnetic experiments

2.1.2. 1 power supply

A power supply is a device that outputs electrical energy and is used to provide energy. Usually divided into DC power supply and AC power supply two categories.

(1) AC power

Use the symbol "AC"or ∽"to indicate AC. The common AC is 380V and 220V,

the frequency is 50Hz, AC power with the symbol "" indicates.

(2) DC power

The symbol "DC" or "-" is used in the circuit to indicate direct current. The DC power supply symbol "", the long wire represents the positive pole of the power supply, the short line represents the negative pole of the power supply. The current flows from the positive pole of the power supply and through the wires and electrical appliances into the negative pole of the power supply. Therefore, when using DC power, the positive and negative poles can not be mistaken.

When using the power supply, the circuit must be checked correctly before it is connected, and after the experiment, the power supply should be removed before the circuit is removed. To strictly prevent short circuits, short circuit resistance is very small, so that the current is very large, so that the circuit burned, power damage. When using high-voltage electricity, to prevent electric shock, pay attention to the safety of electricity.

2.1.2. 2 Knowledge of the use of electricity meters

(3) DC ammeter

The purpose of DC ammeter is to measure the current in the circuit, according to its range, can be divided into three categories: ampere meter, milliammeter, microammeter. Its main specifications are:

1) Range

The current value when the ammeter pointer deflects to the full scale. There are both one-range current meters and multi-range ammeters. Multi-range ammeters are commonly used in laboratories. For example, the C31-A multi-range ammeter has a range of 12.

2) Internal resistance

The resistance between the two terminals of the ammeter. The smaller the internal

resistance of the ammeter, the larger its range. The internal resistance of an ammeter is generally less than 1Ω; the internal resistance of milliammeter is generally several to several hundred; the internal resistance of micro ammeter is generally several to several thousand. All of them can be determined by experiments.

(4) DC voltmeter (voltmeter, millivoltmeter)

1) Range

The voltage value when the pointer of voltmeter deflects to full scale. There are not only single range voltmeters, but also single range voltmeters and multi range voltmeters. Multi range voltmeters are commonly used in laboratories.

2) Internal resistance

The resistance between the two terminals of the voltmeter. The larger the internal resistance of the voltmeter, the smaller the influence on the measured voltage. The larger the range of the voltmeter, the greater the internal resistance.

(5) Instrument error of electric meter

In college physics experiments, the basic error of the meter is generally taken as the error limit of the instrument, which will not change with the different users. The basic error of the instrument γ_m (instrument error limit) is defined as the percentage of the ratio between the maximum absolute error Δ_m of the instrument and the range x_m of the instrument.

$$\gamma_m = \frac{maximum\ absolute\ error\ (\Delta_x)}{\text{range}\ (x_m)} \times 100\% \qquad (3)$$

The relationship between the accuracy level *k* of the instrument and the basic error of the instrument is as follows:

$$k\% \geq \frac{\Delta_x}{x_m} \times 100\% \qquad (4)$$

According to the national standard gb776-65 "general technical regulations for

electrical measuring indicating instruments", the accuracy level *k* of instruments is divided into seven levels: 0.1, 0.2, 0.5, 1.0, 1.5, 2.5 and 5.0. The smaller the number is, the higher the accuracy is. Generally, the level of the instrument is marked on the dial when it leaves the factory. According to the accuracy level and range of the instrument, the error limit of the instrument can be calculated as follows:

$$\Delta_{仪} = x_m \times k\% \qquad (5)$$

For example, if the measuring range of an ammeter is 1A and its maximum absolute error is 0.012A after calibration, then;

$$\gamma_m = \frac{0.012A}{1A} = 1.2\% \qquad (6)$$

The accuracy level *k* of this ammeter is defined as 1.5. On the contrary, if we know the accuracy level and range of an ammeter, we can calculate the instrument error limit of the ammeter.

When measuring with an instrument, the maximum relative error of an indication x is

$$E = \frac{\Delta_{仪}}{\mathrm{x}} = \frac{\mathrm{x_m}}{\mathrm{x}} * k\% \qquad (7)$$

It can be seen that in the selection of instrument range, the measured value should be close to the full range value of the instrument as far as possible, and the accuracy of measurement is close to the accuracy of the instrument.

2.1.2.4 Resistor

Resistor can be divided into three types: Fixed resistor, variable resistor and standard resistor. The main technical specifications of resistor include resistance value, accuracy and rated power. In the experiment, variable resistance box and sliding line rheostat are often used to change the resistance, current and voltage in the circuit.

(1) Resistance box

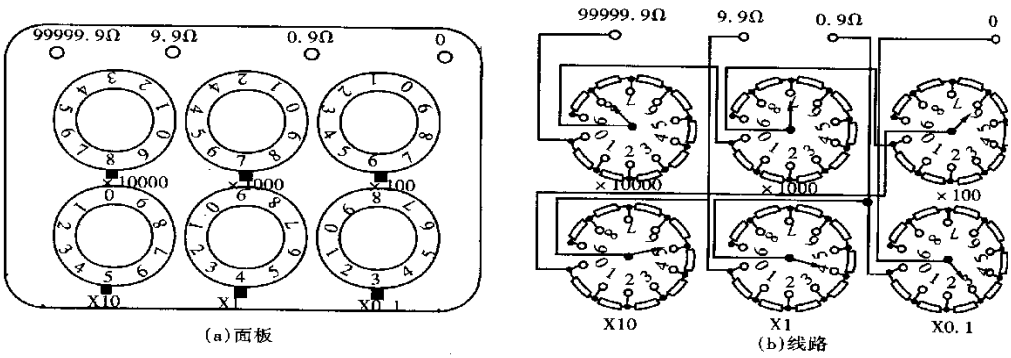


Figure 02-9

3) Accuracy level. The accuracy level of resistance box is divided according to the allowable error percentage of its nominal resistance value. There are usually several levels, such as 0.01, 0.02, 0.05, 0.1, 0.2, 0.5, 1.0, etc. For a class *a* resistance box, the allowable error is k% when the ambient temperature is (20 ± 8) ℃ and the relative humidity is less than 80%.

The brush of each knob in the resistance box has contact resistance, which varies according to its grade. For example, the contact resistance of each knob of ZX21 resistance box (grade 0.1) is not greater than 0.002 Ω. Contact resistance and allowable error constitute the basic error of resistance box, which is usually calculated by the following formula:

$$\frac{\Delta R}{R} = K\% + \frac{bm}{R} \qquad (8)$$

where *K* is the level of the resistance box; R is the indication value of the resistance box; m is the number of knobs in the resistance box between the two lead terminal knobs actually used; B is the coefficient related to the accuracy level (actually the brush contact resistance). It can be seen from equation (8) that when R is large, the contact error is very small; when R is low, the error of contact resistance can not be ignored. This error can be reduced by reducing the number of knobs used in the resistance box. Therefore, when the resistance value is not more than 9.9 Ω or 0.9 Ω, the 0-9.9 Ω or 0-0.9 Ω pair of terminals shown in figure 02-13 (a) should be used.

2.1.2.5 Operating procedures of electromagnetic experiment

(1) Arrange the instruments reasonably. The experimental instruments and devices can be arranged according to the order of the experimental circuit diagram, so as to facilitate the wiring of the circuit diagram, and also take into account the convenience of reading and operation.

(2) Wiring. On the basis of understanding the experimental circuit, according to the order of the circuit diagram wiring. Generally, starting from one pole of the power supply, according to the sequence of the circuit diagram, one instrument is connected with another instrument until the other pole of the power supply. If there is a branch circuit in the circuit, the main circuit can be connected first, and then the branch circuit can be connected. When checking the line, it is also necessary to check the line according to the circuit. When wiring, the power key should be disconnected, and the terminal should be tightened properly to ensure good electrical contact. In addition, in wiring, we should also pay attention to the use of different colors of wires, generally with red power supply positive or high potential, black power supply negative or low potential, so as to facilitate inspection.

(3) Check. After the circuit is connected, it should be carefully checked one by one according to the circuit diagram. Whether the key of the power supply is disconnected; whether the positive and negative poles of the meter and the power supply are correct; whether the range of the meter is appropriate; whether the resistance of the resistance box meets the requirements of the experiment; whether the sliding end of the sliding line rheostat has been placed in a safe position (even if the current and voltage in the circuit are the lowest). After the confirmation, the instructor can check and agree, and then turn on the power to carry out the experiment.

(4) Power on. Before formally connecting the power supply, connect the power supply instantaneously and observe whether the response of each instrument is normal. If there is any abnormality, the power should be cut off immediately, find out the cause and remove the fault. Only when it is normal can the power be turned on and the experiment be carried out.

If it is necessary to pause during the experiment, the relevant power key should

be disconnected. If the battery needs to be replaced, all instruments in the circuit should be adjusted to a safe position, and then the circuit can be replaced by disconnecting the power key. After the circuit is replaced, it still needs to be checked carefully, and the power supply can be connected to continue the experiment only after it is confirmed to be correct.

(5) Put it in order. After the experiment, the instructor can remove the stitches after checking the experimental data. The power supply should be cut off before the wiring is removed. Finally, all instruments should be put in order and wires should be tied together.

Exercises

1. When Ohm's law is used to measure a certain resistance value, the pointer of volt meter is deflected by 80 grids, the pointer of milliammeter is deflected by 100 grids. It is known that the range of voltmeter is 10V, 100 decibels in total, grade 1.0; the range of milliammeter is 150mA, 150 decibels in total, grade 0.5. Q:

(1) what are the graduation values of the two tables? (2) what are the readings of the two gauges?

(3) what are the instrument error limits of the two meters? (4) calculate the experimental results $R \pm \Delta_R$.

Experiment 10

The Measurement of Resistance: Voltmeter - Ammeter Method

Resistance is a common physical quantity in electricity. Measuring a resistance of a conductor by Ohm's law is called "voltmeter-ammeter method". In order to research the electrical conductivity of materials, volt-ampere characteristics are usually displayed. Some materials show a constant resistance over a wide range of applied voltages and are said to be "linear components". Those that do not are said to be "nonlinear components". The resistances of the two categories can both be measured by voltmeter-ammeter method. For more accurate resistance measurement, one must take the internal resistance of electrometer into account.

[Experimental Objectives]

1. Master the voltmeter- ammeter method.
2. Understand the method of analyzing systematic errors and operation of some basic electrical instruments.
3. Represent measurement data in graphical form so as to illustrate experimental data and uncertainty visually.

[Experimental Theory]

1. Volt-ampere characteristics

(1) linear resistors

Some materials show a constant resistance over a wide range of applied voltages and are said to be "linear resistors", such as wirewound resistors, carbon-film resistors and metal film resistors .The I-V characteristic of a linear resistor is shown in Fig.10-1.

(2) nonlinear resistors

If an I-V characteristic curve of a resistor is not a straight line, the resistor is called "nonlinear resistor", such as diode and thermistor.

2. Zener diode

(1) voltage stabilization characteristics of zener diode

A zener diode is virtually a silicon diode. The volt-ampere characteristic of a zener diode is shown in Fig.10-2. Compared with a common diode, the forward characteristic is similar while the reverse characteristic of a zener diode is steeper.

(2) parameters of zener diode

① A zener diode is a typical nonlinear component. The forward resistance is only a few ohms to a few hundred ohms. But the reverse resistance is above thousands of ohms.

② The stable voltage U_z is the actual operating voltage drop across the zener diode under reverse breakdown. For every zener diode, there is an assured stable voltage corresponding to a certain operating current.

③ The stable current I_z is the operating current corresponding to the stable

voltage.

④ Dynamic resistance r_z is defined as, $r_z = \frac{\Delta U_z}{\Delta I_z}$.

⑤ The operating current should be between the maximum stable current $I_{z\max}$ and the minimum stable current $I_{z\min}$.

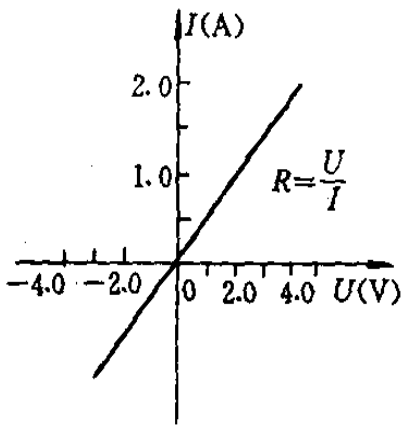


Fig.10-1 I-V characteristic of a linear resistance

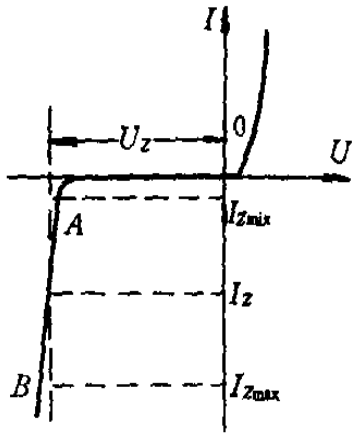


Fig.10-2 I-V characteristic of a zener diode

3. Voltmeter- ammeter methods and errors of the methods

There are two basic arrangements by which resistance is measured with an ammeter and a voltmeter, as shown in Fig.10-3.

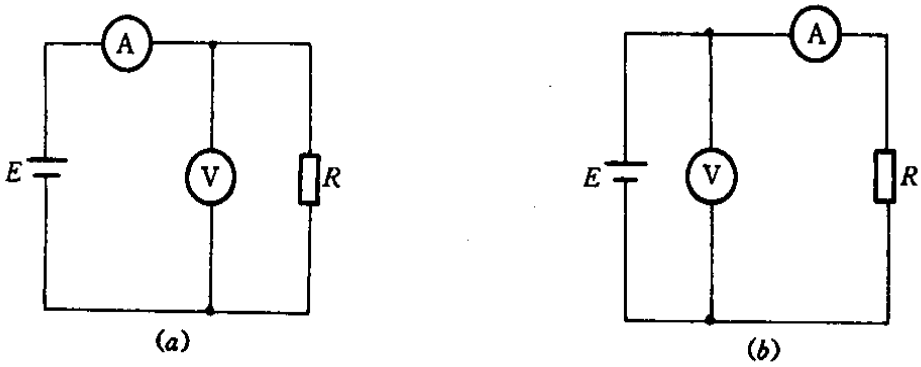


Fig.10-3 two connections of voltmeter- ammeter methods

(1) external connecting of ammeter

One circuit is shown in Fig.10-3(a). The current I though the resistance R is measured with an ammeter, and the voltage drop U across the resistance is measured with a voltmeter. Then, by Ohm's law, $R = U / I$.

Strictly speaking, however, this value of the resistance is not altogether correct, since the current registered on the ammeter divides between the resistance R and the voltmeter in parallel. A voltmeter is a high-resistance instrument and

draws relatively little current, provided that voltmeter resistance R_v is much greater than R. Hence, it is more appropriate to write

$$R \simeq \frac{U}{I} \quad if \;\; R_v \gg R$$

For more accurate resistance measurement, one must take the resistance of the voltmeter into account. The current drawn by the voltmeter is I_v. The ammeter measures the sum of the currents through the resistance and the voltmeter, that is $I = I_x + I_v$, where I_x is the actual current drawn by the resistance. It can be seen that the measured value R is less than the true value R_x, if the measured value is

$$R = \frac{U}{I} = \frac{U_x}{I_x + I_v} = \frac{U_x}{I_x}\left[\frac{1}{1+\frac{I_v}{I_x}}\right] = R_x(1+\frac{R_x}{R_v})^{-1} \quad . \qquad (10\text{-}1)$$

Therefore, the method of external connecting of ammeter is suit for measuring low resistance.

(2) internal connecting of ammeter

Another possible arrangement for measuring R is shown in the circuit diagram in Fig.10-3(b). In this case, the ammeter measures the current through R alone, but now the voltmeter reads the voltage drop across both the ammeter and the resistance. Since the ammeter is a low-resistance instrument, to a good approximation

$$R \simeq \frac{U}{I} \quad if \;\; R_A \ll R \,,$$

where R_A is the resistance of the ammeter.

In fact, the measured value R is greater than the true value R_x, if the measured value is taken to be

$$R = \frac{U}{I} = \frac{U_x + U_A}{I} = R_x + R_A = R_x(1+\frac{R_A}{R_x}) \quad , \qquad (10\text{-}2)$$

where U_x and U_A are the voltage drops across the resistance and the ammeter respectively. Therefore, the method of internal connecting of ammeter is suit for measuring high resistance.

[Equipment Needed]

Power supply, Ammeter, Voltmeter, Rheostat, Resistors, Zener diode, Switch, Connecting wires.

[Experimental Procedure]

1. Measuring the volt-ampere characteristics of linear resistance
   Set up circuits as shown in Fig.10.3. The voltage drop across R should not exceed the maximum value permitted,

$$U_{\max} = \sqrt{PR}\ , \qquad (10-3)$$

   where P is rated power of the resistance. And the maximum current permitted is,

$$I_{\max} = \frac{P}{U_{\max}} \qquad (10-4)$$

   Select ten values between $0 \sim U_{\max}$ at regular intervals, record the data. Also, record the resistances of the ammeter and the voltmeter.
2. Measuring the volt-ampere characteristics of zener diode
   (1) Measuring the forward characteristics of zener diode

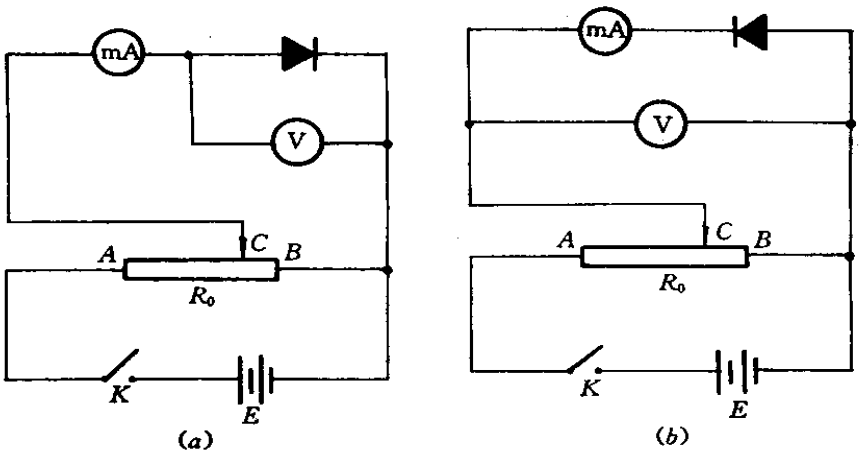


Fig.10-4 Circuit diagrams for measuring the volt-ampere characteristics of zener diode

① Use the ohms scale (×100 or ×1K scale)of the multimeter to distinguish the positive and negative direction of the zener diode.

② Set up a circuit as shown in Fig.10.4(a). The voltage of the source is set to be

1.5V. Attention should be given to adjusting the rheostat in order to make the voltage drop across the voltmeter to zero.

③ Close the switch. Adjust the rheostat slowly so that the readings of the voltmeter increase. Investigate the changing of the resistance of the zener diode.

④ Measure the forward characteristics of zener diode. Be attention to select the appropriate values.

(2) Measuring the reverse characteristics of zener diode

① Set up a circuit as shown in Fig.10.4(b). The voltage of the source is set to be 7.5V. Adjust the rheostat to make the voltage drop across the voltmeter to zero.

② Close the switch. Adjust the rheostat slowly so that the readings of the voltmeter increase. Investigate the changing of the current through the zener diode.

③ Measure the reverse characteristics of zener diode. Be attention to select the appropriate values.

[Data Analysis]

(1) Draw the volt-ampere characteristics curves of the linear resistance under two connections on the same coordinate paper. Compute the resistance respectively.

(2) Revise the results basing on the resistance of the meter which will be found on the meter face or will be supplied by the instructor.

(3) Draw the complete volt-ampere characteristics curve of the zener diode on another coordinate paper.

[Notes]

(1) Set the voltage of the source to zero before connecting. Don’t adjust the power supply at random.

(2) Do not activate the circuit until your laboratory instructor has checked it.

(3) When measuring the volt-ampere characteristics curve of the zener diode, the current through the diode should not exceed the maximum value permitted.

Preview Questions

1, What is the difference of the two connections when we use the voltmeter-ammeter methods? What are the application conditions of the two connections?

2, How to check the diode by using a multimeter?

Discussions

What's the feature of the forward and reverse characteristics of a zener diode?

Why the method of external connecting of ammeter is used in measuring the forward characteristic, while the method of internal connecting of ammeter is used in measuring the reverse characteristic?

Experiment 13 Measuring Young's modulus of metal by tensile method

All bodies will deform more or less under the action of external forces. When the deformation does not exceed a certain limit, the external force is withdrawn, and the deformation can disappear. This deformation is called "elastic deformation". When elastic deformation occurs, the internal stress of the object is restored to its original state. Young's modulus (also called Young's modulus of elasticity) is a physical quantity that describes the resistance of solid materials to deformation. It is one of the basis for selecting mechanical components and an important parameter commonly used in engineering. In this experiment, Young's modulus is measured by tensile method and the relationship between tensile normal stress and strain is studied.

The principle of optical lever method, which is a measurement method of small length variation involved in this experiment, is widely used in many measurement techniques. The optical lever device is also used in many highly sensitive measuring instruments such as impulse galvanometer and photoelectric galvanometer.

[Purposes]

1. Learn to measure Young's modulus of metal wire by tensile method, master the principle of measuring small length variation with optical lever device;

2. Learn to deal with experimental data by difference method and drawing method.

[The experimental principle]

According to Hooke's law, within the elastic limit of the material, the magnitude of the normal stress σ is proportional to the strain ε, i.e

$$\sigma = E\varepsilon \qquad (1)$$

In the formula, the proportional coefficient E is called the elastic modulus, also known as Young's modulus. For a uniform wire or rod of length L and cross-sectional area S, elongated by an external force F along the length direction, there

is $\sigma = \frac{F}{S}$, $\varepsilon = \frac{\Delta L}{L}$,

$$E = \frac{FL}{S\Delta L} \qquad (2)$$

Young's modulus has nothing to do with the external force applied, the length of the object and the size of the cross-sectional area of the material. It is an important physical quantity reflecting the property of the solid material itself. Its engineering unit is Newton/meter^2(N / m^2).

According to Equation (2), if the Young's modulus of the wire is measured experimentally, the value of F, S, L and L can be obtained by calculating Young's modulus E. F, S and L can be easily measured by ordinary measuring instruments. Usually very small, with ordinary measuring instruments, commonly used measurement methods, not only more difficult to measure, and the accuracy of measurement is very low. The optical lever method can solve this problem well.

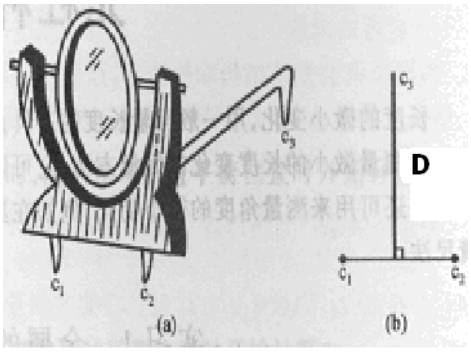


Figure 13-1 Schematic diagram of the smooth rod

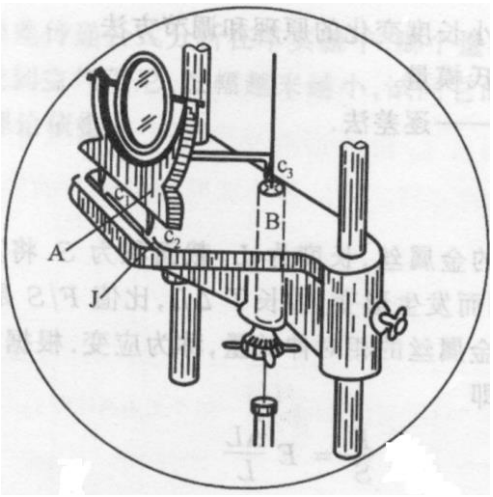


Figure 13-2

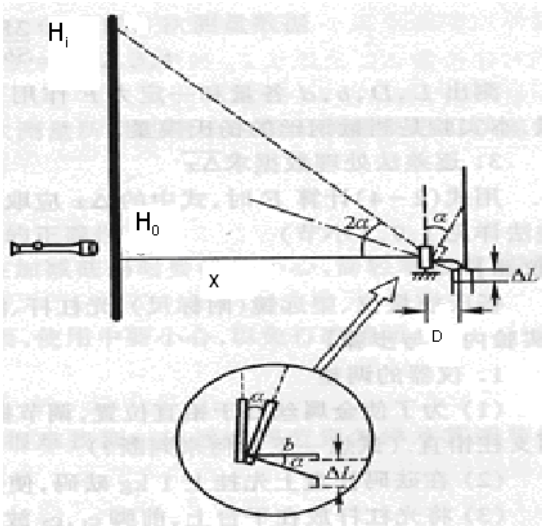


Figure 13-3 Schematic diagram of optical lever amplification principle

The smooth rod is composed of A small circular plane mirror and three pointed feet fixed on the frame A. The length of the perpendicular segment to D is called the optical lever constant (as shown in Fig. 13-1). To measure, the two front feet are placed in the groove J of the platform, and the rear feet are placed on top of the cylinder clip B (as shown in Figure 13-2). The upper end of the steel wire to be measured is clamped in the middle of the clamp on the beam, and the lower end is clamped in the clamp that can slide up and down. The lower end of the steel wire is provided with a hook to hang the weight. Adjust the plane mirror to be roughly straight. Put a ruler vertically in front of the mirror and a telescope next to the ruler. After proper adjustment, you can see clearly the ruler reflected by the small mirror from the telescope, and you can read the ruler scale value coincidences with the cross line of the telescope fork wire. (See Figure 13-3)

Let the reading of the ruler from the telescope be when the weight is not added, and when the weight is added, the wire is extended and the rear leg of the optical lever is lowered accordingly. At this time, the plane mirror rotates the Angle alpha and the mirror normal also rotates the Angle alpha. According to the law of reflection of light, the reflected ray will rotate around the Angle 2α, that is, the scale on the ruler can be seen from the telescope after the mirror reflection, then

$$tg2\alpha = \frac{|H_i - H_0|}{x} = \frac{\Delta H_i}{x}$$

Where, x is the distance between the optical lever mirror and the ruler. But as you can see from a larger version of Figure 13-2 $tg\alpha = \frac{\Delta L}{D}$

Because it's a small change in length, $\Delta L \ll b$, α is small

$$tg2\alpha \approx 2\alpha, \quad tg\alpha \approx \alpha \approx \frac{\Delta L}{D}$$

$$2\frac{\Delta L}{D} = \frac{\Delta H_i}{x}$$

$$\Delta L = \frac{D\Delta H_i}{2x} \tag{3}$$

It can be seen from Equation (3) that the function of the optical lever mirror ruler is to convert the small length change into the small Angle change through the optical lever. At the same time, it is converted by the telescope and the ruler into a large reading variation on the ruler. For the same reason. the bigger x is, the bigger ΔH_i it is, the smaller the relative error of the measurement. The ratio of $\beta = \frac{\Delta H_i}{\Delta L} = \frac{2x}{D}$

It's the magnification of the optical lever.

$$E = \frac{2xFL}{SD\Delta H} = \frac{8xFL}{\pi d^2 D\Delta H} \tag{4}$$

Where D is the diameter of the wire. Under the action of L, X, D, D and a certain force F, the Young's modulus of the wire can be indirectly measured by Equation (4). This experiment is to measure the Young's modulus of steel wire.

［Experimental apparatus］

Young's modulus measuring instrument, reading telescope, vernier caliper, micrometer, steel tape, etc.

A ruler reading telescope consists of a scale and a telescope. Turn the eyepiece of the telescope and you can see the image of the cross wire clearly. Adjust the telescope focusing handwheel and see the image of the scale through the plane mirror of the light lever. The axis of the telescope can be adjusted through the telescope axis adjustment screw, loosen the telescope scale fastening screw, the telescope and scale can be moved up and down along the column respectively.

[The experiment content]

1. Adjustment of instrument

(1) In order to make the wire in the straight position, adjust the bottom screw of the tripod of Young's modulus meter to make the two pillars straight. (How do you figure that out?)

(2) hang 1kg weight on the weight tray to straighten the wire (this weight is not included in the added force F).

(3) the light lever is placed on the platform, the forefoot is placed in the groove J in front of the platform, and the rear foot is placed on the cylindrical chuck B, so that the mirror is roughly straight. The telescope and ruler are placed approximately in front of the light lever mirror. Position the telescope up and down so that it is at the same height as the light lever.

(4) Adjust the telescope to see the ruler reading. Including the following three links of regulation:

1) adjust the eyepiece and see the cross wire clearly. This can be done by rotating the eyepiece.

2) Adjust the objective lens to see the ruler reading clearly. First, aim the telescope at the light lever mirror, and then look at the outside of the telescope along the direction of the mirror cylinder, to observe whether there is a ruler image in the light lever mirror. If so, you can observe it through a telescope; If not, the light lever or ruler should be slightly moved until the ruler image is seen in the telescope, adjust the

distance between the eyepiece and objective lens to see a clear ruler reading.

3) Eliminate parallax and carefully adjust the distance between eyepiece and objective lens until there is no relative movement between ruler image and fork wire when human eyes move up and down slightly.

2. measurement

(1) After all the instruments are adjusted, write down the reading on the ruler in the telescope at the beginning, and then add 1 kg weight to record the reading on the ruler. Add another 1 kg weight, regardless of the reading, and then successively reduce 1 kg weight, each decrease, accordingly record the reading on the ruler $H_i'(i=7,6,\cdots,1,0)$ 。 Take the average of two readings under the same load.

$$\bar{H}_i = \frac{H_i + H_i'}{2}, (i=0,1,2,\cdots,7)$$

(2) Repeat step (1) Do it again.

(3) Use steel tape to measure the wire length L and the distance between the optical lever mirror and the scale X.

(4) Use micrometer to measure wire diameter D (6 times at different places).

(5) Remove the light lever and put it on a flat piece of white paper to press out three points. Then measure the distance D between the rear sharp feet and the two front sharp feet with a vernier caliper.

The list records all the data, and the form is self-drawn.

[Data processing and analysis]

In the experiment, two methods are used to deal with the data and the Young's modulus of the wire is obtained respectively.

1. Using the difference method processing: using the difference method to calculate the corresponding 4kg load of the wire elongation: $\Delta H_i = \bar{H}_{i+4} - \bar{H}_i$ （i=0,1,2,3）

and average elongation per kilogram

$$\overline{\Delta H} = \frac{\sum_{i=0}^{3} \Delta H_i}{16}$$

Substitute the measurement results of $\overline{\Delta H}$, L, X, D and D into Equation (4) to calculate the Young's modulus and its uncertainty of the metal wire to be measured.

2. Graph

$$\Delta H = \frac{8Lx}{\pi d^2 DY} F = KF \qquad (5)$$

Under the given experimental conditions, K is constant, and if $\Delta H_i = \overline{H}_i - \overline{H}_0$ (I =1,2... 7) is the ordinate and F is the abscissa. A straight line can be drawn, and then the Young's modulus of the metal wire to be measured can be obtained by calculating the slope of the line K.

$$Y = \frac{8xL}{\pi d^2 DK} \qquad (6)$$

[Precautions]

1. In telescope adjustment, attention must be paid to the elimination of parallax, otherwise it will affect the accuracy of the reading.

2. During the experiment, the instrument shall not be collided with, and the position of the main rod of the optical lever shall not be moved. Add or subtract weight must be handled gently, to be stable after the system can be read.

3. The steel wire to be tested shall not be bent, and the steel wire that cannot be straightened by adding background weight and is seriously corroded must be replaced.

4. Light lever plane mirror is a fragile article, in order to maintain a good reflection of the mirror shall not be touched by hand, nor shall it be wiped at will, but not to fall to the ground, so as not to break the mirror.

Preview the question

1. What is the function of mounting background silicon code?

2. How does a light lever measure small changes in length? What are the advantages?

To discuss the topic

Can you calculate the magnification of the optical lever used based on the data measured in the experiment? How to increase the magnification of the optical lever to improve the sensitivity of the optical lever to measure the small length variation? What is the indexing value of the optical lever in your instrument?

Experiment 17 The resistance measurement by DC bridge

Bridge method is one of the commonly used resistance measurement methods. The balance bridge is measured by the comparative method, that is, under equilibrium conditions, the resistance under test is compared with the standard resistance to determine its resistance value. It has the characteristics of sensitive, accurate and convenient testing. Bridge circuits have many uses in automatic instruments and automatic control processes.

The bridge is divided from the structure: a single arm bridge and two arms bridge; From the indicating state to divide: there are balance bridge and non-balance bridge; From the use of power properties to divide: direct bridge and AC bridge. DC bridge is divided into two kinds of single-arm bridge and double-arm bridge. The former is Wheatstone bridge, which is mainly used for accurate measurement of median resistance (tens to hundreds of kilo ohms). The latter, called a Kelvin bridge, is suitable for measuring low resistance (below a few ohms). AC bridge can also measure capacitance, inductance and other physical quantities.

Practice 1 Measuring the median resistance with a Wheatstone bridge

[Experimental Purpose]

1. Master the structure characteristics of Wheatstone bridge and the principle of resistance measurement;
2. Learn to use box type (finished product) current bridge to measure resistance;
3. Understand the sensitivity concept and measurement method of the bridge.

[Experimental principle]

The unknown resistance R_x can be measured by voltammetry. However, due to the influence of the internal resistance of the ammeter, there is always a systematic error whether the ammeter is connected inside or outside. The point of conflict is that the meter has an internal resistance, a current flowing through the meter. How to make the table without current flow, but also the Rx resistance value accurate measurement? An efficient circuit is the bridge circuit shown in Figure 17-1. The basic components are: the bridge arms (resistors R_1, R_2, R_o and R_x), the "bridge" -balance indicator (also known as the ZERO indicator), and the working power supply (E).

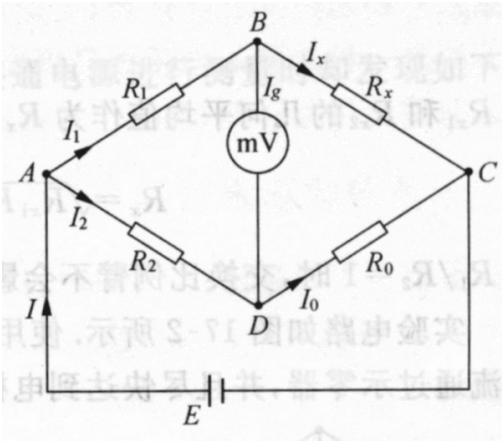


Figure 17-1 Principle circuit diagram of Wheatstone Bridge

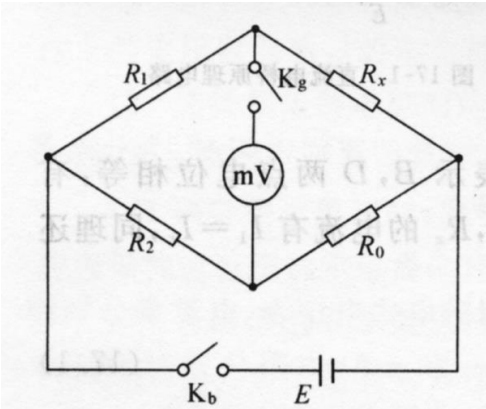


Fig. 17-2 Experimental circuit diagram of Wheatstone Bridge

1.Wheatstone bridge principle

The circuit principle of a Wheatstone bridge is shown in Fig. 17-1. Four resistors, R_1, R_2, R_o and R_x, are connected to form A quadrilateral. Each edge is called an arm of the bridge, and the connection points, A, B, C and D, are called the vertices of the bridge. The diagonal points A and C are connected to the working power supply E, and the other diagonal points B and D are connected to the millivoltmeter. The branch connected to the power supply, AC, is called the power supply diagonal, and the branch connected to the millivoltmeter, BD, is called the measurement diagonal. The "bridge" refers to this diagonal line, which is used to compare the potential of the two vertices of the bridge, B and D. In the four bridge arms, the resistors R_1 and R_2 are called proportional arms. R_0 is called the comparison arm and R_x is called the measuring arm.

In use, the proportional arm resistors R_1 and R_2 are selected to adjust the comparison arm R_0. If the ZERO indicating device refers to zero, it means that the potential of B and D is equal, $U_{AB} = U_{AD}$, $U_{BC} = U_{DC}$. When the bridge is balanced, the current flowing through the resistors R_1 and R_x is I_1= Ix, and similarly, I_2= I_0. That is

$$\frac{R_1}{R_2} = \frac{R_x}{R_0} \qquad (1)$$

So we get $$R_x = \frac{R_1}{R_2} R_0 \qquad (2)$$

If R_1, R_2, and R_0 are all known (or R_1/R_2 and R_0 are known), then R_x can be solved from Equation (2). In this experiment, the DC voltage range of 200 mV (internal resistance is 10 MΩ) of the digital avometer is used as the ZERO indicating device. Its minimum resolution is 0.1 mV, and it can display the positive and negative property of the voltage.

2. Bridge sensitivity

The principle formula (2) for measuring resistance of a bridge is derived under the condition of bridge equilibrium. In actual measurement, whether the bridge is balanced is judged by observing the ZERO indicator. But the sensitivity of the ZERO indicator and the resolution of the human eye are always limited, small to a certain degree of current, it will not detect. Assuming the bridge is balanced at $R_1/R_2=1$, then $Rx=R_0$. At this time, if you change R_0 by a quantity $\triangle R_0$, the bridge should be out of balance, so that current Ig flow through the ZERO indicator. But if Ig is so small that the ZERO indicator can't detect it, then we would assume that the bridge is still in equilibrium, and therefore $R_x= R_0 \pm \triangle R_0$. $\triangle R_0$ is the measurement error of $\triangle R_x$ caused by the insufficient sensitivity of the bridge. In order to quantitatively analyze the relationship between bridge sensitivity and measurement error, we introduce the concept of bridge relative sensitivity S, which is defined as

$$S = \frac{\Delta n}{\Delta R_x / R_x} \qquad (3)$$

$\triangle R_x$ is the small change in Rx after the bridge is balanced. $\triangle n$ is the change in the ZERO indicator caused by the bridge being out of balance. The larger S is, the more sensitive the bridge is, and the smaller the measurement error will be. For the measurement of the bridge sensitivity, since the resistance to be measured, R_x can not actually be changed, it is usually used to change the comparison arm R_0 to measure S. After the bridge reaches equilibrium, a small change in $R_0(\triangle R_0)$ will cause the bridge indicator to deviate from the equilibrium position $\triangle n$, then the bridge sensitivity will be approximately

$$S \approx \frac{\Delta n}{\Delta R_0 / R_0} \qquad (4)$$

It has been proved experimentally and theoretically that bridge sensitivity is related to the following factors:

(1) It is proportional to the sensitivity S of the galvanometer. However, if S value is too large, the bridge is not easy to balance and the balance adjustment is difficult. S value is small, the measurement accuracy will be reduced. Therefore, it is very important to choose a ZERO indicator with appropriate sensitivity when measuring resistance with a Wheatstone bridge.

(2) It is proportional to the output voltage of the power supply. However, the

increase of supply voltage should be limited by the current allowed by the resistance of each bridge arm.

(3) It is related to the configuration of the bridge arm ratio R_1/R_2.

(4) It is related to the position of the zero indicator and the power supply and the internal resistance of the power supply.

3. Exchange measurement method

In Equation (2), if the values of R_1 and R_2 are not accurate, there will be systematic errors in the measurement results, which can be eliminated by the exchange measurement method. Switch the positions of R_0 and R_x, do not change R_1 and R_2, adjust the balance of the bridge again, write down the value of the resistance box at this time, set it as R'_0, then there is

$$R_x = \frac{R_2}{R_1} R'_0 \qquad (5)$$

It can be obtained from Equations (2) and (5)

$$R_x = \sqrt{R_0 R'_0} \qquad (6)$$

The above equation shows that R_1 and R_2 do not appear in the measurement formula of R_x by using the exchange measurement method. Therefore, if the self-organizing bridge has only one standard resistor and two resistors whose values are stable but do not cause accurate measurement, the accurate value of R_x can be obtained.

4. Brief introduction of QJ23 electric bridge

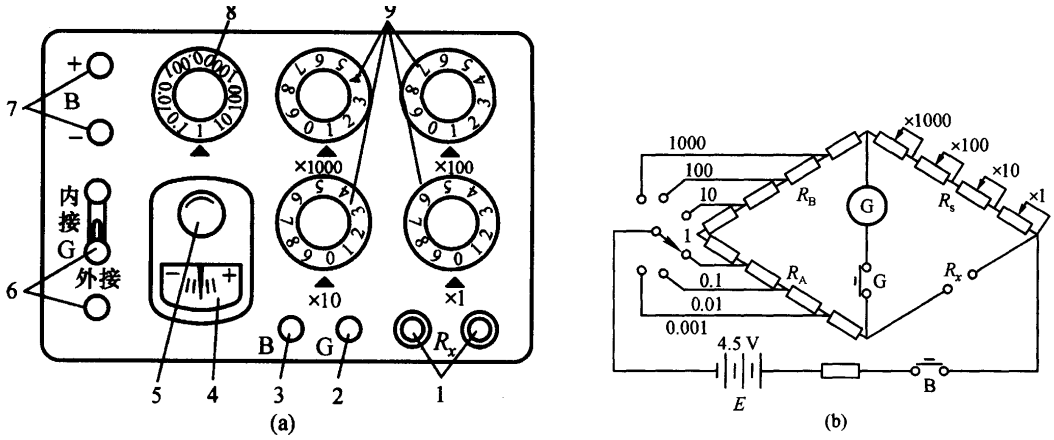


Figure 17-3 Diagram of QJ23 type bridge panel (a) and internal wiring (b)

1. Resistance Rx terminal to be tested; 2. Galvanometer button switch G; 3. Power button switch B; 4. Galvanometer. 5. Zero-setting knob of galvanometer; 6. External galvanometer terminal; 7. External power terminal; 8. The ratio arm, the ratio of R_A/R_B in the bridge circuit, is directly engraved on the turntable. 9. The comparison arm, i.e., the resistance box (R_S) in the above bridge circuit, is here four turntables.

The circuit of QJ23 type bridge is basically the same as Figure 17-1, but the whole device is put in the box. Its panel shape is shown in Figure 17-3 a, and its internal wiring is shown in Figure 17-3 b. The proportional arm R_1/R_2 is adjusted by a knob, which adopts the decimal fixed value, and is divided into seven grades: 0.001, 0.01, 0.1, 1,10,100,1000. Standard variable resistor R_S for a four-knob resistor box. The minimum change is 1 Ω. The galvanometer of the QJ23 bridge has three terminals, which are respectively marked "internal connection" and "external connection". When in use, the copper sheet should be provided to short circuit the "external" two terminals, then QJ23 type bridge connected to the internal galvanometer. After use, the copper sheet should be short-circuited to the two "internal" terminals to protect the galvanometer. If necessary, a more sensitive galvanometer can be added to the two "external" terminals to improve the sensitivity of the bridge. The keys B and G correspond to K_b and K_g in Figure 17-3. The two terminals marked "RX" are used to connect the measuring resistors. The instrument's internal power supply 1 battery 3, about 4.5 V. If the internal power supply of the instrument is not used, an external power supply can be connected to the "+" and "-" terminals at the upper left of the bridge as required.

On the one hand, the accuracy of the bridge depends on the accuracy of the proportional arm R_A/R_B and the comparison arm R_s, on the other hand, it also depends on the sensitivity of the galvanometer and the value of the working power supply voltage. The accuracy of QJ23 DC single bridge is shown in the instrument panel. The uncertainty of the measured value varies with the use of different proportional arms. If the proportional arm is 1 and Rx=9680 Ω, it can be found that the instrument error limit under the proportional arm is 0.2%. So there are:

$$\Delta \boldsymbol{R_x} = 9680 \times 0.2\% \approx 19\Omega$$

5. Discussions related to resistance measurement of Wheatstone bridge

The experimental circuit is shown in Figure 17-2. When using the bridge method to measure the resistance, it is necessary to master the correct method to avoid excessive current passing through the ZERO indicator and to achieve the bridge balance.

(1) Use the digital avometer to measure the resistance value of the resistance R_x to be measured, and select the appropriate proportional arm R_1/R_2 value according to the measured value, determine the initial value of the comparison arm R_0, so that the bridge is as close to balance as possible when the power is on.

(2) Attention should be paid to the switching sequence. When switching on the circuit, K_b should be closed first, then K_g; When disconnecting the circuit, it is necessary to disconnect K_g first and then K_b. Special attention should be paid when measuring the inductive resistance.

(3) In the process of adjusting R_0, the deflecting direction of ZERO indicator may be the same or opposite for two consecutive times. Whether the value of R_0 is too large or too small should be analyzed, and how to adjust R_0 in the next step should be considered to avoid blindness.

(4) When the bridge is balanced, the following steps should be taken to determine whether the bridge is truly balanced:

① repeatedly on and off K_g (using the jump method, to see whether there is a change in the number of zero indicator);

② A small increase or decrease of R_0 to destroy the balance of the bridge. When the polarity of the corresponding zero indicator is different, the original balance judgment is true.

(5) Correctly judge the effective digits of the measurement results. After the bridge is balanced, change the lower resistance value of the resistor box as R_0, so that R_0 has a relative change $\triangle R_0/R_0$. If this bit changes from 0 to 0.9 and the indicator value of the zero indicator remains the same, it indicates that the bridge sensitivity is low and this bit cannot be used as the digit of the significant number. At this time, the change of R_0 should continue to increase until there is a significant change in the zero indicator, and the significant number of R_0 should be recorded to this digit.

[Experimental Instrument]

3 resistance boxes, QJ23 bridge, digital avometer, power supply, switch and resistance to be measured, etc.

[Experimental contents]

1.Use resistance boxes to form a Wheatstone bridge circuit to measure unknown resistance (a few hundred ohms).

(1) Use the ohmic gear of the digital avometer to measure the resistance value of the resistance Rx to be measured.

(2) Self-organizing bridge circuit (see Figure 17-3). Three resistance boxes are used as the bridge arm resistors, and the 200 mV DC voltage range of the digital avometer is used as the zero indicator. The circuit should be carefully checked after connection, and ask the teacher to check before the power is connected.

(3) R_a: R_b =1:1 (for example R_a: R_b =500Ω:500Ω), so that the measurement result has four significant digits. The power supply voltage is E=3 V, and the bridge balance is judged according to the correct experimental steps.

(4) Exchange R_0 and R_x for another measurement. Calculate the resistance value of the resistance Rx to be measured according to Equation (6).

2. Select appropriate conditions and measure the resistance with R_x of tens of ohms and thousands of ohms respectively.

3. Measurement of unknown resistance with QJ23 bridge circuit (one each for tens, hundreds and thousands of ohm resistances)

3 resistors to be measured, each resistor measured once. Since the comparison arm resistance has four knobs, the measurement should have four significant digits. Therefore, the proportional arm should be selected according to the resistance value to be measured, so that R_0 can have four significant digits. For example, when the

resistance to be measured is tens of ohm, the proportional arm should be 0.01.
4.Understand the factors affecting the sensitivity of DC Bridges.

Variate the relevant factors, such as the value of supply voltage E, R_1 and R_2, the sensitivity of the zero indicator (e.g. 2 V or 20 mV DC voltage range), and observe the change in the sensitivity of the bridge.

[Data Processing and Analysis]

1. According to the ratio of R_1/R_2 and R_0, Rx and △Rx values measured by the exchange method are calculated according to the equation, and the result is expressed as Rx±△Rx.

2. Calculate the values of R_{x1}, R_{x2}, R_{x3} measured by QJ23 bridge and their errors △R_{x1}, △R_{x2}, △R_{x3}, and express the results as Rx±△Rx.

3. The above results are analyzed and discussed. The difference between self – assembled bridge and box - type bridge is analyzed.

4. Analyze the influence of the change of DC voltage range of digital avometer on the sensitivity of the bridge, and measure the sensitivity of QJ23 bridge.

[Precautions]

When box type bridge is used, the power supply connection time should be very short, that is, B and G two knobs can't be pressed down for a long time at the same time. When measuring, press B first and then G; Disconnect, must first disconnect G and then disconnect B, and develop a habit, (think why?)

The question of preview

1. What is the principle of resistance measurement by bridge method? How to judge bridge balance? How to achieve bridge balance during the specific operation?
2. What is the principle of selecting the multiplier value of the proportional arm of QJ23 bridge?

DISCUSSIONS

1. Will the following factors increase the measurement error?
   (1) The power supply voltage drops greatly;
   (2) The power supply voltage fluctuates slightly;
   (3) In the measurement of low resistance, wire resistance can't be ignored;
   (4) The sensitivity of zero indicator is not high enough.
2. If the circuit is connected according to Figure 17-2, and the following situation is found after the power is switched on, what is the fault of each circuit?
   (1) The indicator value of the zero indicator does not change;
   (2) The indicating polarity of the zero indicator remains unchanged.

Experiment 19 Digital oscilloscope

Purpose

1. Understand the main components of digital oscilloscope and the waveform display principle of oscilloscope.
2. Learn to observe electric signal and Lissajous figure with digital oscilloscope.
3. Learn how to measure electrical signals by comparison.

Experimental principle

1. Principle of digital oscilloscope

The diagram of a typical digital oscilloscope is shown in Fig. 1. The analog input signal is amplified or attenuated properly, and then converted into binary digital signal through A/D converter. After the analog signal is converted into digital signal, the digital signal is written into the memory under the control of the microprocessor. In the graphic display working stage, the digital signal is read out from the memory, converted into analog signal by D/A converter, amplified by the amplifier and added to the Y deflector of the display. At the same time, the address signal given by the microprocessor is converted by D/A converter to generate sawtooth wave, amplified by the amplifier and added to the X deflector of the display. And then the analog signal is reproduced on the display.

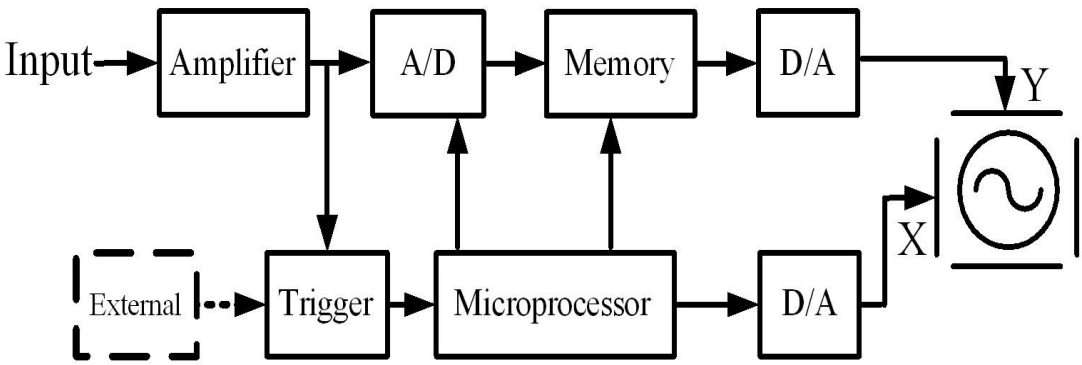


Figure 1. Digital oscilloscope diagram

2. Scanning principle

Apply linear voltage and sinusoidal voltage to x-axis and y-axis deflection plate of oscillograph at the same time in figure 2. If their cycles are the same, a cycle can be divided into four same time intervals. The values of U_x and U_y correspond to the positions where the light points deviate from the x-axis and y-axis respectively. Connect the projection light points of U_x and U_y to obtain the waveform (sinusoidal) of the measured voltage.

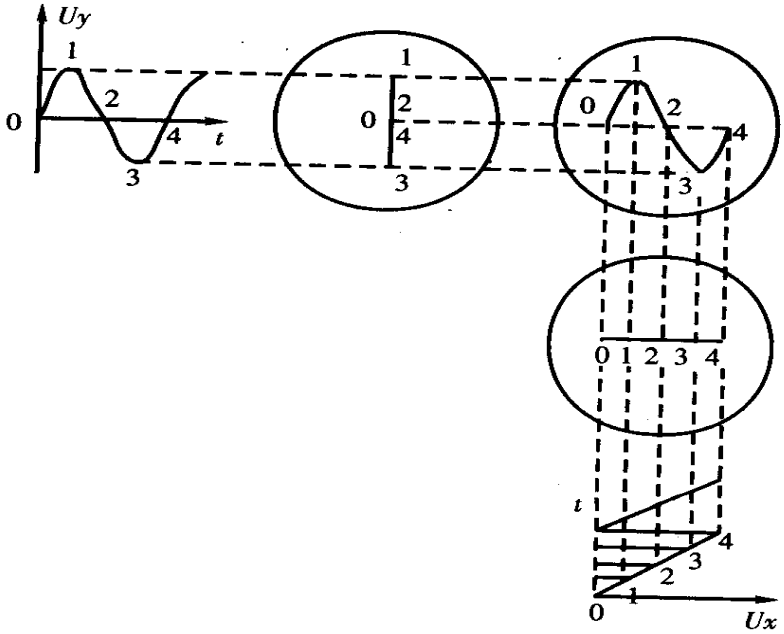


Figure 2. Scanning principle

3. Synchronization

If the cycles are slightly different between the sine wave and sawtooth wave, a moving unstable figure will appear on the screen. This can be illustrated by Figure 3. Let the period of sawtooth wave be slightly smaller than that of sine wave, for example. In the first scanning cycle, the curve segment between 0 and 4 points of sinusoidal signal is displayed on the screen; in the second cycle, the curve segment between 4 and 8 points is displayed with the starting point at $4'$; in the third cycle, the curve segment between 8 and 11 points is displayed with the starting point at $8'$. In this way, the waveforms displayed on the screen do not overlap each time, as if the waveforms are moving to the right. Similarly, if the ratio is slightly larger, it seems to be moving to the left. The situation described above often appears in the use of oscilloscope. The reason is that the period of the scanning wave is not equal to or integral multiple of the period of the measured signal, so that the starting point of the waveform curve is not the same at the beginning of each scanning.

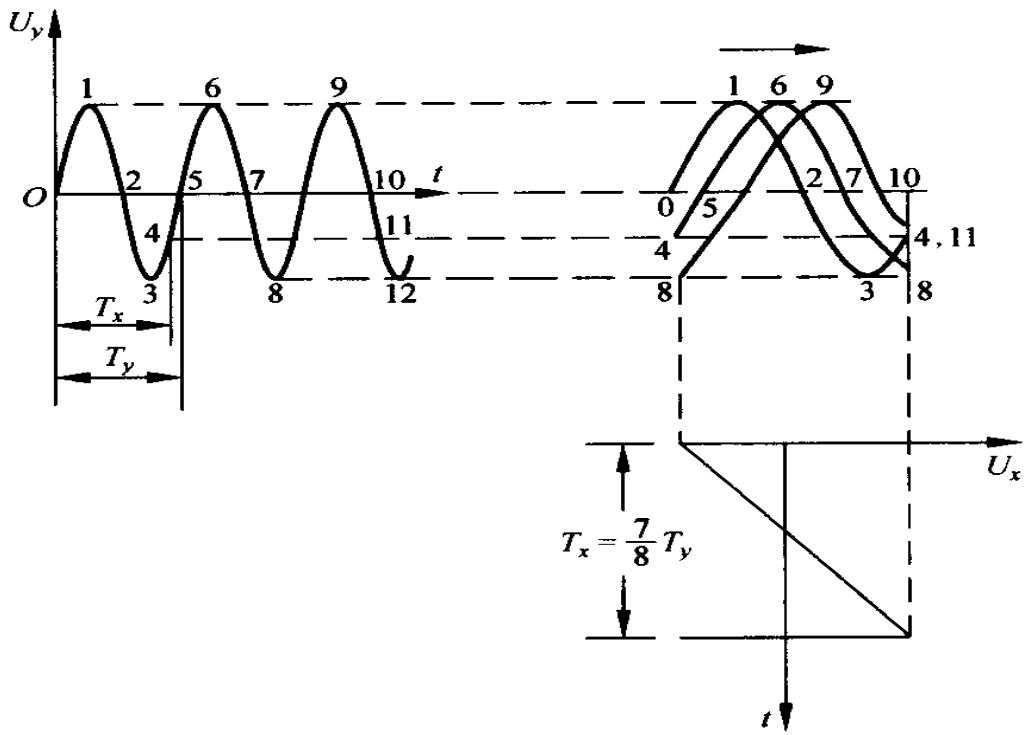


Figure 3. $T_x = (7/8)T_y$ waveform

4. Application of oscilloscope

(1) measure the voltage and period of AC signal

When measuring AC voltage with oscilloscope, it is generally to measure its peak-to-peak value U_{PP}, that is, the value from peak to trough, as shown in Figure 4. In the experiment, the scale in front of the screen shows the distance y in the vertical direction corresponding to the peak-peak voltage and the distance x in the horizontal direction corresponding to a periodic waveform.

Peak-to-peak voltage: $U_{PP} = y(cm) \times Y_{Scale}(V/cm)$ or (mV/cm)

Signal period: $T = x(cm) \times X_{Scale}(s/cm)$或$(ms/cm)$、$(\mu s/cm)$

Frequency: $f = 1/T$

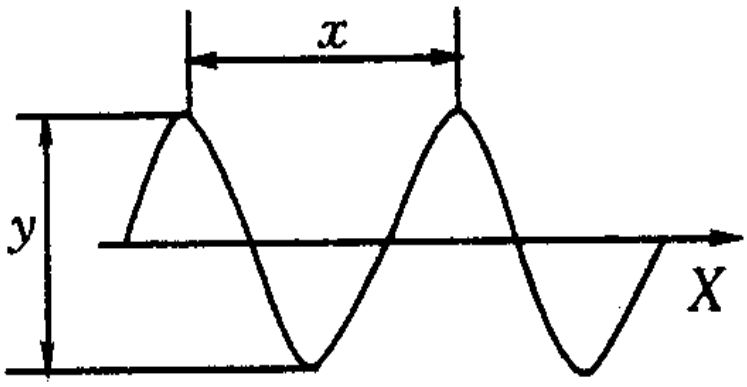


Figure 4

When measuring the voltage of the measured signal, its amplitude should be enlarged as much as possible by adjusting the y-axis scale (V/DIV), but not beyond the display screen. When measuring the period and frequency of the measured signal, the horizontal distance between the two peaks of the measured signal should be enlarged as much as possible by adjusting the x-axis scale（t/DIV）, but it should not exceed the display screen.

(2) Lissajous figure

When the x-axis input scanning signal, the oscilloscope can display the transient process of the y-axis input electrical signal. Now the x-axis input sinusoidal signal and the y-axis input another sinusoidal signal. When the frequencies of the two sinusoidal signals are equal or in a simple integer ratio, the composite pattern of the electron beam subjected to two mutually perpendicular resonant motions is observed. The synthesis track of the bright spot on the screen is a stable closed curve, which is called Lissajous figure, as shown in Figure 5.

We use f_x and f_y as the frequency of sine type in X direction and Y direction, respectively. For example, when the stable Lissajous figure is displayed on the screen, make two straight lines tangent to the figure in the horizontal and vertical directions respectively, and count the number of tangent points between the two straight lines and the figure. We can get

$$\frac{f_y}{f_x} = \frac{\text{X-dierection tangential points}}{\text{Y-dierection tangential points}}$$

As shown in Figure 5, the number of tangent points between horizontal line and

figure is 1 (*a*); the number of tangent points between vertical line and figure is 2 (*b*, *c*).

$$\frac{f_y}{f_x} = \frac{1}{2}$$

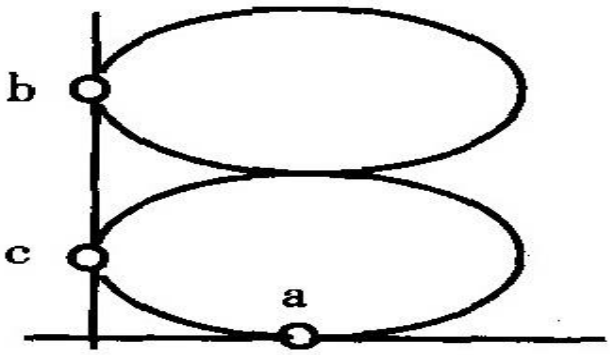


Figure 5

By counting the number of tangent points between horizontal line and figure and the number of tangent points between vertical line and figure on the screen, another frequency f_x (or f_y) can be obtained from a known frequency f_y (or f_x).

Experimental apparatus

oscilloscope, signal generator, etc.

Experiment content

1. Measurement of the calibration signal waveform

Press *Power* key, and the indicator light is on. Press *default* to restore the initial setting. Connect the calibration signal output end of oscilloscope with Ch1 (or CH2) input socket by using probe. Press *Run* and *Autoset*. The stable square wave waveform can be displayed on the screen. Adjust the X and Y shift knobs to make the waveform in a proper position for easy reading.

2. Measurement of DC voltage

The first step is to determine the position of the 0V baseline on the screen. Press *CH1* key if the probe is connected with Ch1 input socket. Press the leftmost bottom menu key (AC-GND-DC) to select GND. Adjust the CH1 *Position* knob to make the scan line coincide with the central scale line of the screen, and determine the zero-level baseline. In this measurement, the "CH1 *Position* " knob cannot be changed in the following part, otherwise the reading results is inaccurate.

Using 1.5V dry battery as DC voltage source, the oscilloscope panel switch is changed as follows: The "AC-GND-DC " selection switch is changed to "DC", and

CH1 *Scale* switch is adjusted to "0.5v/div". Connect the two output terminals of dry battery with CH1 input socket of oscilloscope by using probe. Observe the direction and amplitude of scanning line jump. Record the data. After measuring, remove the connecting probe from the dry battery.

3. Measurement of AC voltage:

The function signal generator is used as AC voltage source.

The " AC-GND-DC " selection switch is changed to "AC", and the "CH1 Scale" switch of attenuator is adjusted to 1V / Div. When the power supply of the signal generator is "off", the probe is used to connect the output end of the signal source with the CH1 socket of the oscilloscope. Turn on the power switch of the signal generator, adjust the signal to a certain frequency, adjust the output voltage of the signal, and correspondingly adjust the oscilloscope scanning speed "t/div Scale" switch and "level" trigger level knob, so that the stable waveform of 6 DIV height can be displayed on the oscillograph screen. Adjust the signal generator "output attenuation" to 40dB, adjust the oscilloscope " CH1 Scale V/div" switch and "level" trigger level knob accordingly, display stable waveform, measure output voltage and record data.

4. Frequency measurement

On the basis of the above AC voltage measurement, the frequency of signal generator is changed to 0.1khz, 1kHz and 10kHz in turn, and the scanning speed "t / div" switch and "level" trigger level knob of oscilloscope are adjusted accordingly to display stable waveform on oscillograph screen. Read out the horizontal distance L between two adjacent peaks or troughs on the waveform, then calculate the signal period T and frequency f, and record the data.

5. Frequency measurement of sinusoidal AC signal by Lissajous figure

First，set f_x as 50Hz using Digital Oscilloscope. 1.Press *Option* key. 2. Select *Arbitrary Wave Generator (AWG).* 3. Press *Output setup*. 4. Press *Output* to select channel on. 5. From AWG menu press *GEN1 setup* to select the waveform for generator 1. 6. Press *Waveform* from the bottom menu, and select *Sine* using variable knob. 7. Press *Frequency* to set the frequency rate of the waveform. 8. select *50Hz* using variable knob.

To observe Lissajous figure, press *Acquire*, press *XY*, and turn on *XY Open*. At this time, channel 1 of the oscilloscope becomes a horizontal input. Then connect the 50Hz signal to CH1 input socket as the X-axis input signal. Connect the output signal of the signal generator to the CH2 input socket as the Y-axis input signal, and adjust the output voltage to the appropriate size.

Adjust the y-axis "Scale V / div" level and the output signal strength of the signal

generator properly to make the figure moderate, and get the Lissajous figure one by one to determine the frequency ratio on the oscilloscope. To determine the frequency of the measured signal. In fact, it is impossible to adjust to an accurate simple integer ratio during operation, and the graph is difficult to be completely stable, so it can be adjusted to the slowest graph change.

Notice: To achieve a stable waveform. 1.The trigger channel is consistent with the input signal channel. 2.The trigger level value should be between the maximum value and the minimum value of the input signal, and pressing *50%* key is a shortcut.

Experiment data and analysis

1. Measurement of calibration signal waveform

$Y_{p\text{-}p}$ Peak-to-peak		Period T	
Y Scale		X Scale	
Height		Length	
$Y_{p\text{-}p}$		T	

2. Measurement of DC voltage

Y Scale	Y Height	Voltage

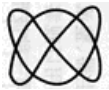
3. Measurement of AC voltage

Signal generator output attenuation		0dB	40dB
	Y Scale		
	Height		
	$Y_{p\text{-}p}$		

4. Frequency measurement

Frequency/KHz		0.1	1	10
	X Scale			
	Length			
	Period			
	Frequency			

5. Frequency measurement with Lissajous figure

Pattern				
n_x				
n_y				
$n_x : n_y$				
Signal generator frequency f_y				
f_x				

Precautions

1. According to the instrument manual provided by the laboratory, we must make clear the function of each knob on the oscilloscope and signal generator panel, and then start the experiment.

2. All switches and knobs on the oscilloscope have a certain strength and adjustment angle. When using, they should be rotated gently and slowly. They should not be rotated too hard or arbitrarily.

Preview thinking questions

1. If there is no X-axis scanning signal, what waveform is displayed on the screen?

Discussion questions

1. If the waveform always moves horizontally, how to adjust it?

20 Adjustment of spectrometer and measurement of refractive index of prism

Reflection, refraction, and diffraction occur when light incident onto optical elements such as plane mirrors, prisms, and gratings. Spectrometer is an instrument used to accurately measure the angle of deflection between incident light and outgoing light. It can also be used to measure refractive index, wavelength of light, dispersion index and many other physical quantities. There are many similarities between the basic components and adjustment principle of spectrometer and other more complex optical instruments (such as spectrograph, monochromator, etc.). Learning and using spectrometer will lay a good foundation for the future use of precision optical instruments.

［Purpose］

1. Understand the structure of spectrometer and the function of each component, and correctly master the requirements and methods of adjusting spectrometer；

2. The vertex angle of prism was measured and the dispersion of mercury lamp was observed；

3. The refractive index of glass prism to monochromatic light was measured.

Practice 1 Adjustment of spectrometer and determination of vertex angle of prism

［Experimental principle］

The prism is shown in Figure 20-1. AB and AC are two transparent optical surfaces, which are called refraction surface, and their included angle α is called the vertex angle of the prism; BC is ground glass surface, which is called the bottom surface of the prism.

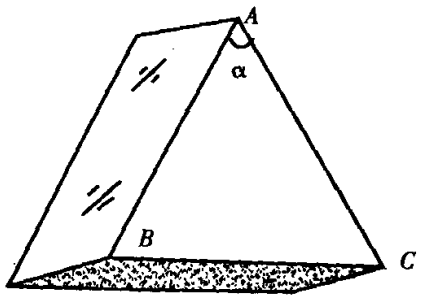


Fig. 20-1

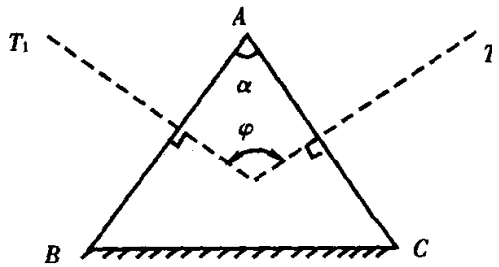


Fig. 20-2

1．Measurement of vertex angle of prism by autocollimation method

Figure 20-2 is a schematic diagram of measuring the vertex angle of a prism by self collimation method. When the light is reflected back along the original path perpendicular to the AB plane, record the incident direction $\boldsymbol{T}_1$, then make the light incident perpendicular to the AC plane, and record the reflected direction $\boldsymbol{T}_2$ along the original path, then the angle $\varphi = \left|\boldsymbol{T}_2 - \boldsymbol{T}_1\right|$ and the vertex angle α＝180°－φ, that is,

$$\alpha = 180^0 - \left|\boldsymbol{T}_2 - \boldsymbol{T}_1\right| \qquad (1)$$

2．Spectrometer

（1）structure

There are many types of spectrometers with basically the same structure. They are all composed of four parts: collimator, autocollimator telescope, loading platform and reading device (see Fig. 20-3). The lower part of the spectrometer has a vertical axis, which is called the central axis of the spectrometer.

1）Auto-collimating telescope (Abbe type). Abbe auto-collimating telescope has three parts: eyepiece, reticule and objective lens, just like the general telescope. A "+" directrix is depicted on the reticle, and a small total reflection prism is glued on the edge. Its surface is coated with an opaque film, and a hollow cross window is carved on the film, After the light emitted by the small electric bead enters from the tube side, adjust the front and rear position of the eyepiece, and you can see the scene

as shown in fig. 20-4a in the field of view of the eyepiece of the telescope. A plane mirror is placed before the objective lens to adjust the distance between the eyepiece (together with the reticle) and the objective lens before and after the objective lens, so that when the reticle is on the focal plane of the objective lens, After passing through the objective lens, the light emitted by the small electric bead passes through the hollow ten ampere window and forms a parallel light to the plane lens. After passing through the objective lens, the reflected light forms an image of a cross window on the reticle. If the plane mirror is perpendicular to the optical axis of the telescope, the image will fall on the intersection above the guide line, as shown in fig. 20-4 (b).

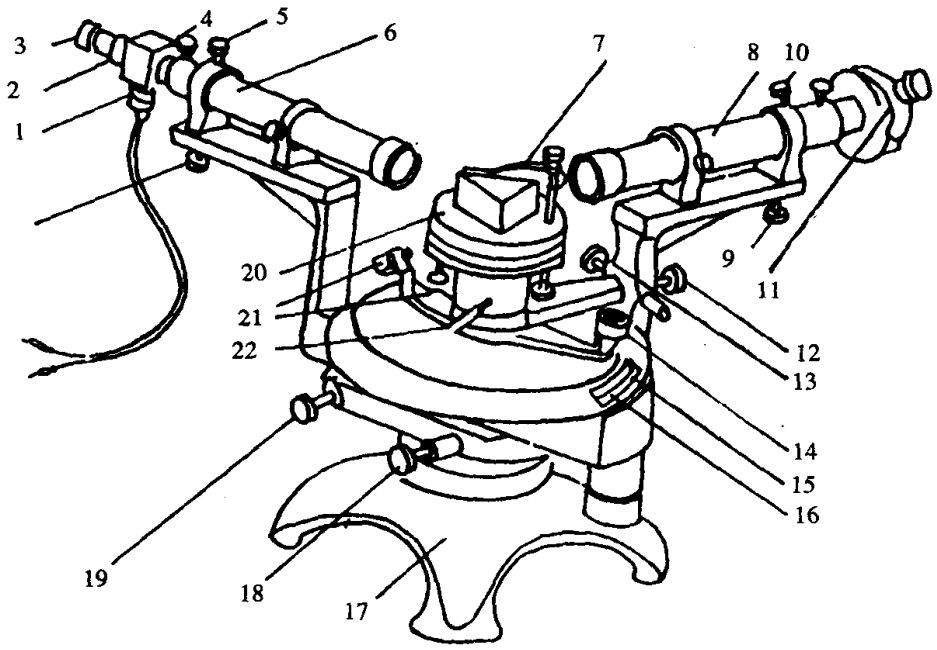


Fig.20-3 1. Little lamp; 2. Refining board sleeve; 3. eyepiece; 4. Eyepiece cylinder brake screw; 5. Telescope tilt adjusting screw; 6. Telescope barrel; 7. Clamp the spring plate of the workpiece to be tested; 8. Collimator; 9. Collimator tilt adjusting screw; 10. Slit sleeve brake screw; 11. Slit width adjusting handwheel; 12. Vernier disc brake screw; 13. Vernier disc adjusting screw; 20. magnifier; 15. Vernier disk; 16. Dial; 17. base; 18. Dial brake screw; 19. Micro adjustment screw of scale disc; 20. Loading platform; 21. Level adjusting screw of stage; 22. Table fastening screw

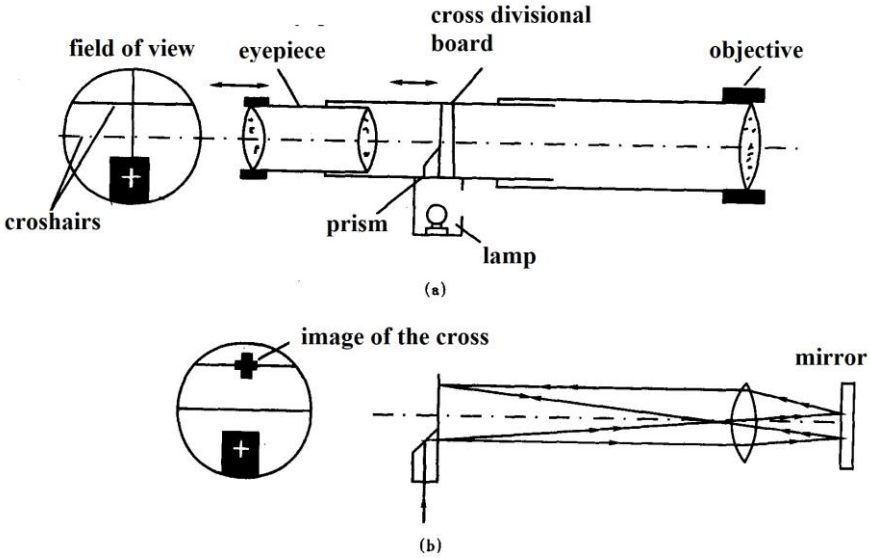


Fig.20-4 Abbe autocollimation telescope

2）Collimator Which is composed of a slit with adjustable width and position and a convergent lens. As shown in figure 20-5. When the slit is located on the focal plane of the lens, all the light coming into the collimator from the slit will become parallel beam after passing through the lens.

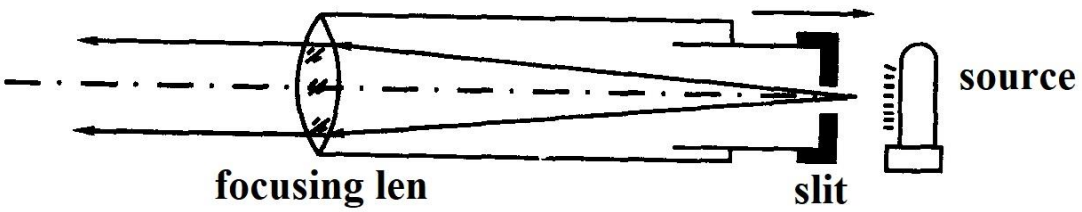


Fig.20-5 Collimator

3）**objective table.** The objective table is set on the vernier disk and can rotate around the central axis. It is set for placing prism, grating or other optical elements to be measured. There are three screws under the table to adjust the level of the platform.

4）**Reading device.** The reading device is composed of a scale disc and two cursors T and T' symmetrically 180^0 apart along the edge of the disc. The

difference between the two dials is 360^0, The minimum division is 0.5^0 （$30'$）, The readings less than 0.5^0 are read by vernier. There are 30 squares on the cursor, so the reading unit on the cursor is $1'$. The reading method of angle vernier is the same as that of vernier. As shown in fig. 20-6. The reading is:

$$87^0(30'+15')=87^0 45'$$

The two cursors are placed symmetrically to eliminate the eccentricity between the center of the dial and the center axis of the spectrometer. Record the readings of the two cursors at the same time.

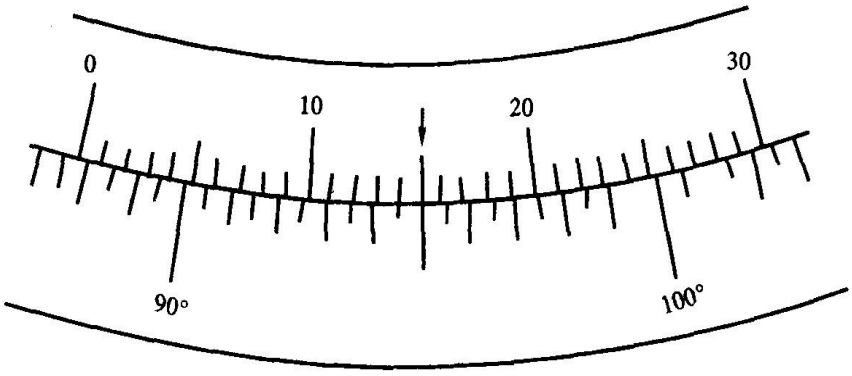


Fig.20-6 Angle vernier reading

（2）adjustment In order to measure the angle accurately, the function of each part on the spectrometer should be understood before measurement so as to adjust it. A spectrometer that has been adjusted must have the following three conditions: ① Telescopes focus at infinity, or are suitable for observing parallel light; ② The light emitted by the collimator is parallel light - that is, the position of the slit is just at the focal plane of the collimator lens (objective). ③ The central optical axis of telescope and collimator must be perpendicular to the central axis of spectrometer.

［Experimental apparatus］

Spectrometer, mercury lamp, prism, plane mirror.

［Experiment content］

1. Adjustment of spectrometer

Various types of spectrometers are composed of the four parts mentioned above, but the position of the adjusting screw of each part is different. Therefore, before adjusting, you should be familiar with the position of the following screws in the spectrometer:

①Eyepiece focusing (see the reticle alignment) handwheel;②Telescope focus (see objects) adjusting hand wheel;③Screw for adjusting the high and low inclination of telescope;④The brake screw for controlling the rotation of the loading table;⑤Screw for adjusting the level of the stage;⑥The brake screw for controlling the rotation of the loading table;⑦A screw for adjusting the width of the slit on the collimator;⑧A screw for adjusting the high and low inclination of the collimator;⑨Slit sleeve brake screw for collimator focusing.

(1) Visual rough adjustment According to the rough estimation of the eye, adjust the telescope and collimator to be roughly horizontal; Adjust the three horizontal adjusting screws under the stage to make the stage roughly horizontal. This step of coarse tuning is the premise of the following fine tuning, and also the guarantee of the success of fine tuning.

(2) Adjusting telescope focus to infinity by autocollimation

1） Adjust the eyepiece focusing hand wheel until you can clearly see the "alignment line" of the reticle.

2） Connect the power supply of the small lighting lamp, turn on the switch, and you can see the "guide line" as shown in fig.20-4 (a) and the small cross window with green in the field of view of the eyepiece.

3） Place the flat mirror on the stage according to the orientation shown in fig. 20-7, This arrangement is based on the following consideration: to adjust the elevation of the flat mirror, you only need to adjust the screw a or C under the stage, while the adjustment of screw B has nothing to do with the elevation of the flat mirror.

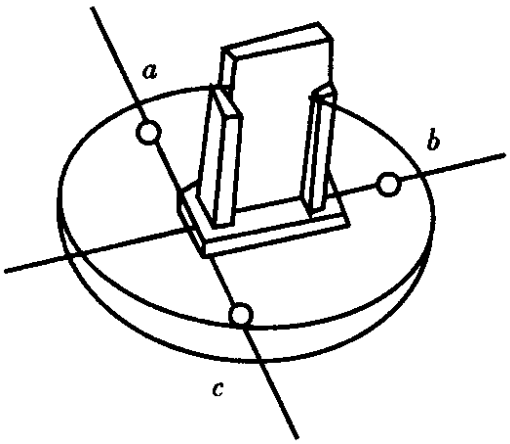


Fig.20-7 Position the plane mirror

4）From the outside view of the telescope, we can see that there is a bright cross in the plane mirror. When the stage rotates gently, the bright cross will also rotate. But if you look at the plane mirror with a telescope, you can't see the bright cross, which means that the light from the telescope is not reflected into the telescope by the plane mirror. We still aim the telescope at the plane mirror on the stage, adjust the elevation of the mirror, and rotate the stage to let the reflected light return to the telescope, so that the light emitted by the transparent cross passes through the objective lens (at this time, the light from the objective lens is not necessarily parallel light) and then reflects through the plane mirror, and the objective lens focuses again, thus forming a fuzzy image spot on the reticle（Note: the above steps are the key to the smooth adjustment）. Then adjust the distance between the objective lens and the reticle plate, and then adjust the distance between the reticle plate and the eyepiece, so that you can see the collimation line and the reflection image of the clear cross from the eyepiece. Pay attention to whether there is parallax between the guide line and the reflected image of the bright cross. If there is parallax, it needs to be adjusted repeatedly to eliminate it. If there is no parallax, the telescope is focused at infinity.

（3）Adjust the optical axis of the telescope to be perpendicular to the central axis of the spectrometer

The optical axes of collimator and telescope represent the direction of incident

light and outgoing light respectively. In order to measure the angle accurately, their optical axes must be parallel to the dial. The dial is perpendicular to the central axis of the spectrometer when it is manufactured. Therefore, when the telescope is perpendicular to the central axis of the spectrometer, it meets the requirement of being parallel to the dial. The specific adjustment methods are as follows:

The plane mirror is still vertically placed on the stage, so that the telescope is aimed at the front and rear mirrors of the plane mirror respectively, and the reflection images of two bright crosses can be observed by using the autocollimation method. If the optical axis of the telescope is perpendicular to the central axis of the spectrometer, and the reflecting surface of the plane mirror is parallel to the central axis, when the stage is rotated, it can be observed twice from the telescope that the bright cross image reflected from the front and rear surfaces of the plane mirror completely coincides with the upper cross line of the guide line of the reticle, as shown in figure 20-8 (c). If the optical axis of the telescope is not perpendicular to the central axis of the spectrometer, and the reflecting surface of the plane mirror is not parallel to the central axis, the two bright cross reflections observed from the telescope will not coincide with the upper cross of the guide line of the reticle at the same time when the stage is rotated, but one is low, one is high, and even only one can be seen. At this time, we need to carefully analyze and determine the adjustment measures. We must not adjust blindly. It's important to make coarse adjustment first, that is, to make visual observation from the outside of the telescope, so that two bright cross images can be observed from the outside of the telescope; Then fine tuning: when observing from the field of view of the telescope, no matter which reflector of the plane mirror is used to align the telescope, ten bright images can be observed, If you can see from the telescope that the collimator does not coincide with the bright cross image, their intersection points differ by a certain distance h in the high and low methods, as shown in figure 20-8 (a). At this time, adjust the telescope tilt screw to reduce the gap to $\frac{1}{2}h$, as shown in figure 20-8 (b); Then adjust the horizontal adjusting screw under

the loading table to eliminate the other half of the distance, so that the upper cross line of the guide line coincides with the bright cross line, as shown in fig. 20-8 (c). Then rotate the stage 180^0 so that the telescope faces the other side of the plane mirror and adjust it in the same way. Repeat the adjustment until the bright cross image reflected from the front and rear surfaces of the plane mirror coincides with the upper cross line of the reticle guide line when the stage is rotated. At this time, the optical axis of the telescope is perpendicular to the central axis of the spectrometer. This method is often called successive approximation half adjustment method.

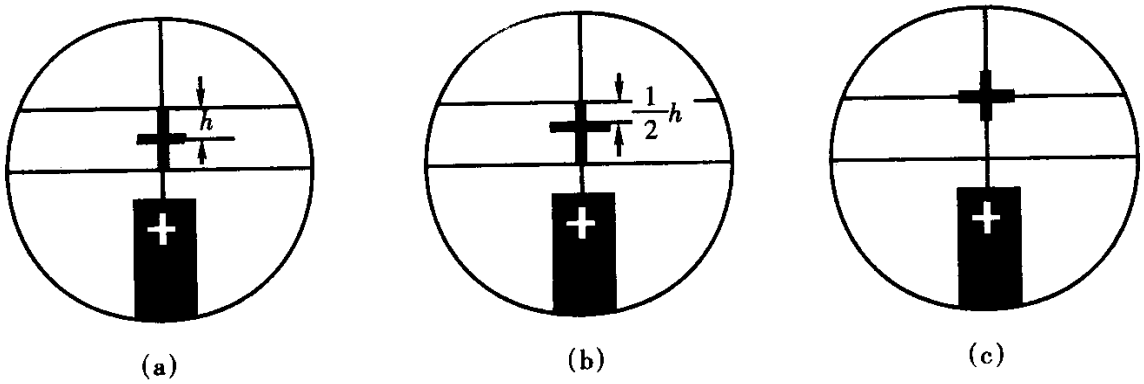


Fig.20-8 Successive approximation and half adjustment method.

（4）Adjust collimator

Adjust the collimator with the previously adjusted telescope. When the collimator emits parallel light, the slit is imaged on the focal plane of the objective lens of the telescope, and the slit image can be clearly seen in the telescope without parallax.

1) Adjust the collimator to produce parallel light. Take the flat mirror on the downloading stage, turn off the small light in the telescope, illuminate the slit with mercury lamp, observe the slit image from the telescope, and adjust the distance between the parallel tube and the lens, until the clear slit image is seen, then adjust the width of the slit to make the width of the seam in the telescope field 1mm.

2) Adjust the optical axis of the collimator to be perpendicular to the central axis of the spectrometer. After seeing the clear slit image, turn the slit (but it can't move

back and forth) to a horizontal state, adjust the inclined screw of the collimator, so that the slit horizontal image is bisected by the horizontal line of the central cross on the reticle, as shown in Fig. 20-9 (a). At this time, the optical axis of the collimator is perpendicular to the central axis of the spectrometer. Then turn the slit to the vertical position and keep the slit image clear and parallax free, as shown in fig. 20-9 (b).

So far, all the spectrometers have been adjusted. When using the spectrometers, it must be noted that the other screws on the spectrometers can not be rotated at will except for the brake screw of the dial and its fine adjustment screw. Otherwise, the working conditions of the spectrometers will be damaged and the spectrometers must be readjusted.

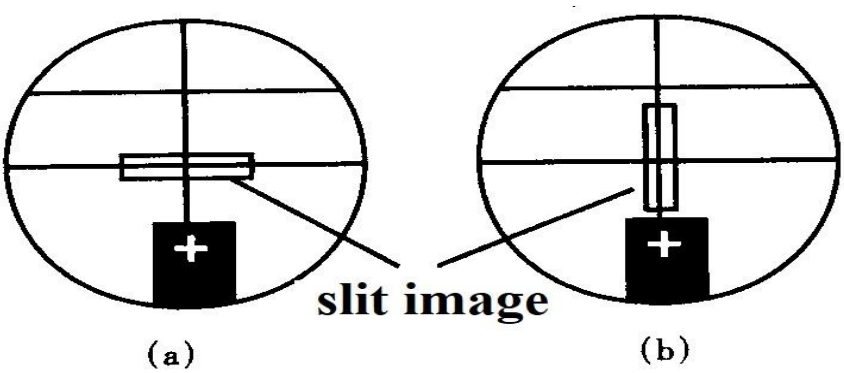


Fig.20-9 Schematic diagram of collimator adjustment

2． Measure

Before formal measurement, please find out the position of the following screws in the spectrometer you are using:

①A brake screw for controlling the rotation of the telescope (together with the dial); ②Control telescope micro screw;③ A stop screw and a fine adjustment screw for controlling the rotation of the cursor disk (together with the stage).

（1）Measurement of vertex angle of prism by autocollimation method α

1 ）The adjustment of prism. The normals of the two refraction surfaces of the prism should be perpendicular to the central axis of the spectrometer. The adjustment method is based on the principle of autocollimation, and the telescope has been

adjusted. For the convenience of adjustment, the prism should be placed on the stage according to the position shown in fig. 20-10, so that the connecting lines of each two of the three screws A, B and C under the platform are orthogonal to the mirror surface of the prism. Adjust the stage so that a refracting surface AC of the prism is aligned with the telescope, and adjust the horizontal adjusting screw B under the stage below the AC surface so that the horizontal line on the upper part of the alignment line coincides with the bright cross image (note that the telescope has been adjusted and can not be adjusted again); Then rotate the stage so that the other refraction surface ab of the prism is aligned with the telescope and adjusted to coincide in the same way. After repeated adjustment, the bright cross image reflected from AC and ab surfaces can coincide with the upper cross line of the reticle guide line. At this time, the normals of the two refraction surfaces of the prism are perpendicular to the central axis of the spectrometer.

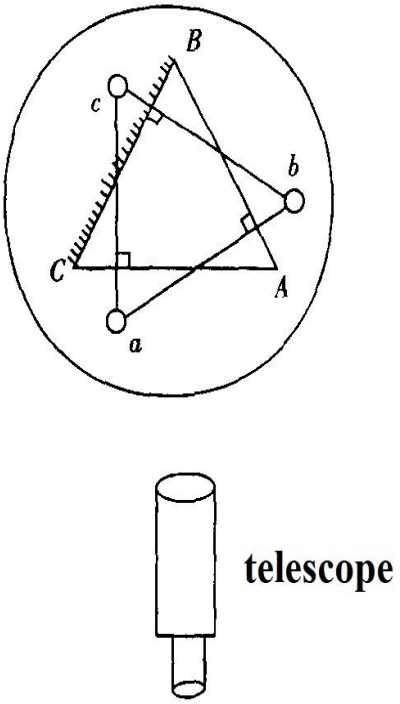


Fig. 20-10 Measurement of vertex angle of prism by autocollimation method α

2）Measurement of vertex angle. According to fig. 20-10, the optical axis of the telescope is perpendicular to the AB plane, and the angle position values θ_1 and θ_2

of the telescope are read out by two cursors; then the optical axis of the telescope is perpendicular to the AC plane, and the positions θ_1' and θ_2' of the telescope are read out by two cursors, then the vertex angle of the prism is:

$$\alpha = 180^0 - \frac{1}{2}(|\theta_1' - \theta_1| + |\theta_2' - \theta_2)|)$$

Repeat the measurement for 4 times and calculate the average value of vertex angle α.

Practice 2 Measurement of refractive index of prism

【Experimental principle】

The refractive index is calculated from the minimum deflection angle of each monochromatic light;

The refractive index can be obtained by the law of light reflection and refraction n:

$$n = \frac{\sin i_1}{\sin i_2} = \frac{\sin\frac{1}{2}(\delta_{min} + \alpha)}{\sin\frac{1}{2}\alpha} \qquad (2)$$

Where i_1, i_2 and δ_{min} are the incident angle, refraction angle and minimum deflection angle of a monochromatic light respectively, and α is the vertex angle of the prism.

By measuring the vertex angle α and the minimum deflection angle β_{min} with a spectrometer, the refractive index n of the prism to this kind of light can be calculated from the above formula. The refractive index is a function of the wavelength of light wave. After the light from a non monochromatic light source (such as a mercury lamp) is refracted by a prism, each monochromatic light component (spectral line) will have different deflection angles, which can be measured separately.

【Experiment content】

Measure the minimum deflection angle of monochromatic light in low pressure mercury lamp

1. When the collimator slit is aligned with the light source and the prism, telescope and collimator are in the relative position as shown in Fig. 20-11, the color spectral line (monochromatic image of slit) can be seen in the telescope. Adjust the slit width so that the spectral lines can be imaged on the plane of the dividing plate of the telescope.

2. A kind of monochromatic spectral line is identified to make the spectral line close to the direction of the incident light, that is, reduce the angle β, continue to rotate the stage, and rotate the telescope to track the spectral line until the spectral line reverses when the prism continues to rotate in the same direction. This turning point is the position of the minimum deflection angle corresponding to the spectral line.

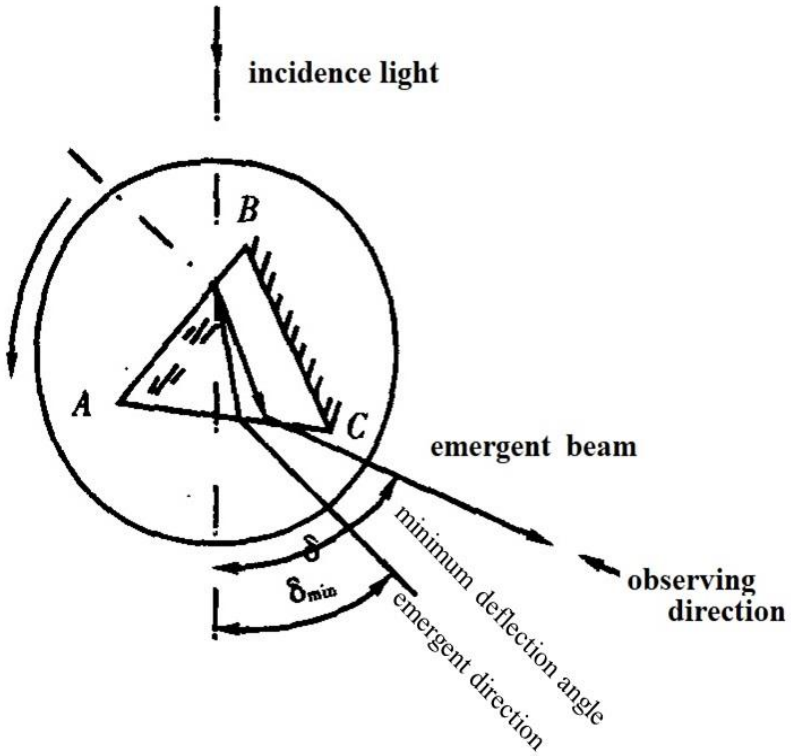


Fig.20—11 Schematic diagram of minimum deflection angle

3. Observe the identified monochromatic spectral line through the telescope, and carefully re observe the movement of the spectral line when the stage rotates (note that the center of the collimator in the field of view of the telescope should always track the spectral line), so that the monochromatic spectral line just reverses at the collimator in the field of view of the telescope (the position of the telescope is the minimum deviation position of the monochromatic spectral line).

4. Read the corresponding readings θ_1 and θ_2 on the double vernier at this time.

5. The azimuth reading of the minimum deviation angle of the four single spectral lines of yellow, grass green, blue and purple in the mercury lamp spectrum was measured in turn.

6. Remove the prism, align the telescope with the collimator, align the collimator with the midpoint of the slit, and read the corresponding readings θ_1' and θ_2' of the two cursors.

7. According to the formula $\delta_{\mathbf{min}} = \frac{1}{2}(|\theta_1' - \theta_1| + |\theta_2' - \theta_2|)$, calculate the minimum deflection angle $\beta_{\min}$, repeat the measurement for four times, and calculate the average value of $\beta_{\min}$.

［Data processing and analysis］

1. All measurement data are recorded in the list. Please make the form yourself.

2. Substituting the measured vertex angle α and minimum deflection angle $\beta_{\min}$ into formula (2), the refractive index of each monochromatic light is calculated, and the law of the refractive index of prism changing with wavelength is analyzed.

［Matters needing attention］

1. Do not touch or wipe the lens on the telescope, parallel lamp tube, prism or

plane mirror. If dust is found, wipe it gently with lens paper. It is not allowed to bump or drop the prism and plane mirror to avoid damage.

2. The spectrometer is a precision instrument. All moving parts should be operated carefully. When the rotatable parts (such as telescope and vernier disk) can't be rotated by gently pushing them, remember not to force them to rotate, so as not to wear the rotating shaft of the instrument. In order to avoid this situation, check whether the stop screw is loosened before turning the telescope and the cursor disk each time.

3. When adjusting the width of the slit, it must not be closed to avoid serious damage to the slit.

4. In the process of vernier reading, since the telescope may be in any position, it should be noted whether the telescope has crossed the zero point of the scale in the process of rotation. If it crosses the zero point of the scale, the telescope angle must be calculated according to formula $\varphi = 360^0 - |\theta' - \theta|$. For example, when the telescope turns from position I to position II, the readings of the double cursors are as follows:

Telescope position	Cursor (left)	Cursor (right)
I	$\theta_1 = 175^0 45'$	$\theta_2 = 355^0 48'$
II	$\theta'_1 = 295^0 43'$	$\theta'_2 = 115^0 44'$

From the reading of the left vernier, the telescope angle can be obtained as follows:

$$\theta_{\text{left}} = \theta'_1 - \theta_1 = 119^0 58'$$

From the right vernier reading, we can get the telescope angle as follows:

$$\theta_{\text{right}} = 360^0 - |\theta'_2 - \theta_2| = 119^0 56'$$

If $\theta_{\text{left}} \neq \theta_{\text{right}}$, there is eccentricity, so the actual rotation angle of the telescope is:

$$\theta = \frac{1}{2}(\theta_{\text{left}} + \theta_{\text{right}}) = 119^0 57'$$

5. In the dark room, when observing the image and reticle cross line through the telescope, the eyes are prone to fatigue, so one must be patient, and the other must adjust themselves in time. That is to say, after observing for a long time, you should leave the telescope, let your eyes rest, and do eye exercises when necessary.

Preview thinking questions

1. What are the main components of the Spectrometer? What are their functions?

2. What are the requirements for spectrometer adjustment?

3. When the plane mirror on the stage is rotated, the green image reflected by the mirror can not be seen in the telescope. How to adjust it?

4. How to make the Green Cross coincide with the cross line on the upper part of the reticle?

5. How to judge the minimum deviation angle?

Discussion questions

1. When the optical axis of the telescope is adjusted perpendicular to the central axis of the instrument, two kinds of phenomena may be observed; (a) The Green Cross images reflected by the two mirrors of the plane mirror are above the guide line; (b) An image reflected by two surfaces, one above and one below. The analysis shows that the two are mainly caused by the tilt of the telescope return stage, and how to quickly make the horizontal lines of the images reflected by the two planes coincide with the horizontal lines above the collimator.

2. If the distance between the upper horizontal line and the lower horizontal line is a, and the other is 5a, how to adjust it?

3. If the minimum deflection angle of one kind of monochromatic light has been found, is the other monochromatic light also at the minimum deflection angle? What is the general relationship between β_{min} and λ? What color is the refractive index of glass?

Experiment 30 Application of Interference Phenomenon of Light-Newton's Rings and Air Wedge

If the light emitted from the same point light source is divided into two beams, and they meet again after going through different distances, interference phenomenon will generally occur when the optical path difference is less than the coherence length of the light source. The interference phenomenon confirms the wave aspect of light. Interference phenomenon is widely used in scientific research and industrial technology, such as measuring the wavelength of light waves, accurately measuring the length, thickness and angle, checking the surface smoothness of tested pieces, studying the distribution of internal stress in mechanical parts, and measuring the thickness of oxide layer on silicon wafers in semiconductor technology.

[experimental purpose]

1. Observe the phenomenon of interferences with equal thickness, and understand the characteristics of interferences with equal thickness.
2. Learn the measurement method of physical quantities such as the curvature radius and film thickness of a plano-convex mirror using equal thickness interference method.
3. Get familiar with the reading microscope.

[experimental principle]

1. Newton's rings

Newton's rings are composed of a plano-convex lens with a large curvature radius. Its convex surface is placed on an optical flat glass (flat crystal), as shown in figure 30-1. The thickness of the air layer between the convex surface of a plano-convex lens and the flat crystal gradually increases from the center to the edge. If the parallel monochromatic light strikes the Newton's rings vertically, there will be an optical path difference between the two beams reflected by the upper and lower surfaces of the air layer. The interference will occur when they meet on the convex surface of a

plano-convex lens. The interference patterns seen from the lens are a series of bright and dark rings centered on contact point of the glasses, as shown in figure 30-2, which is called Newton's rings. It is equal thickness interference, and the thickness of air at equal distance from the contact point is the same.

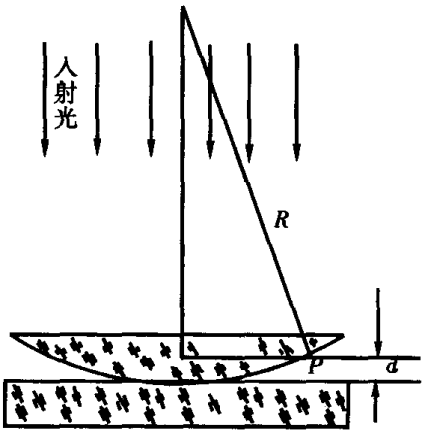


Figure 30-1

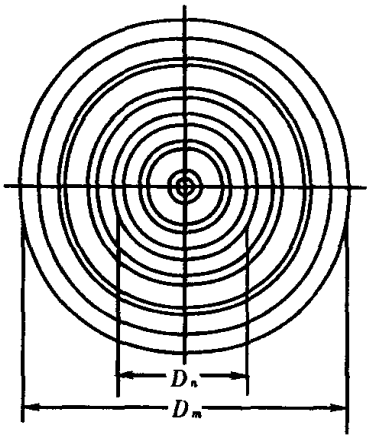


Figure 30-2

As shown in Fig. 30-1, if the radius of curvature of the lens is R, and the thickness at the distance r from the O point (contact point) is d, the geometric relationship is:

$$R^2 = (R-d)^2 + r^2 = R^2 - 2Rd + d^2 + r^2$$

For R>>d, d^2 can be omitted, thus:

$$d=r^2/2R \tag{30-1}$$

The light should be incident vertically. When calculating the optical path difference, we should also consider the additional path difference (λ/2) caused by half-wave loss when reflected on the optical flat crystal. Thus, the total distance difference is:

$$\Delta = 2d+\lambda/2 \tag{30-2}$$

For dark rings:

$$\Delta = (2m+1)\lambda/2 \quad (m=0, 1, 2, \cdots) \tag{30-3}$$

Where m is the order of interference fringes, and the radii of the mth dark rings can be described as follows:

$$r_m = \sqrt{\mathrm{mR}\lambda} \tag{30-4}$$

It can be seen from formula (30-4) that if the λ value of the monochromatic light source is known, the curvature radius R of the plano-convex lens can be obtained by measuring the mth dark rings. On the contrary, if R value is known, the wavelength λ of incident monochromatic light source can be calculated after the r_m value is measured. However, a serious error will occur when using this measurement formula because the convex surface and plane cannot be the ideal point contact. Contact pressure will cause local deformation and make the contact position a round surface, leading to a dark spot at the center of the interference rings. In addition, the possible existence of dust in the air gap layer can add an optical path difference, then the center of the interference rings will be a bright (or dark) spot, and the geometric center of the ring cannot be determined under this circumstance. In fact, the radius of curvature R can be calculated by measuring the square difference of the radius of r_m and r_n dark rings located far away from the center.

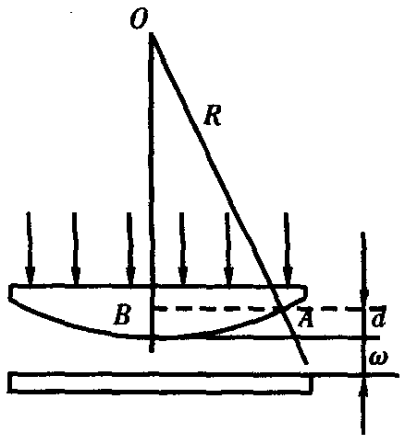
Due to: $$r_m^2 = \mathrm{mR}\lambda\ , \quad r_n^2 = \mathrm{nR}\lambda$$

Subtract the above two formulas:

$$r_m^2 - r_n^2 = R(m-n)\lambda$$

So: $$R = \frac{r_m^2 - r_n^2}{(m-n)\lambda}$$

Or: $$R = \frac{D_m^2 - D_n^2}{4(m-n)\lambda} \tag{30-5}$$



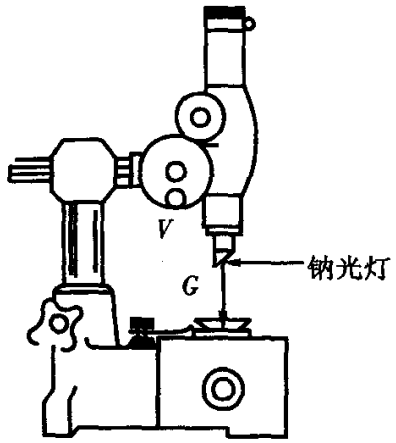


Figure 30-3 Figure 30-4 Experimental device

According to the above formula, as long as the values of dark rings D_m and D_n are measured, the values of R or λ can be calculated. In this way, the difficulty that the series and the center of rings cannot be determined in the experiment is avoided. This is also noticeable in the experimental work.

Because of the elastic deformation of glass at the contact position, the ring near the center will be moved, so it is generally advisable to measure the rings far from the center.

In this experiment, the diameter of Newton's rings is measured using a reading microscope. The construction principle and usage of reading microscope can be referred in Section 2.1.3 (Basic Instruments and Preliminary Knowledge of Optical Experiment).

2. Wedge interference

As shown in Fig. 30-5, two optical glass sheets are laminated together. A polyester film with a thickness of d is inserted at one side to form an air wedge between the lower and upper surfaces of the two glass sheets. When irradiated vertically with monochromatic light, the two beams reflected on the upper and lower surfaces of the wedge-shaped thin film will interfere with

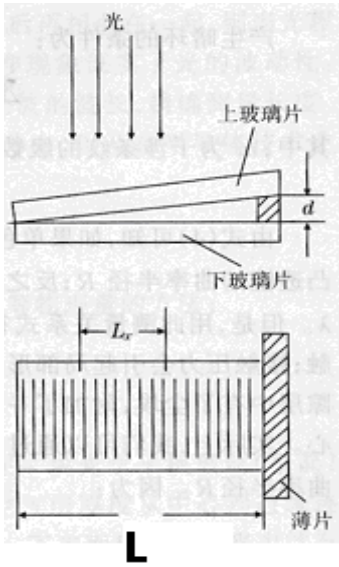


Figure 30-5 Wedge interference

each other. The resulted interference images are alternate bright and dark straight-lines, parallel to the boundary edge of two slides.

It is clear that when the optical path difference is an odd multiple of half wavelength, the reflected lights interfere to form dark lines, namely:

$$2d + \frac{\lambda}{2} = (2m+1)\frac{\lambda}{2} \quad (m = 0, 1, 2, \cdots)$$

Simplified as:

$$d = m * \frac{\lambda}{2} \quad (m = 0, 1, 2, \cdots) \tag{30-6}$$

According to the formula (30-6), as long as the total number of dark lines of m from the edge of the boundary between two glass sheets to the edge of the inner sheet of the air wedge are known, the value of d can be calculated.

[Experimental instruments]

Reading microscope, Newton's rings device, air wedge and sodium lamp, etc.

The instrument of Newton's rings is composed of a plano-convex lens with a large curvature radius laminated with an optical flat glass (flat crystal), sealed in a metal frame with three spirals. It should be noted that the spirals cannot be rotated too tightly to avoid the elastic deformation or even lead to the failure of the lens caused by excessive contact pressure.

The experimental device is shown in figure 30-4. A Newton's rings device is placed on the stage of the reading microscope, and a glass sheet G with an angle of 45° to the horizontal is installed under the objective lens of the microscope. The sodium yellow light with a wavelength of λ_D=589.3 nm emitted from the sodium lamp is reflected by the glass sheet and perpendicularly travels to the lens. The interference fringes of the reflected light-Newton's rings can be observed in the microscope.

[Experimental contents]

1. Measure the curvature radius of the plano-convex lens with the Newton's rings

device

(1) Turn on the power of a sodium lamp, and the lamp tube will be lighted up stably after a few minutes. Be careful not to switch the power supply of the sodium lamp on or off.

(2) The Newton's rings device can be adjusted by sunlight. Adjust the three screws on the device evenly and lightly until the central stripe of Newton's rings the smallest size appears in the center of the lens without distortion, and then, place it under the microscope.

(3) Gently rotate the reflective glass G on the lens barrel to make it face the sodium lamp, until the microscope field of view is bright and yellow.

(4) Observe interference fringes with a microscope. First, reduce the microscope barrel to the lowest level, then slowly raise the barrel. When seeing the fringes, slightly adjust it back and forth until the interference fringes can be clearly observed in the whole field of view.

(5) Move the eyepiece to see the cross hair in the eyepiece barrel clearyly, adjust the micrometer drum wheels V to move the microscope along the horizontal direction, so that the crosshair will sweeps each annular stripe one by one, as shown in figure 30-2. It should be noted that when the microscope barrel moves, the intersection point of the crosshair should pass through the center of Newton's rings and always be tangent to each ring, so that the diameter of each ring can be measured correctly. Otherwise, turn the eyepiece barrel to correct the direction of the crosshair. In addition, the interference fringes should be continuously observed while moving the Newton's rings device until it is well-adjusted.

(6) Measure the diameter of interference ring.

On the basis of the above steps, the measuring range of interference rings is firstly determined. In order to reduce the measurement error, the value of m-n should not be too small, so that several values of $(D_m^2 - D_n^2)$ can be measured and averaged to calculate its uncertainty. If the ring pattern near the center is fuzzy, it can be counted from the clear ring. Assuming that the 5th ring has met the measurement requirements, if m-n=10 is agreed, the measurement range is determined from 5th ring to 20th ring.

In the measuring process, in order to avoid the deadpath error caused by the thread clearance of the rotating parts, it is required to rotate the micrometer drums so that the

crosshair exceeds 25 rings to the left of the center, and then return to the 20th ring to start reading (in the measuring and reading process, it cannot be reversed to avoid errors). In the process of rotating the micrometer drums, each dark ring is read once, so that the crosshair is tangent to the outer sides of the 20th, 19th, 18th, 17th, 16th, 15th, 10th, 9th, 8th, 7th, 6th and 5th rings in sequence, and the corresponding readings are recorded. Continue to move the micrometer drums, so that the crosshair is tangent to the inner sides of 5th, 6th, 7th, 8th, 9th, 10th, 15th, 16th, 17th, 18th, 19th and 20th rings on the right side of the center, and record the readings of each ring in sequence. When measuring, it should be noted that the micrometer drums should be rotated along one direction, and should not be reversed in the middle to avoid errors caused by deadpath motion. Thus, 28 readings about position can be read on both sides of Newton's rings. Based on these parameters, the value of diameter D_i from the 20th ring to the 15th ring and the corresponding 10th ring to the 5th ring can be calculated.

According to formula (30-5), R is the radius to be measured, λ is the monochromatic light wavelength of the light source, and both R and λ are constants. If m-n is taken as a definite value (for example, m-n=10), D_m^2-D_n^2 is also a constant. That is to say, for two rings whose series are separated by 10 (for example, the 20th ring and 10th ring, 19th ring and 9th ring...), the square difference of their diameters should be constant. Therefore, in order to facilitate the measurement and improve the accuracy, the diameter of each ring can be measured successively, and then the data can be processed by the successive differential method. In this experiment, it is required to measure at least six values of D_m^2-D_n^2, and the average value is calculated to obtain the value of R.

2. Observe the wedge interference and measure the thickness of polyester film

(1) Put the wedge on the stage, adjust the lens barrel from bottom to top to obtain clear interference fringes.

(2) Basically, the total number of stripes m is relatively larger. As shown in Fig. 30-5, to avoid misreading, the lengths L_x of x stripes (x=20) are measured first, then the distance between each two adjacent stripes is L_x/x. If the distance from the place

where the film thickness is d to the boundary edge of two slides (d=0) is L, the total number of interference fringes can be described as $m=L*x/L_x$. when substituted into formula (30-6), the film thickness d is obtained:

$$d = m * \frac{\lambda}{2} = \frac{\lambda}{2} * \frac{L}{L_x} * x$$

[Data processing and analysis]

1. List and record all measurement data.

Series of rings	m	20	19	18	17	16	15
Ring position /mm	right						
	left						
Ring diameter /mm	D_m						
Series of rings	n	10	9	8	7	6	5
Ring position /mm	right						
	left						
Ring diameter /mm	D_n						
D_m^2/mm^2							
D_n^2/mm^2							
$(D_m^2 - D_n^2)/\mathrm{mm}^2$							
$\overline{(D_m^2 - D_n^2)} / mm^2$							

2. Process data by successive differential method. For these twelve diameter data of D_i, six pairs are made according to m-n=10, and the square difference $(D_m^2 - D_n^2)$ values of these six pairs of diameters are calculated respectively, thus the curvature radius r of plano-convex lens is obatined by substituting the average value into equation (30-5).

* 3. Graphical data processing

If the curvature radius R of plano-convex lens is obtained by graphic method, it is also unnecessary to determine the center of Newton's rings. Just to measure the diameter of Newton's ring. For $r^2=mR\lambda$, then:

$$D_m^2 = 4R\lambda m$$

The above formula indicates that D_m^2 has a linear relationship with m, and the slope of the straight line is $4R\lambda$. If the wavelength λ of monochromatic light source is known, the curvature radius R of plano-convex lens can be obtained.

When plotting based on D_m^2-m values, the straight line may not pass through the origin of coordinates. It is suggested that the intercept should be zero due to $r_m^2=mR\lambda$. However, if the straight line is extended, it may not pass through the origin, which is ascribed to the non-ideal point contact between the convex surface of the plano-convex lens and the flat crystal. Thus, the actually observed center of Newton's Rings is often a dark spot or bright spot which is not clear. This is because when the convex surface of plano-convex lens contacts with the flat crystal, it will be deformed due to contact pressure to make the contact position be a round surface, or there is a gap between the convex surface and the plane due to the existence of dust. Considering the general situation with gap as shown in Fig. 30-3, let the R be the curvature radius of the lens, ω be the gap between the lens and the flat crystal, and r_m be the Newton's ring radius pass through point A, thus the optical path difference between two coherent lights at point A can be described as:

$$\Delta = 2(d+\omega) + \frac{\lambda}{2}$$

The condition for dark stripes is:

$$\Delta = 2(d+\omega) + \frac{\lambda}{2} = \left(m + \frac{1}{2}\right)\lambda \quad \text{(m=0, 1, 2, } \cdots\text{)}$$

Put formula (30-1) into the above formula, then:

$$2\left(\frac{r_m^2}{2R} + \omega\right) = m\lambda$$

That is:

$$r_m^2 = \mathrm{mR}\lambda - 2R\omega$$

Or:

$$D_m^2 = 4\mathrm{mR}\lambda - 8R\omega \qquad (30\text{-}7)$$

Based on the data of diameter D_m of each ring measured by experiment and the given wavelength of monochromatic light source, the best estimation values of R and ω can be obtained by least square method according to formula (30-7). Moreover, according to the gap value ω, it can be judged whether there is a gap between the convex surface of the lens and the flat crystal or whether it is deformed due to the pressure.

The data can be processed by graphic method. Take D_m^2 as ordinate and m as the abscissa, and the slope is determined as 4Rλ, then the curvature radius R of plano-convex lens is calculated.

* 4. Use least square method to process data to calculate the radius of curvature r and the gap ω of plano-convex lens. According to the value of the gap ω, judge whether there is a gap between the convex surface of the lens and the flat crystal or whether it is deformed due to the pressure.

* 5. Calculate the thickness d of polyester film.

[Points for attention]

1. In the process of measuring the diameters of the Newton's rings, special attention should be paid to the influence of deadpath error.

2. Touch the optical surface is forbidded when taking the Newton's rings device, and wipe it with special mirror paper if it is dirty.

3. It is forbidden to turn the sodium lamp on and off continuously, it should be more than 5 minutes after being ignited or extinguished to prevent damaging the sodium lamp.

[Preview questions]

1. Why is the diameter of Newton's rings measured instead of the radius in the experiment? If the Newton's rings are not locally round, what does it mean?

2. Because there is always a gap between the screw pitches in the micrometer drum, idling will occur when the micrometer drums start to rotate reversely, resulting in deadpath error. How to avoid it in the experiment.

[Discussion questions]

If the following situations are encountered in this experiment, will they affect the experimental results? Why?

1. The center of Newton's rings is bright spot instead of dark spot.

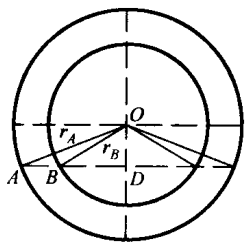


Figure 30-6

2. When measuring each value of the diameter D_m, if the intersection point of the crosshair does not pass through the center, the chord is actually measured instead of the diameter, as shown in figure 30-6.

绪论		
序号	英文	中文
1	Physical quantity	物理学量
2	Measuremrnt error	测量误差
3	Equal accuracy measurement	等精度测量
4	Systematic error	系统误差
5	Random error	随机误差
6	True value	真值
7	Uncertainty	不缺定度
8	Confidence probability	置信度
9	significant figure	有效数字
10	successive difference method	逐差法
11	least square method	最小二乘法
12	exact numbers	准确数
13	unit transformation	单位转换
14	constant	常数
15	vernier caliper	游标卡尺
16	full scale	满刻度
17	graphical representation of data	作图法处理数据
18	linear fitting	线性拟合
19	correlation coefficient	相关系数
20	best empirical formula	最佳经验公式
21	control circuit of rheostat	变阻器控制电路
22	current limiting circuit	限流电路
23	voltage divider circuit	分压电路

直流电桥测电阻		
序号	英文	中文
1	Electrical resistance	电阻
2	DC bridge	直流电桥
3	Wheatstone bridge	惠斯通电桥
4	Galvanometer	电流计
5	Bridge balance	电桥平衡
6	Resistance box	电阻箱
7		

8		

数字示波器		
序号	英文	中文
1	digital oscilloscope	数字示波器
2	waveform	波形
3	Lissajous figure	李萨如图形
4	D/A converter	数/模转换
5	microprocessor	微处理器
6	deflector	偏转板
7	Scanning principle	扫描原理
8	sawtooth wave	锯齿波

杨氏模量		
序号	英文	中文
1	Young’s modulus	杨氏模量
2	Optical lever method	光杠杆法
3	Objective lens	物镜
4	Elastic deformation	弹性形变

电阻元件伏安特性的测定		
序号	英文	中文
1	resistance	电阻
2	voltmeter-ammeter method	伏安法
3	Ohm’s law	欧姆定律
4	zener diode	稳压管
5	external connecting of ammeter	电流表外接法
6	internal connecting of ammeter	电流表内接法

分光计		
1	spectrometer	分光计
2	monochromatic light	单色光
3	prism	三棱镜
4	mercury lamp	汞灯
5	Plane mirror	平面镜
6	telescope	望远镜

光的干涉 牛顿环			
Newton’s Rings	牛顿环	half-wave loss	半波损失
Optical path difference	光程差	Wedge interference	劈尖干涉
coherence length	相干长度	interference fringes	干涉条纹
wavelength of light waves	光波波长	interference ring	干涉环纹
internal stress	应力	deadpath error	空程误差
Interferences with equal thickness	等厚干涉	contact pressure	接触压力
micrometer drum	测微鼓轮	graphic method	图解法
silicon wafers	硅片	bright spot	光斑
oxide layer	氧化层	intersection point of the crosshair	叉丝焦点
semiconductor technology	半导体技术		
curvature radius	曲率半径		